

A



2 OF 55

8 7 6 5 4 3 2 1

FIJI: DIGITAL I/O, BOOTSTRAPPING

D

C

B

A

3 OF 55

8 7 6 5 4 3 2 1

FIJI: DIGITAL I/O, BOOTSTRAPPING

D

C

B

A

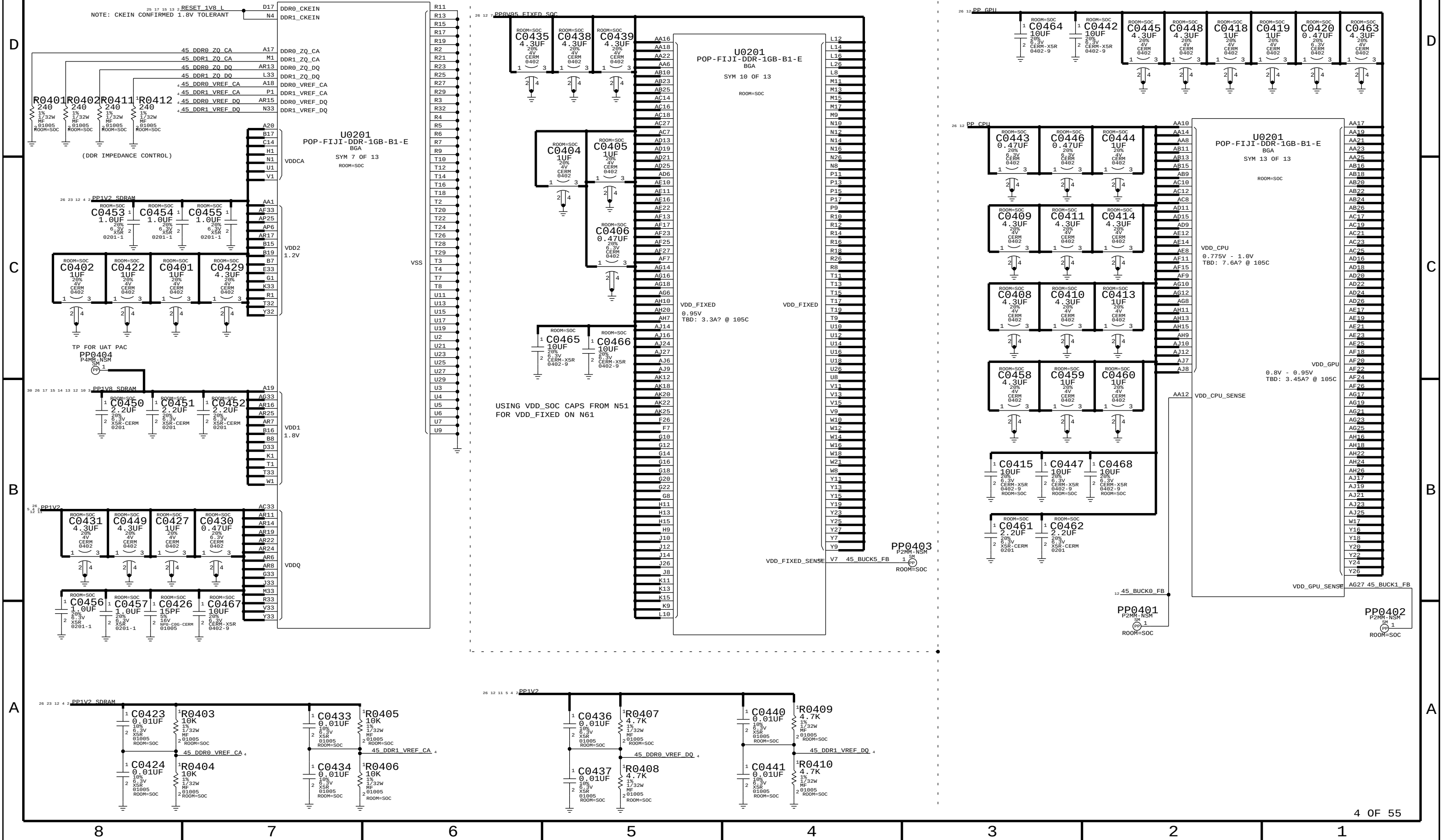
3 OF 55

FIJI: VDDCA, VDD1/2, VDDQ, VDD, VDD_FIXED, VDD_CPU, VDD_GPU

VDDCA, VDD1/2, VDDQ

VDD

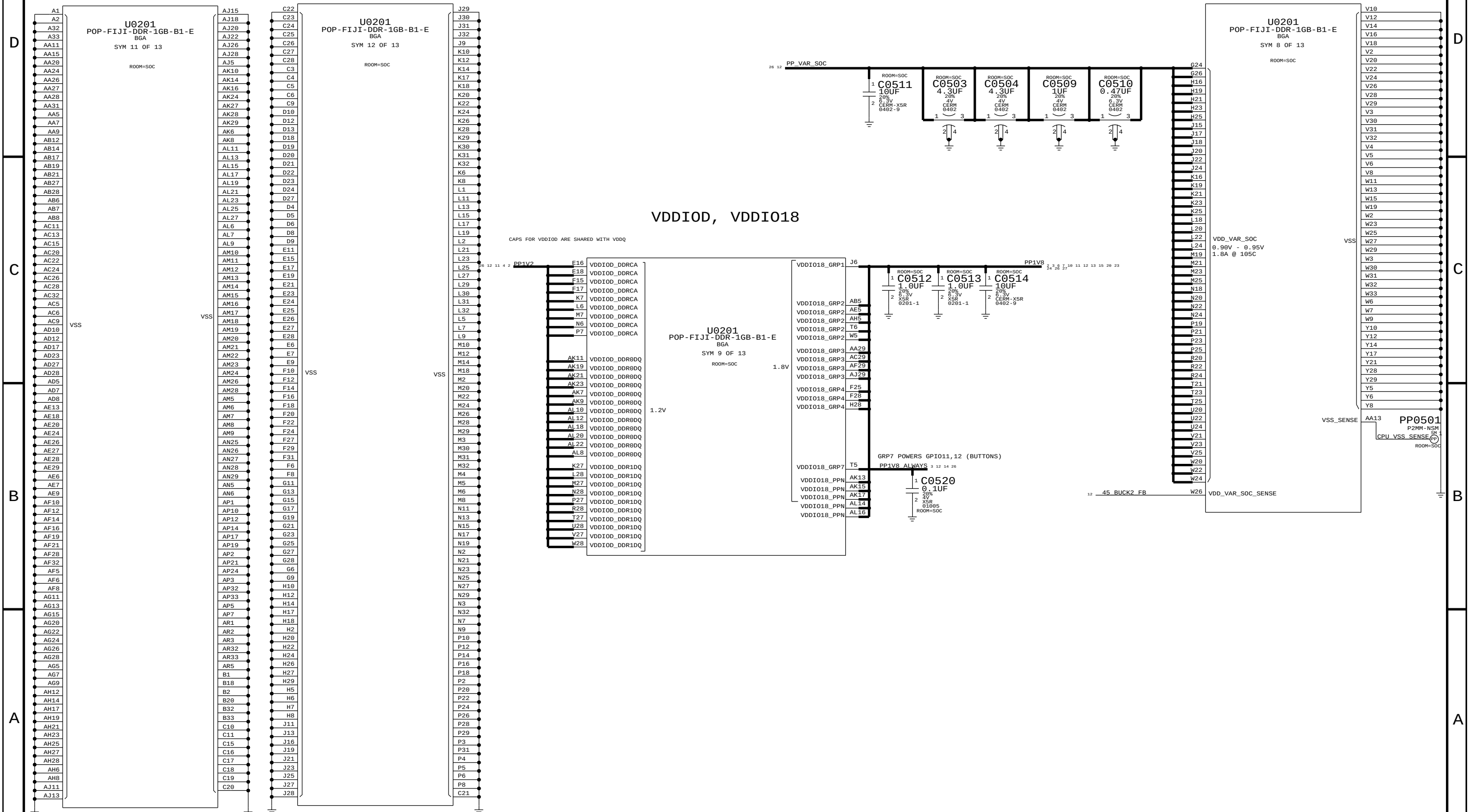
VDD_CPU, VDD_GPU



FIJI: VDDIOD, VDDIO18, VDD_VAR_SOC

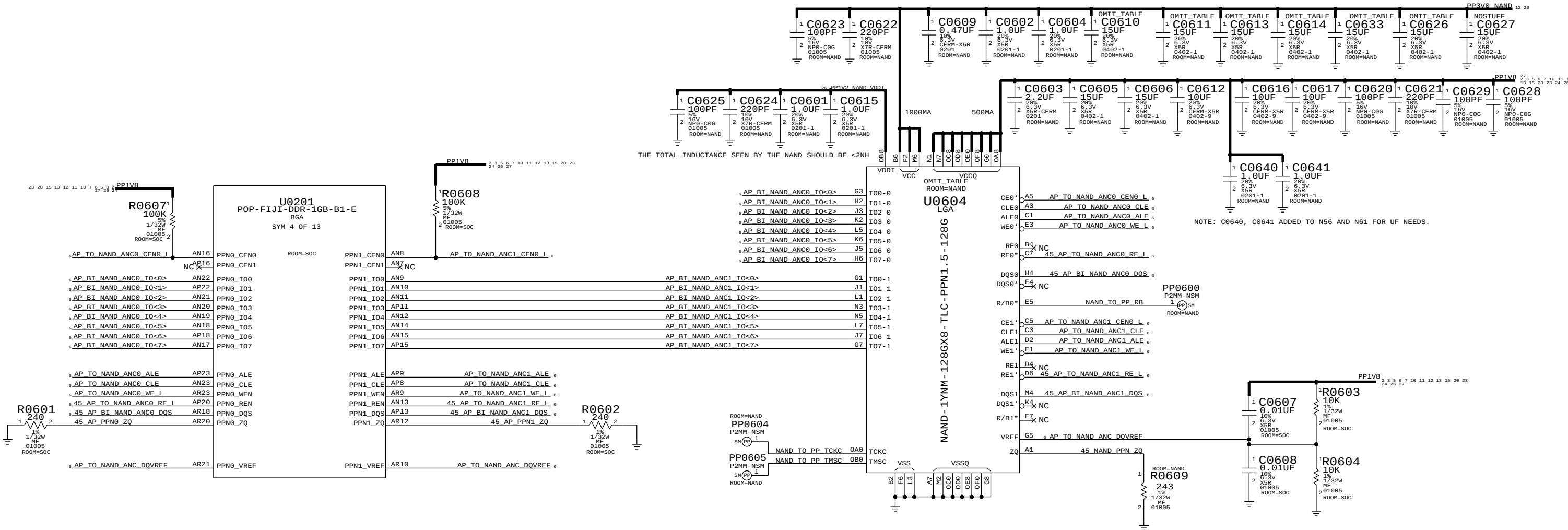
JUST A FEW GNDS

VDD_SRAM, VDD_SOC



FIJI: NAND + 12X17 NAND PKG

SUPPORT FOR PPN1.5 (1.8V IO) ONLY



NOTE: NAND PADS SHOULD BE SHIELDED FROM TRACES WITH A GROUND PLANE

PP0601 P4MM-NSM
ROOM=SOC
NOTE: IO<6> PREFERRED BY MATT BYOM (N51)
(IS A STATUS READY BIT)
1 AP BI NAND ANC0 IO<6> 0

PP0602 P4MM-NSM
ROOM=SOC
1 45 AP TO NAND ANC0 RE L 0

PP0603 P4MM-NSM
ROOM=SOC
1 45 AP BI NAND ANC0 DQS 0

D

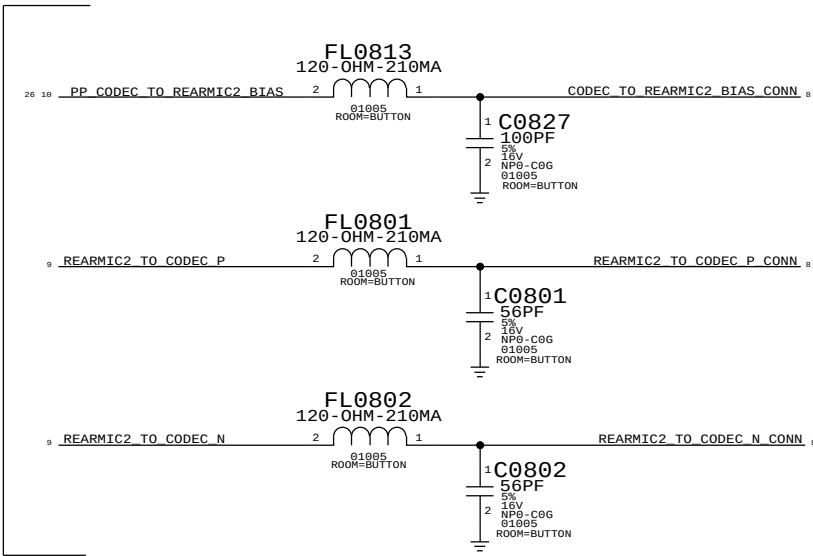


8 7 6 5 4 3 2 1

BUTTON FLEX (BUTTONS, ANC REF MIC, STROBE, STROBE_NTC, WIFI FLEX PAC)

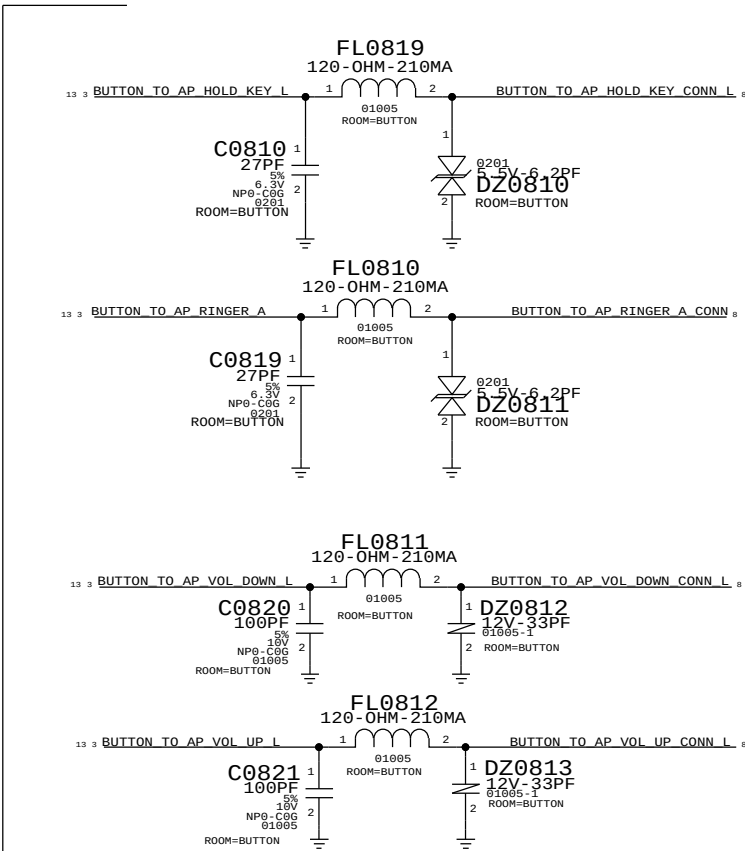
D

MIC2 (ANC REF MIC):
MIC2/4 BIAS,
MIC2_P,_N

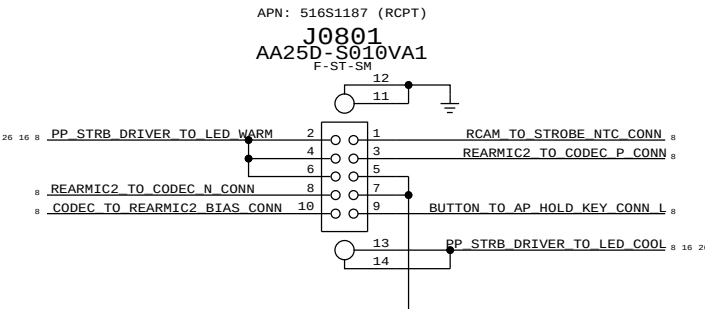


C

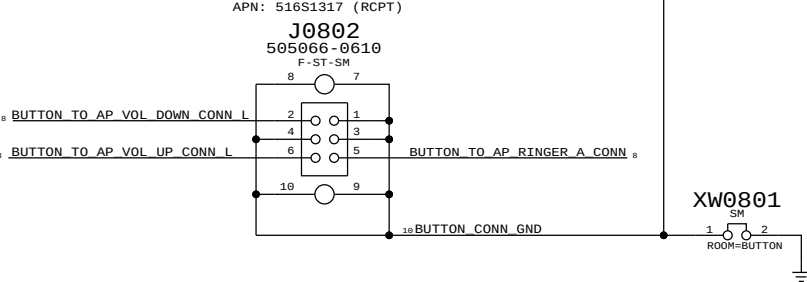
BUTTONS:
RINGER, HOLD,
VOL_UP/DOWN,



RIGHT BUTTON FLEX B2B

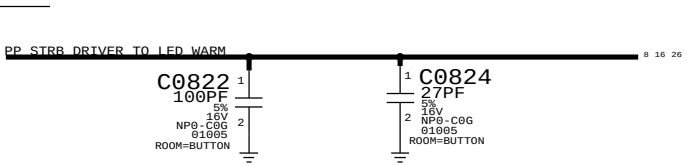


LEFT BUTTON FLEX B2B

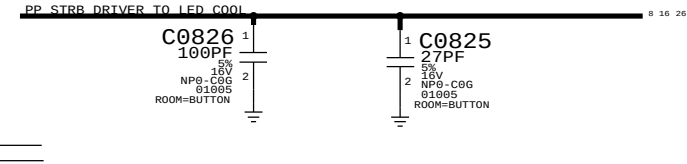


B

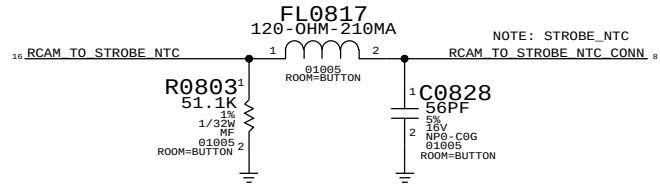
STROBE:
LED WARM



STROBE:
LED COOL



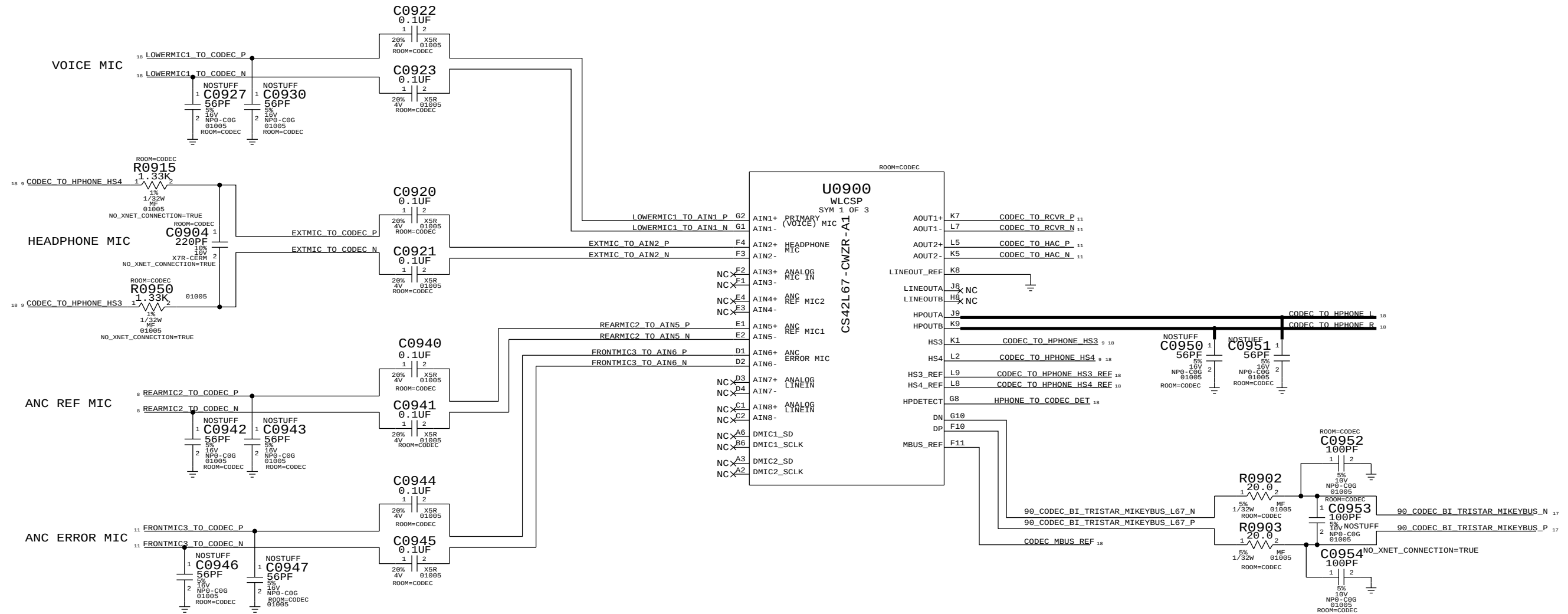
STROBE:
NTC

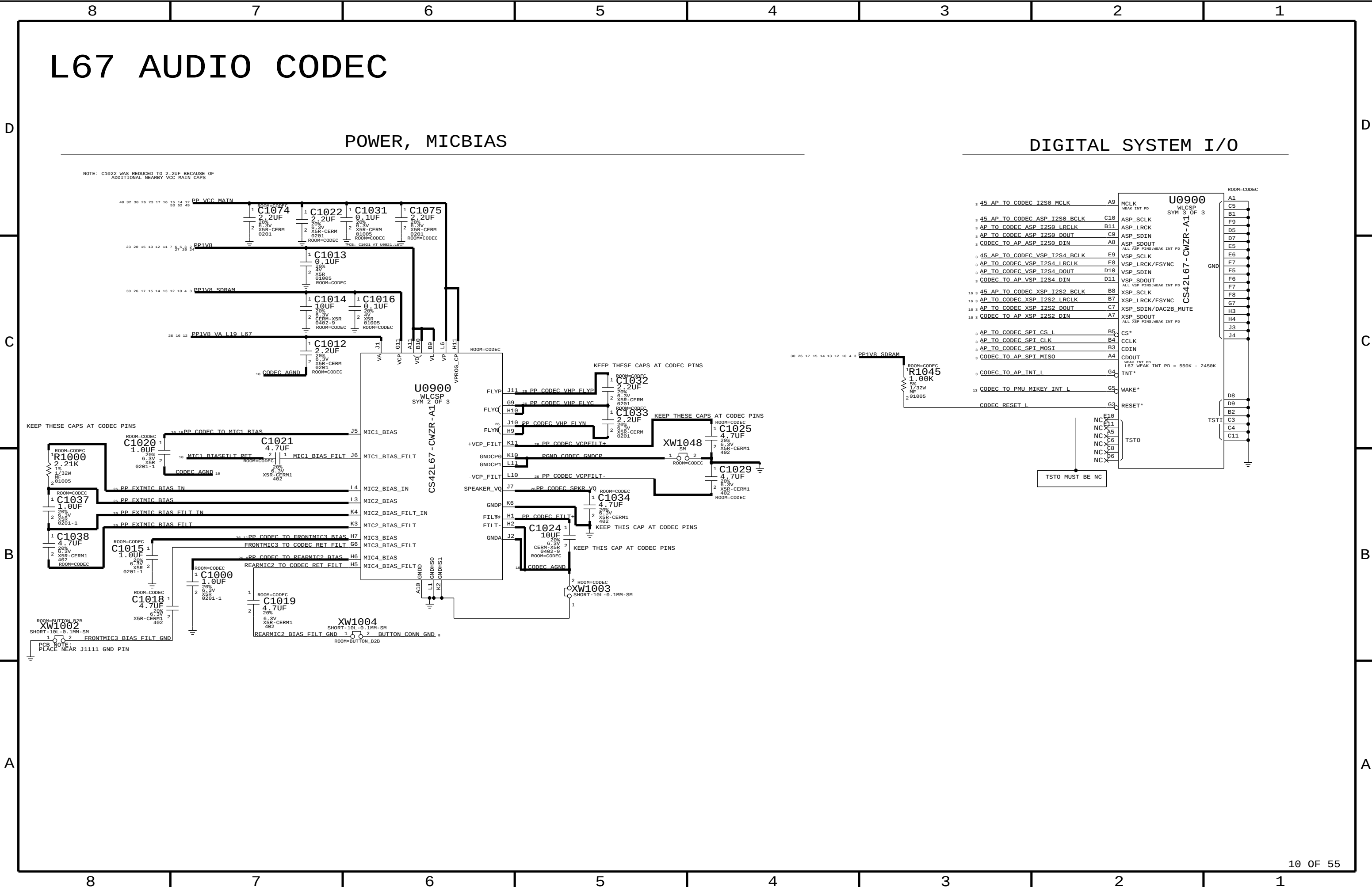
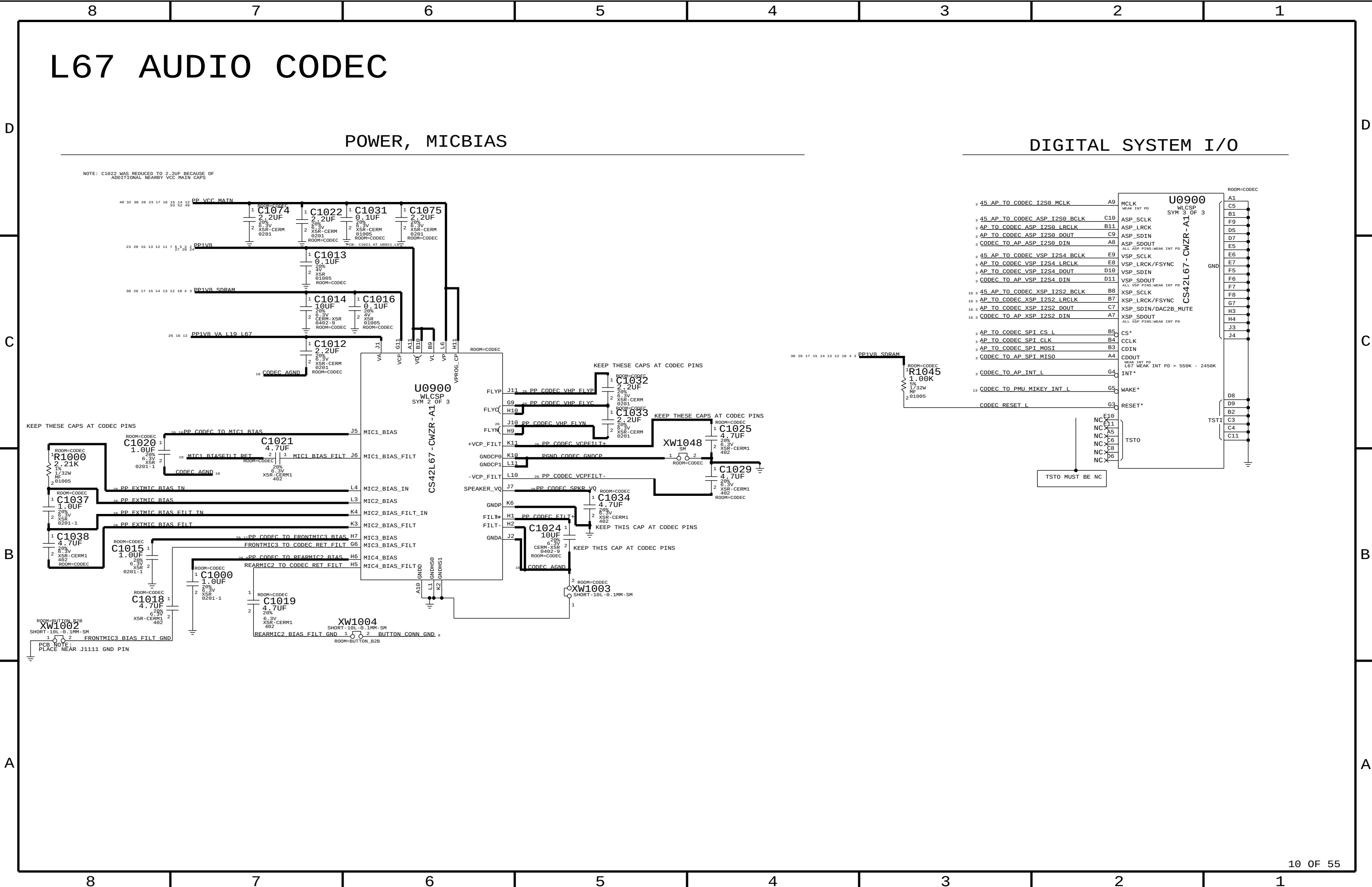


L67 AUDIO CODEC

AUDIO I/O

(ANALOG MIC IN, DIG MIC IN, HPOUT, LINEOUT, RECEIVER OUT, MIKEYBUS)



[illegible][illegible][illegible][illegible]

FRONT CAM FLEX B2B

(FCAM, PROX, ALS, RECEIVER, ANC ERROR MIC)

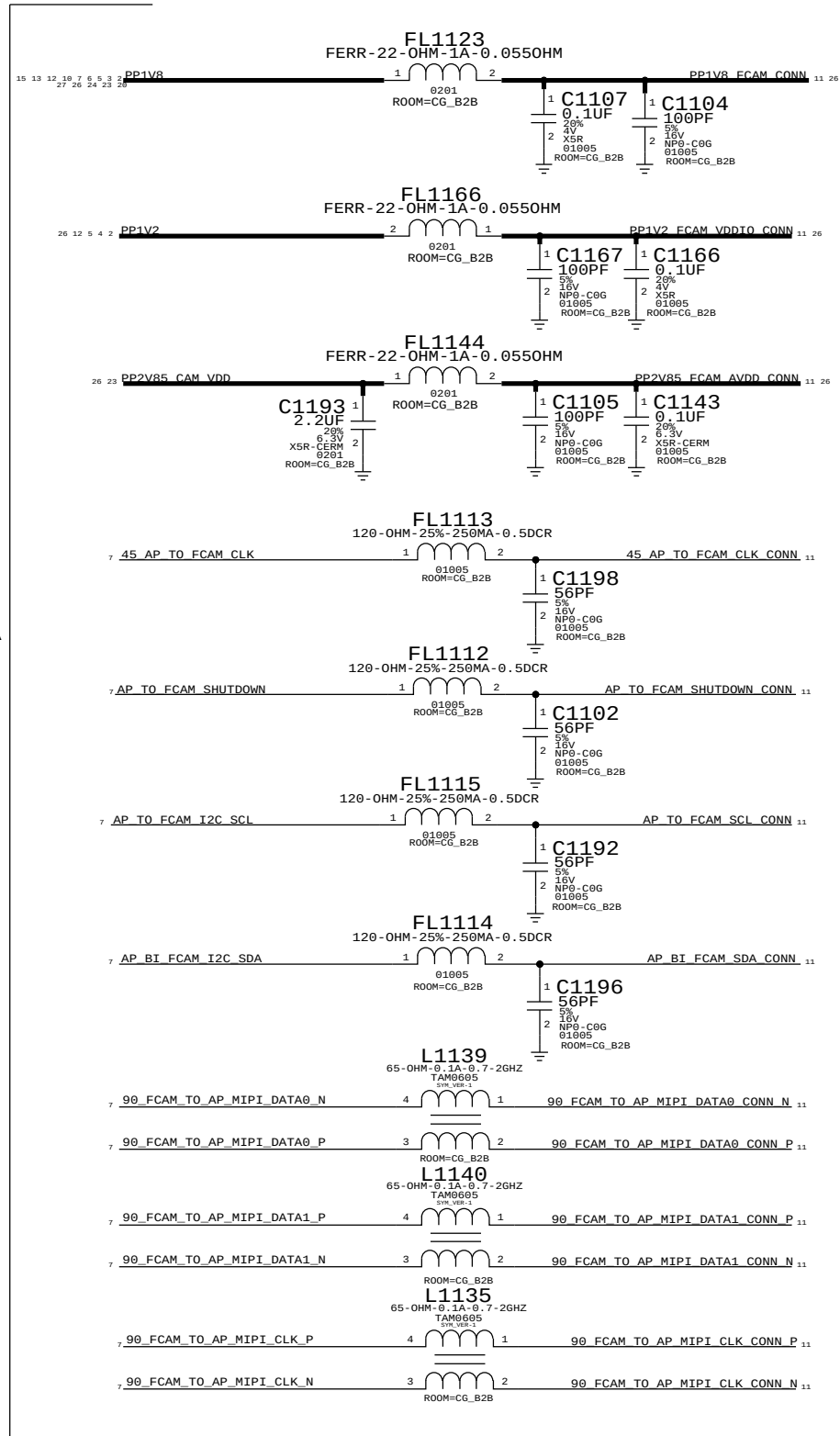
D

C

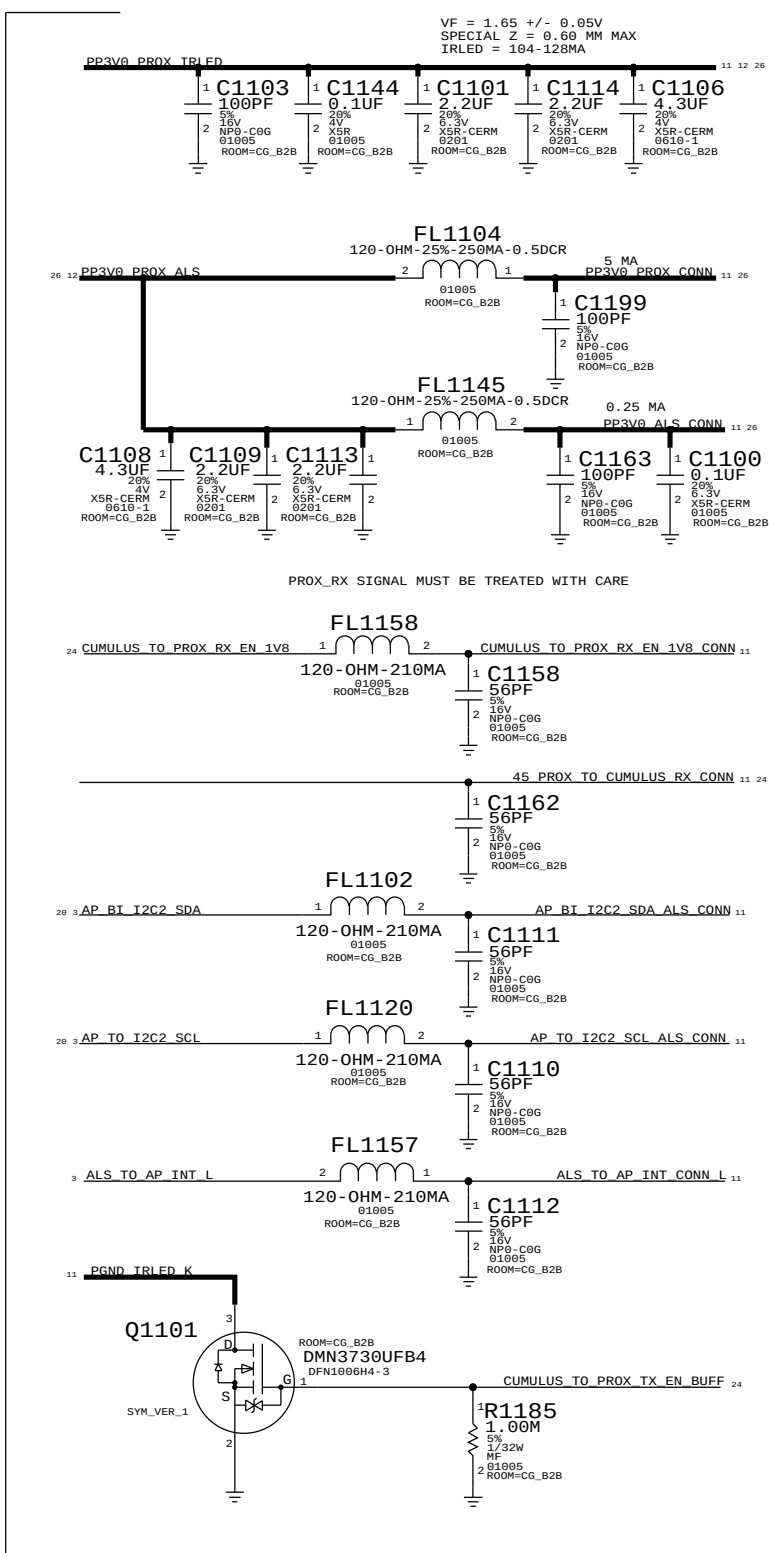
CAMERA

B

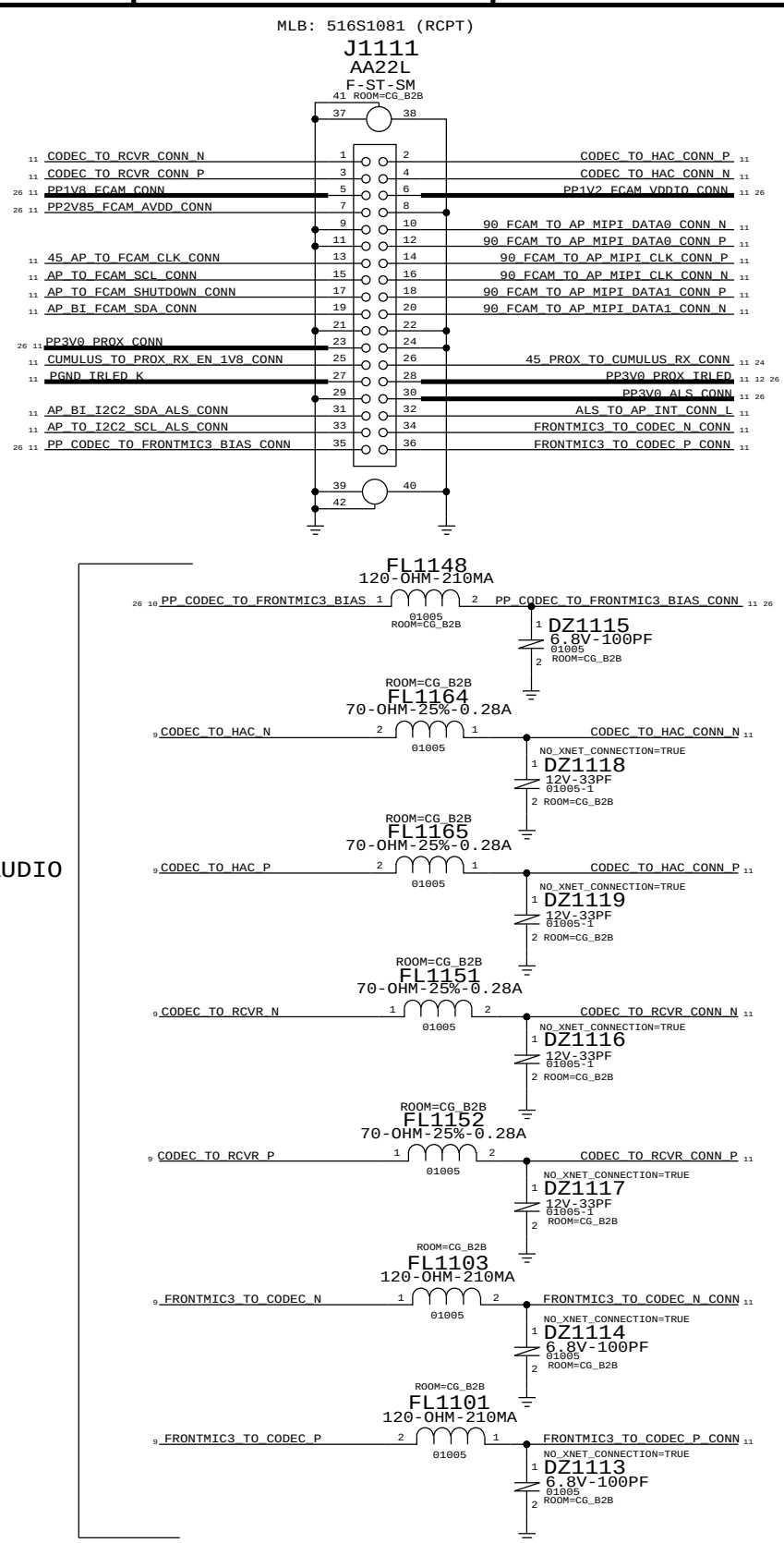
A



ALS,
PROX



AUDIO



ADI PMU

(BUCK, LDO, VIBE DRIVER, 32K, CHARGER)

REVISIT VCC_MAIN CAPS

NOTE: L1210, L1212 BOMPTIONS
CONTROLLED ON PAGE1

APN: 338S1251 (ADI AZ)

U1202 ROOM=PMU
D2186AE0FJAVAC
FCCSP-N56-N61
SYM 1 OF 3

D

C

B

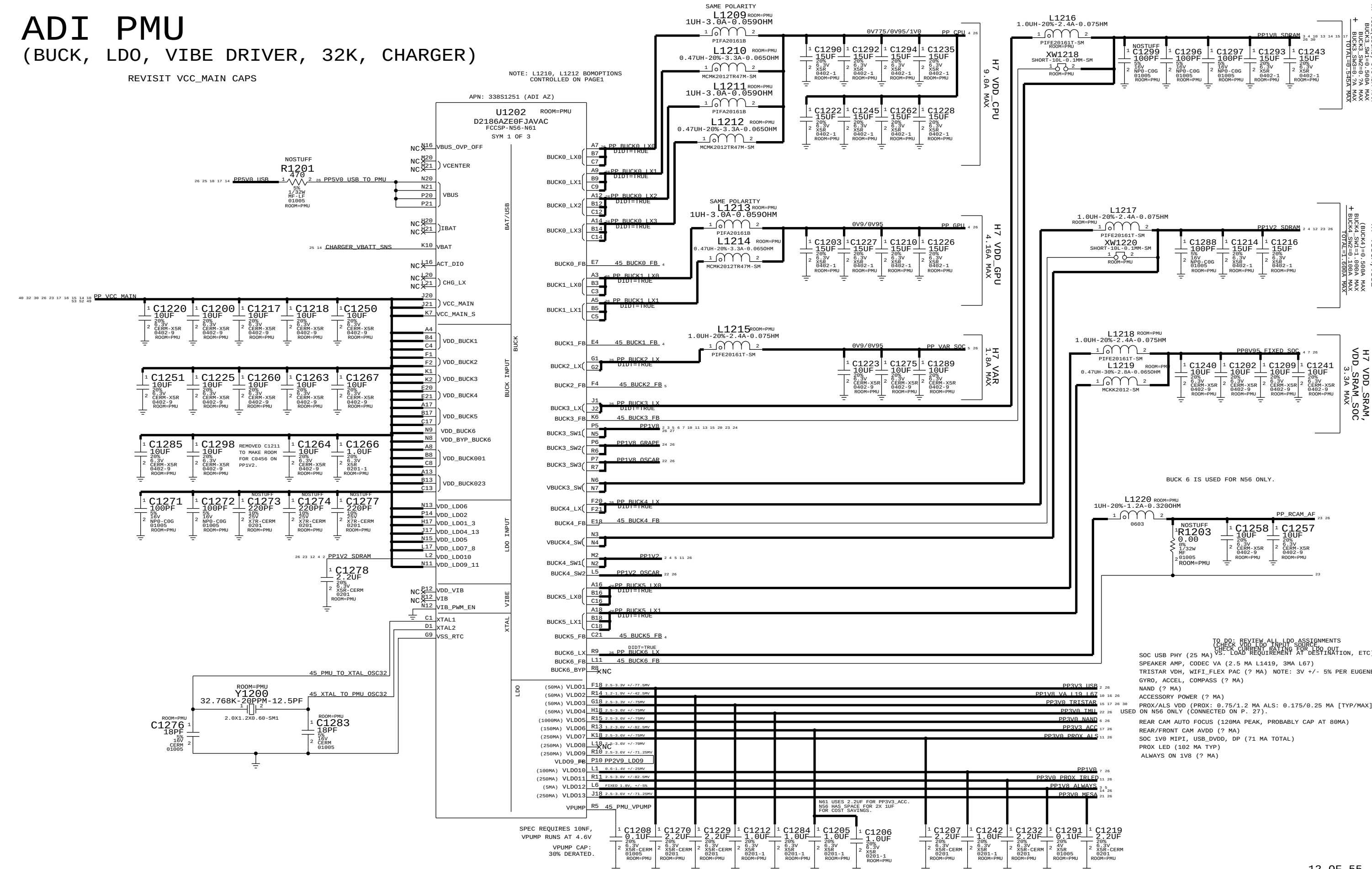
A

D

C

B

A



ADI PMU

(AMUX, GPIO, BUTTONS, ADC, THERMISTORS, SYSTEM I/F, GND)

D

D

C

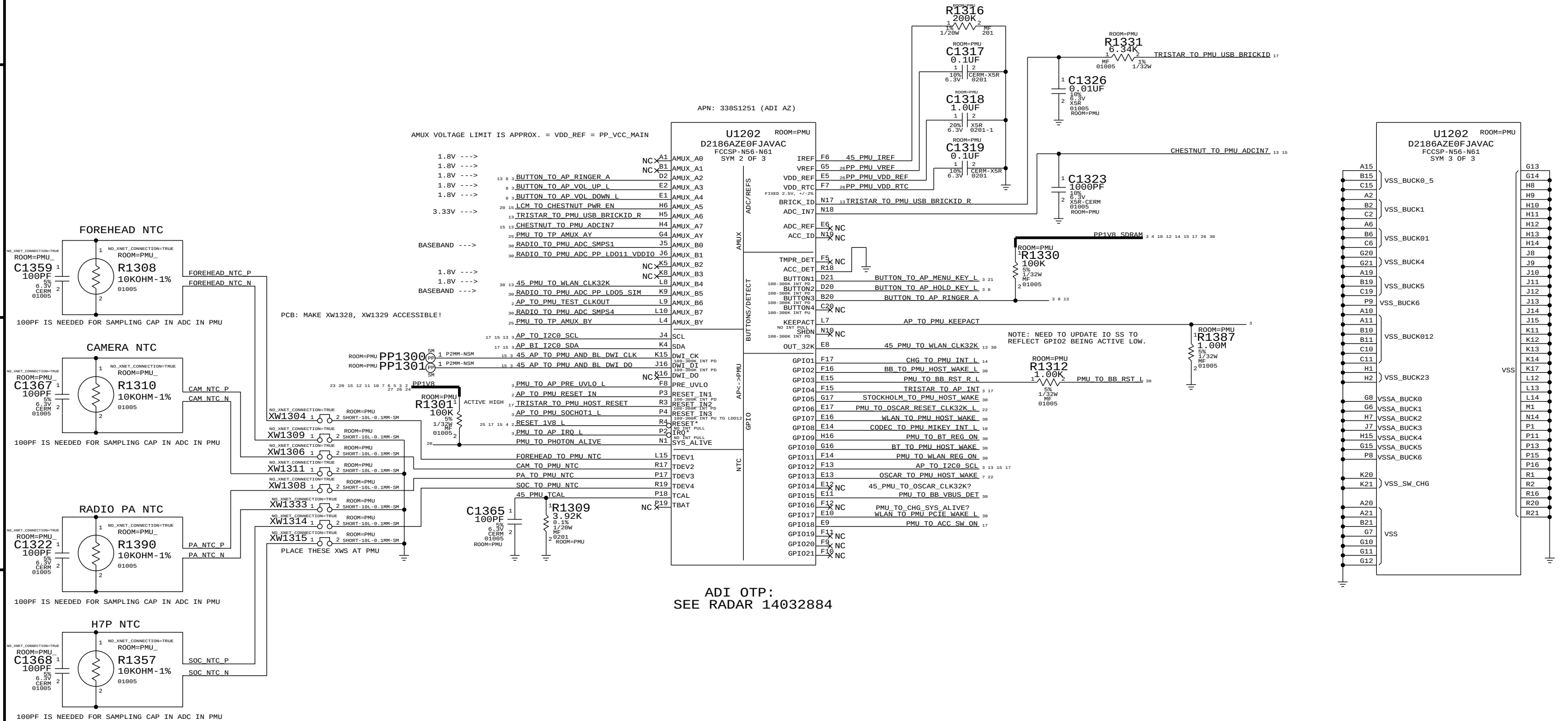
C

B

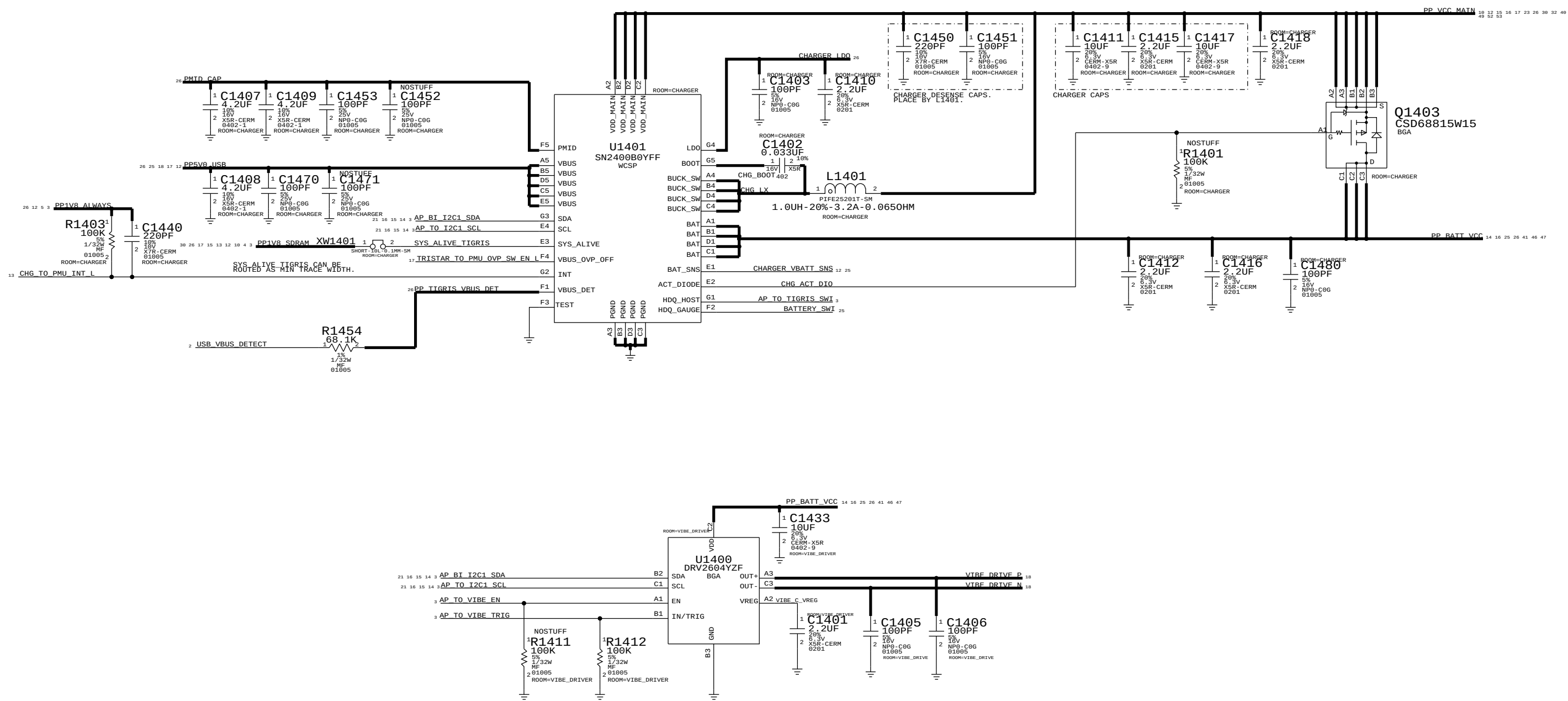
B

A

A

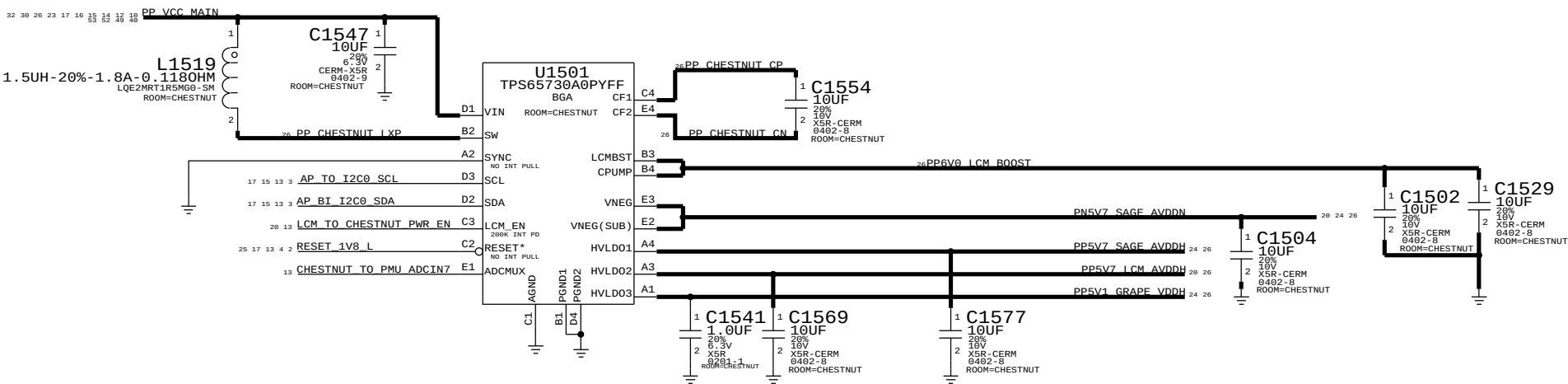


TIGRIS CHARGER & VIBE DRIVER

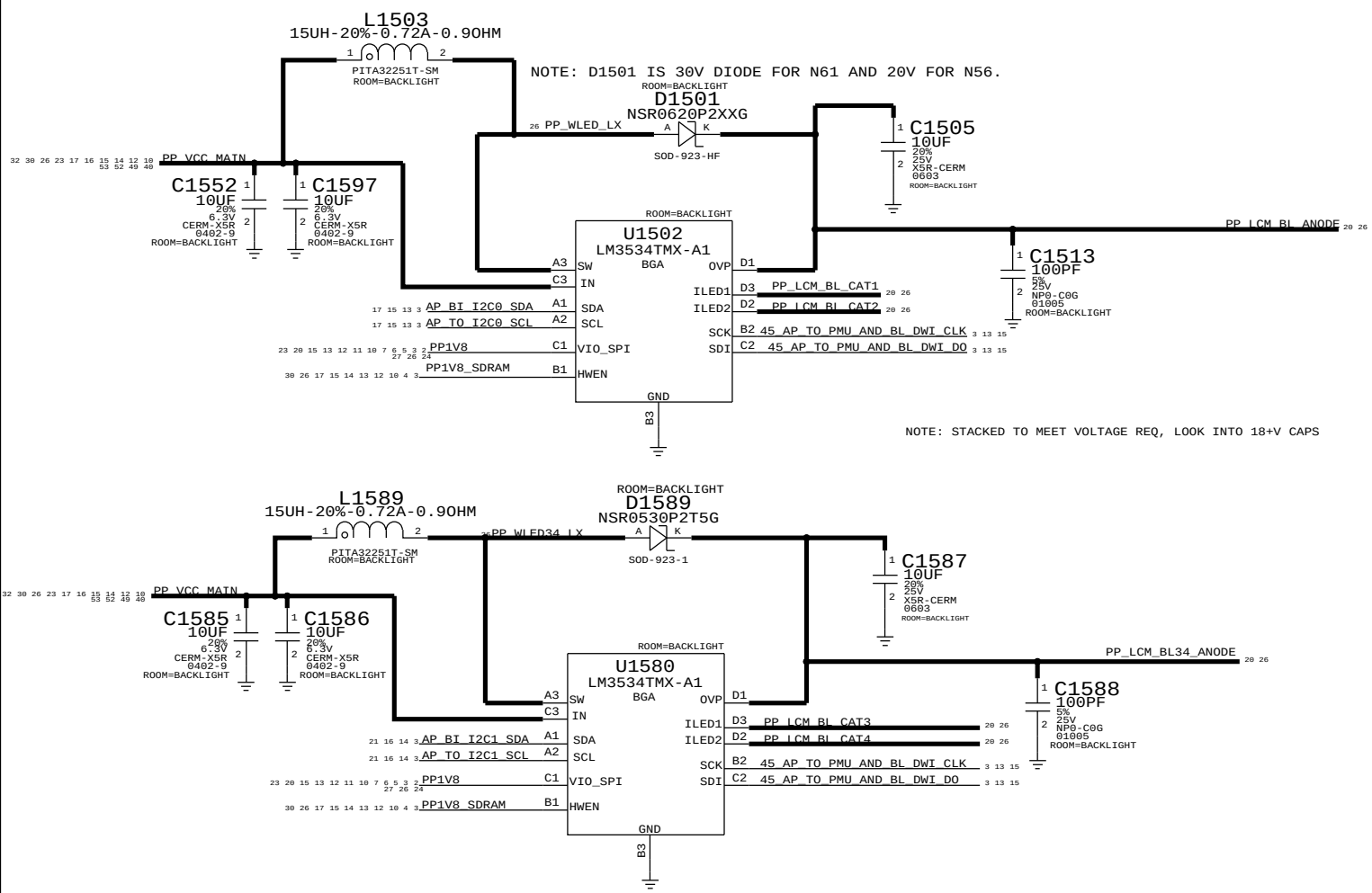


CHESTNUT, BACKLIGHT DRIVER, MESA BOOST

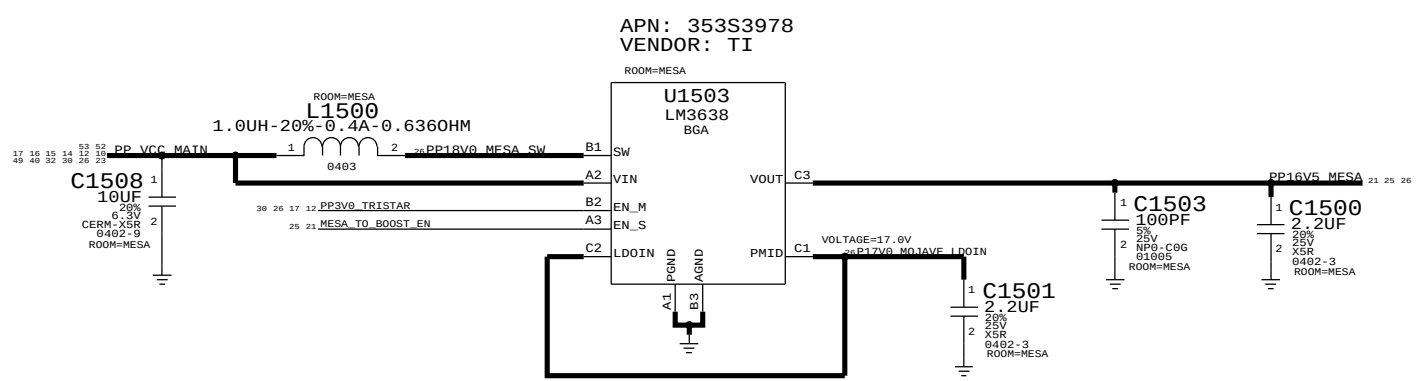
DISPLAY PMU (TI CHESTNUT, 338S1149)



BACKLIGHT DRIVERS



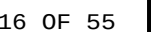
MESA BOOST



D

C

B



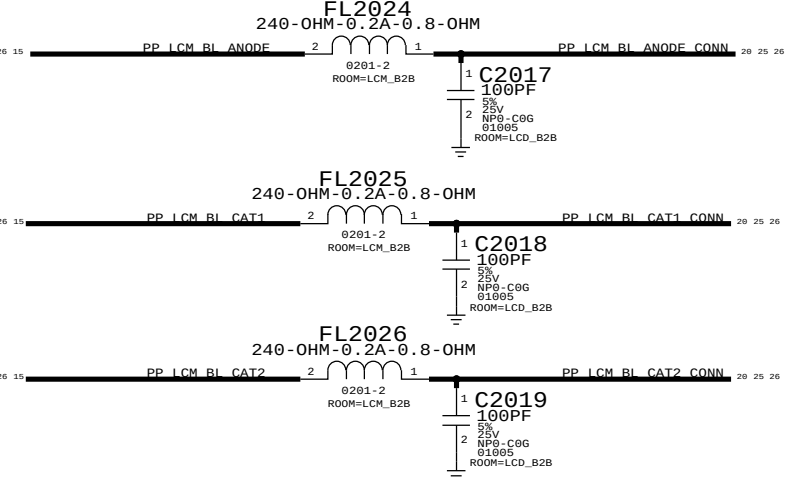
[illegible]

8	7	6	5	4	3	2	1
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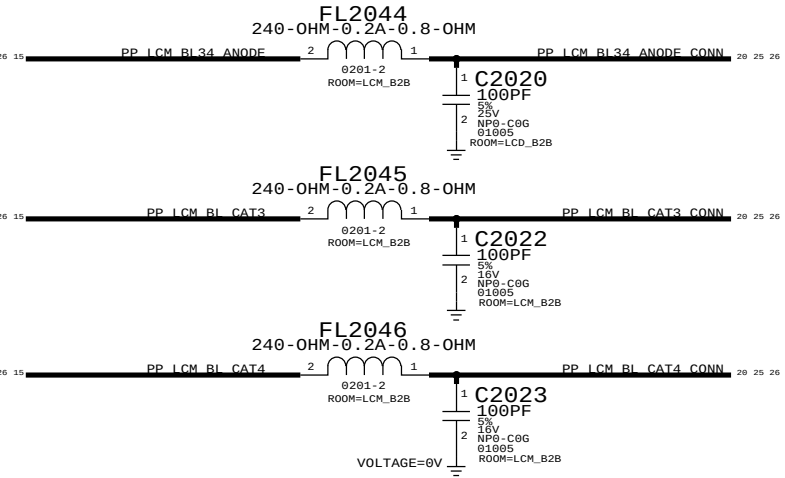


LCD B2B

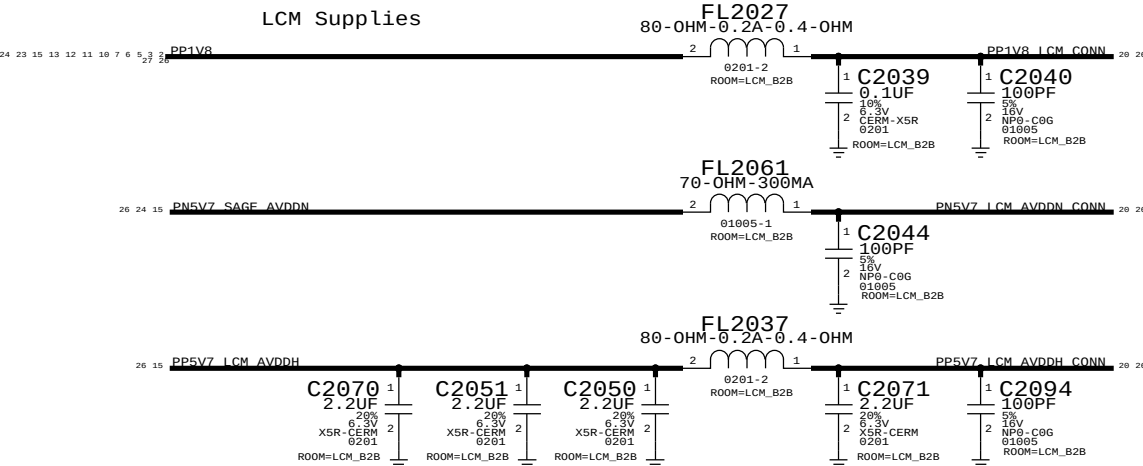
Backlight



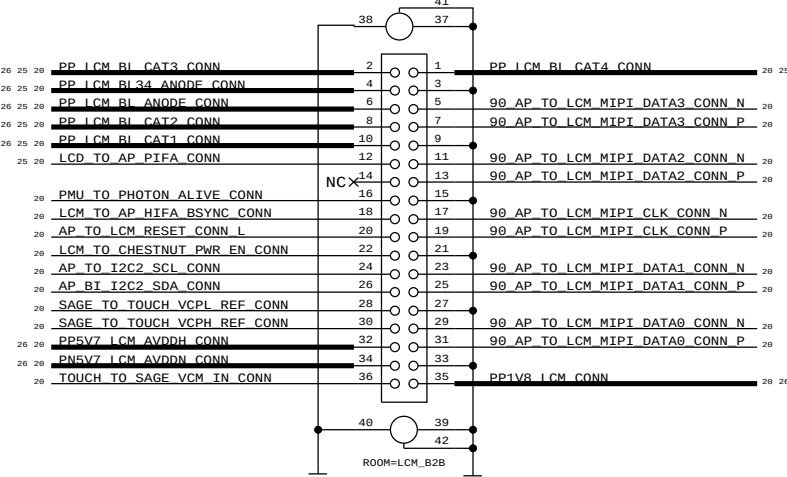
BACKLIGHT 2 (N56 ONLY)



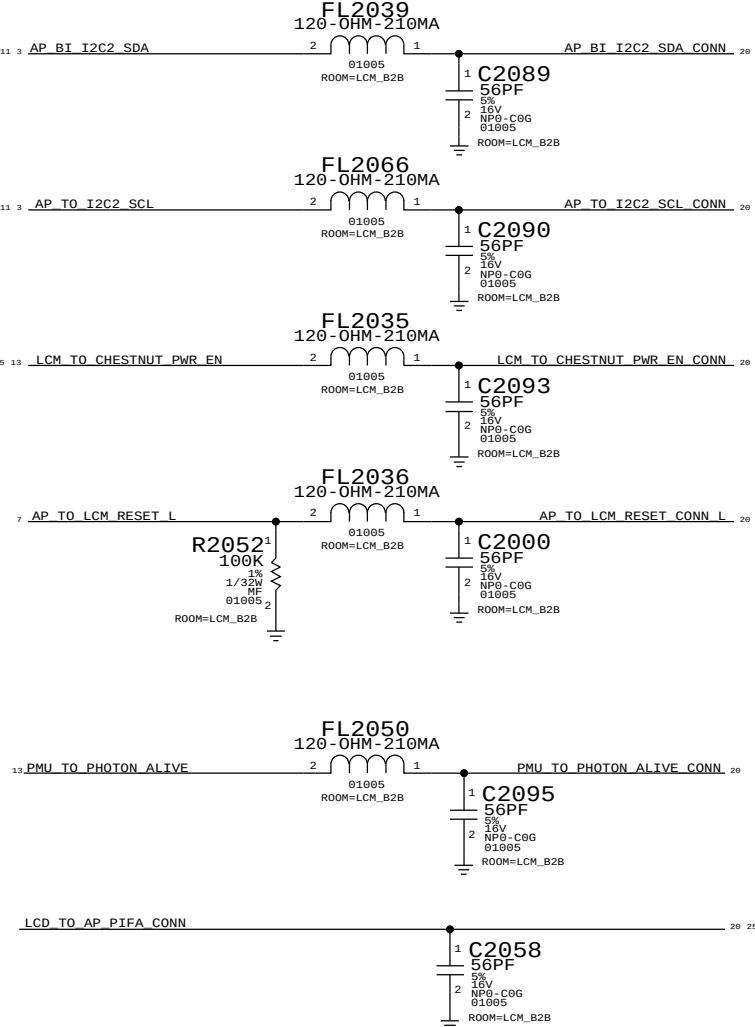
LCM Supplies



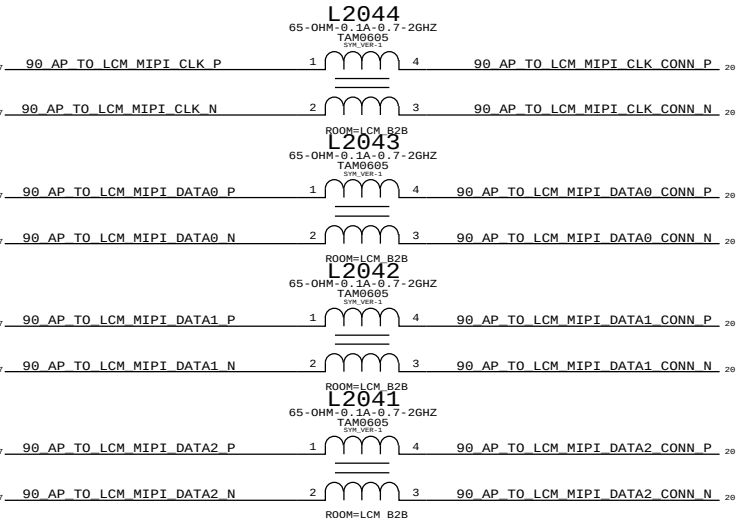
MLB: 516S1185 (RCPT)
(516S1184 ON FLEX)
J2019
20-5857-036-001-829
F-ST-SM



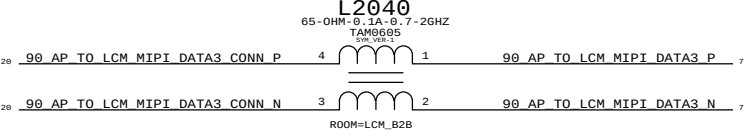
Digital Interfaces



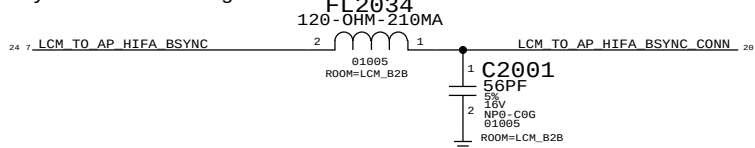
MIPI Common Mode Chokes



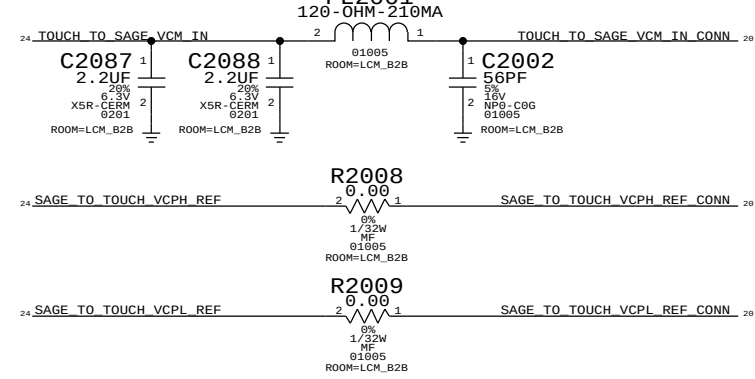
MIPI LANE 3 (N56 ONLY)



Sync/Reset/Debug

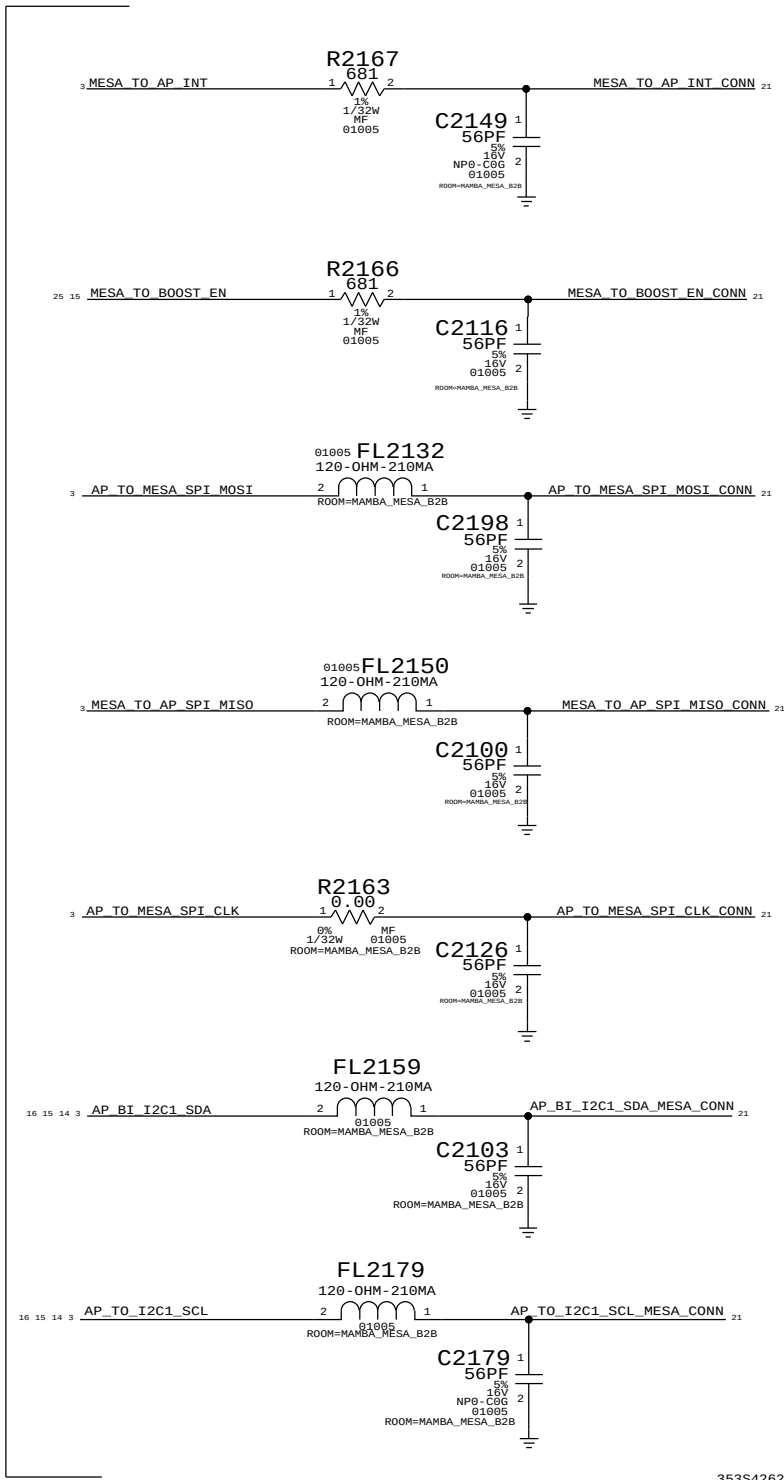


Touch



MESA CONNECTOR

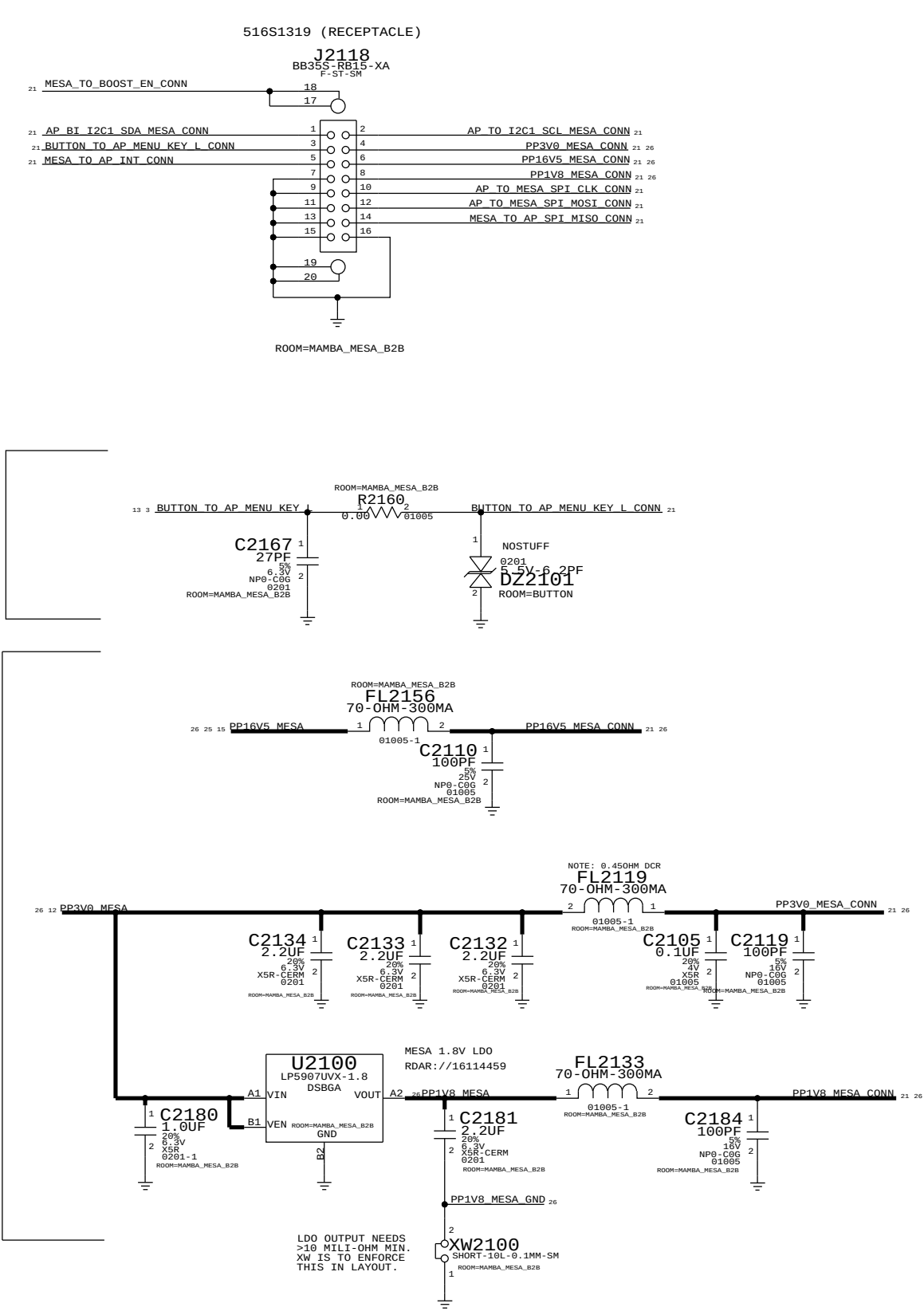
MESA:
INT, EN, SPI, I2C



353S4262

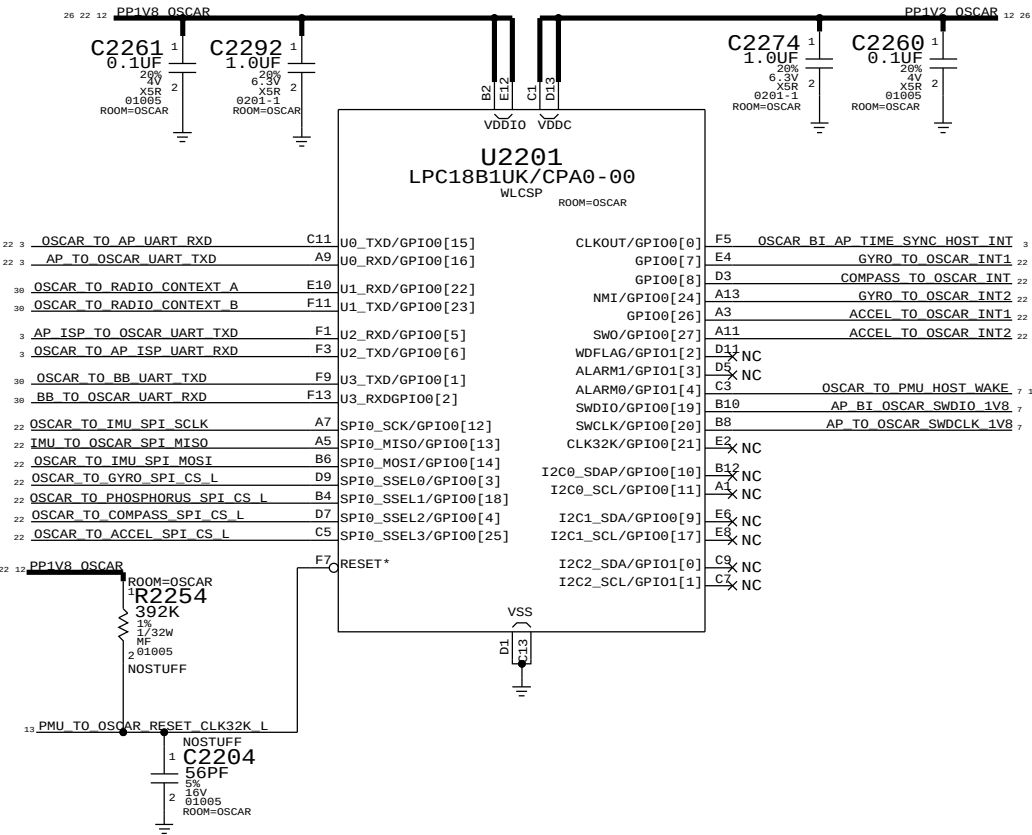
MENU
BUTTON

MESA:
POWER



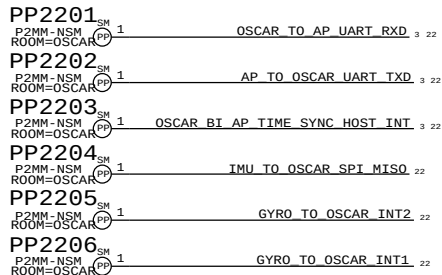
OSCAR + SENSORS

OSCAR VDDIO = 1.8V ALWAYS ON (NEED TO WAKE HOST & RUN PLL)
OSCAR CORE = 1.2V ALWAYS ON (NEED TO RUN IN S2RAM)

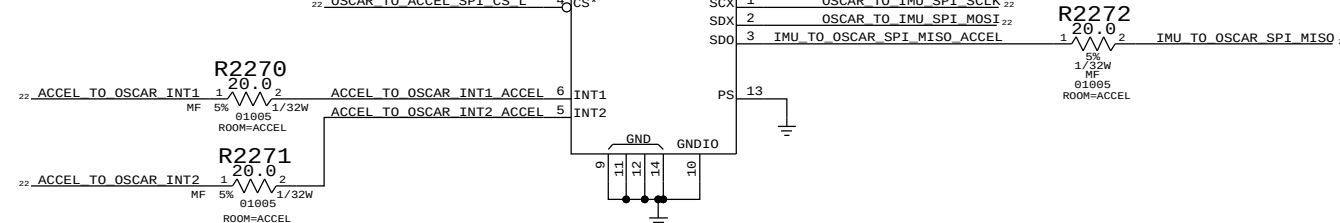


CARBON (ACCEL GYRO COMBO)

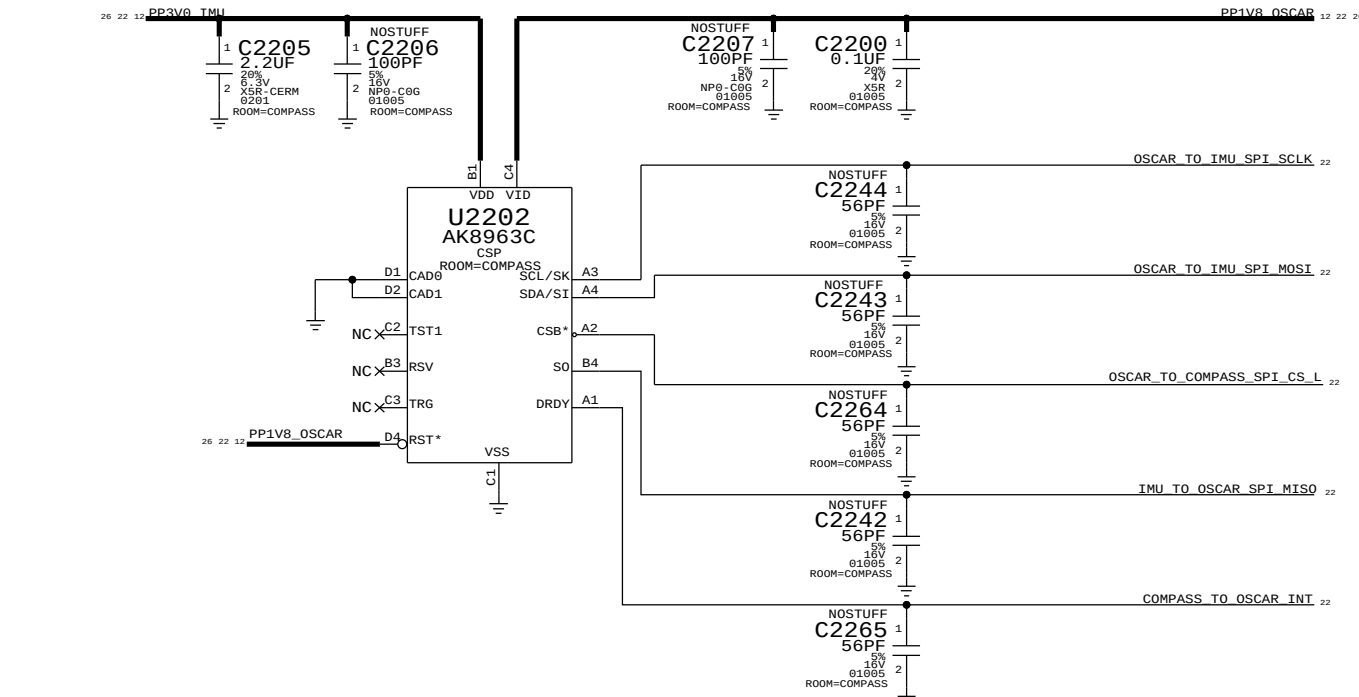
INVENSENSE, APN 338S00017, C2211=0.1UF (132S0395)



DISCRETE ACCEL

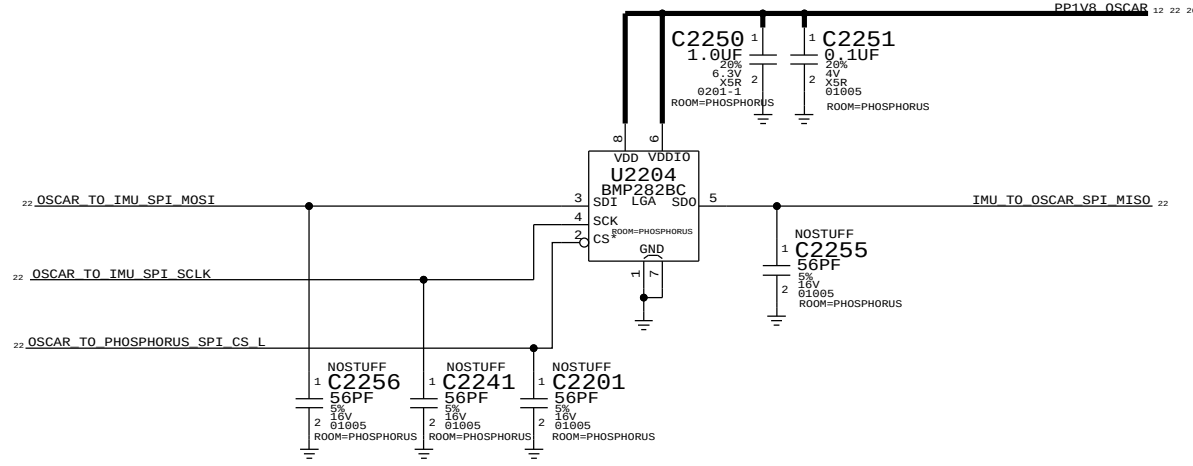


THIS PART OUTSIDE OF SHIELD ON THE PENINSULA COMPASS



THIS IS OUSIDE OF SHIELD TO THE RIGHT OF THE NAND

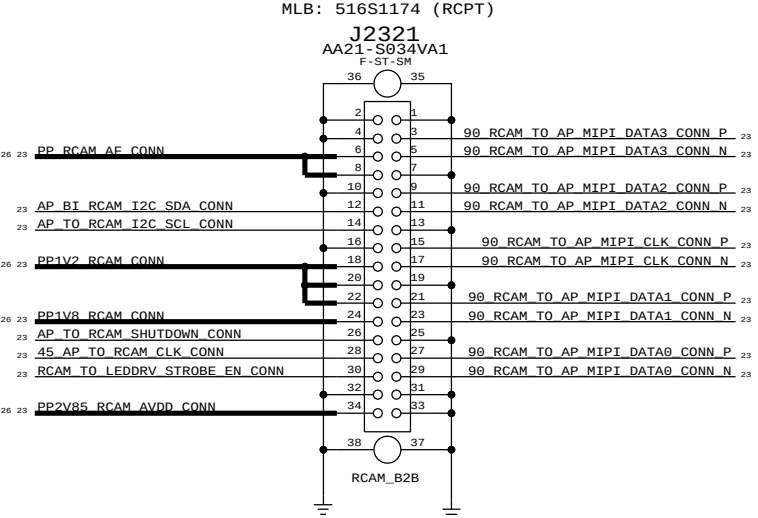
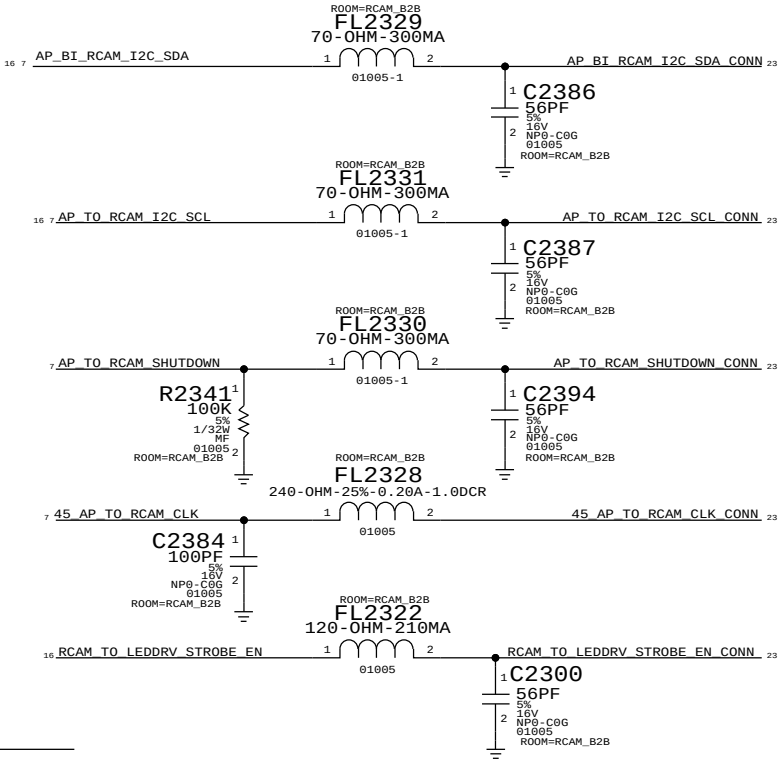
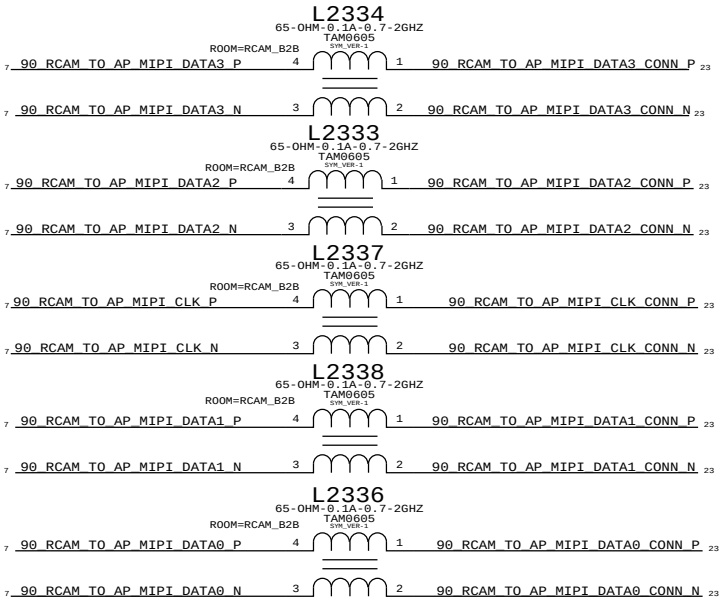
PHOSPHORUS



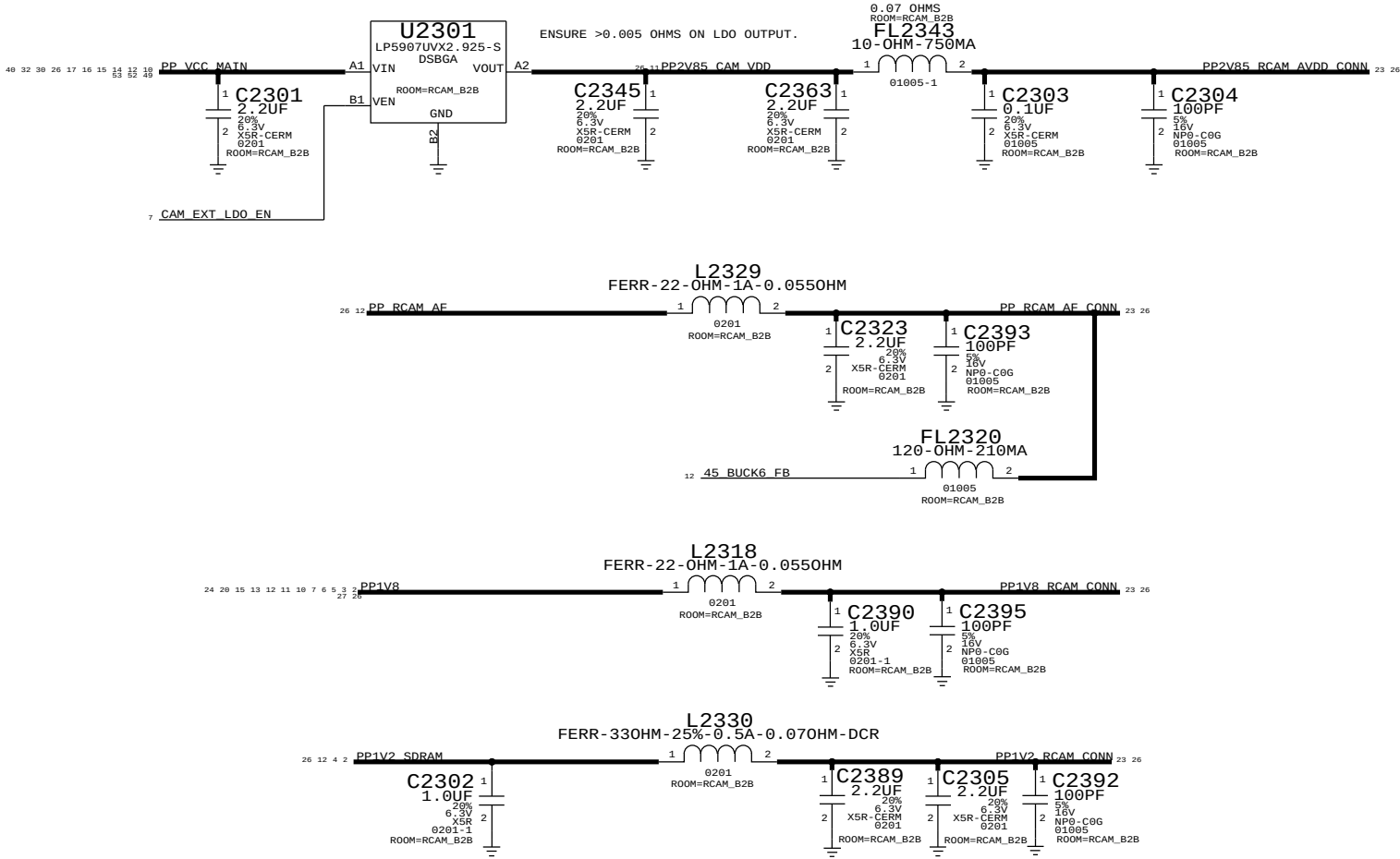
RCAM B2B (REAR CAMERA CONNECTOR)

RCAM:
4-LANE MIPI

RCAM:
DIGITAL I/F
(I2C, CTRL, CLK)



RCAM:
POWER:
(1.8V DOVDD)
(2.9V AVDD)
(1.2V DVDD)
(2V AF)

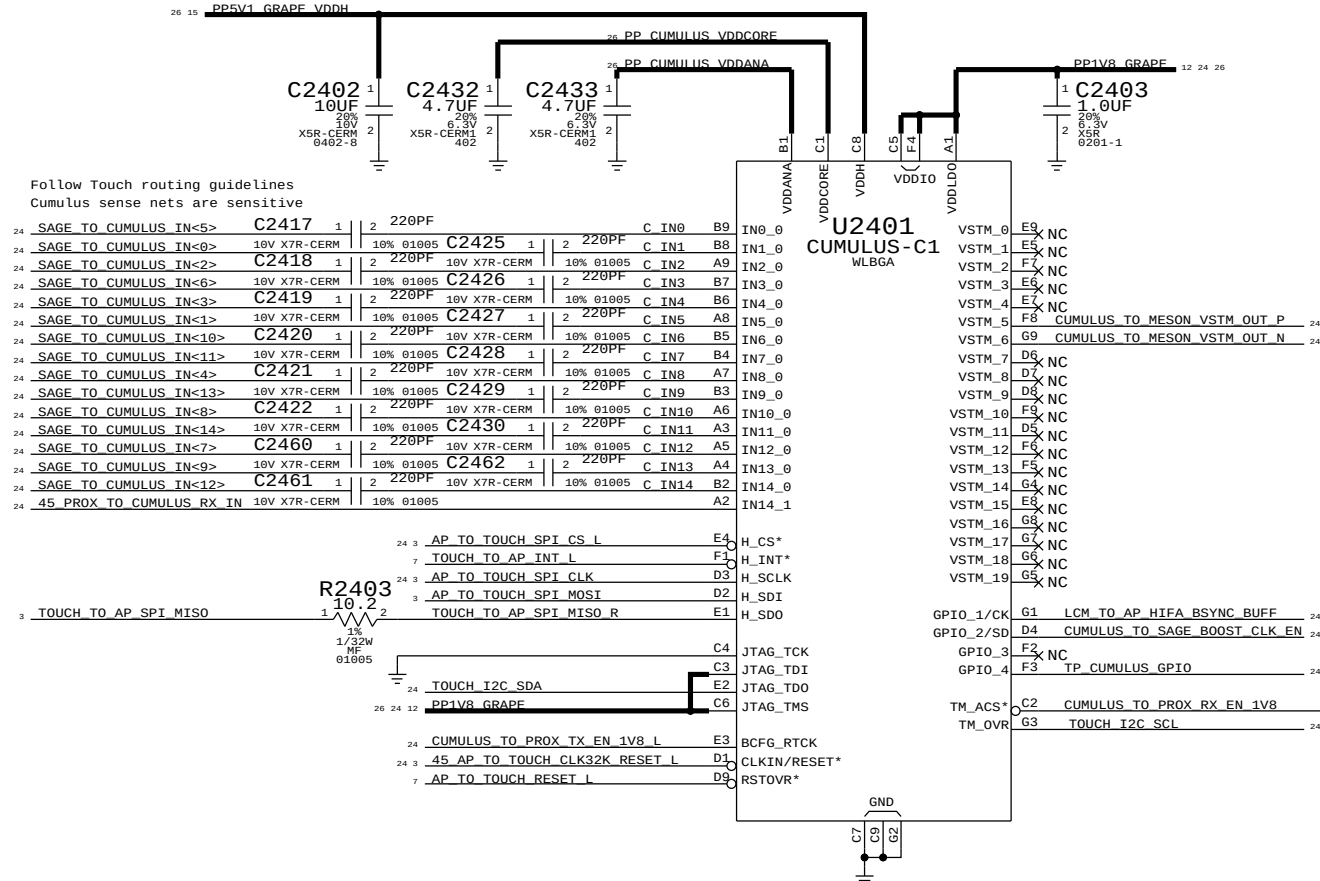


Touch (B2B, Driver ICs)

Cumulus

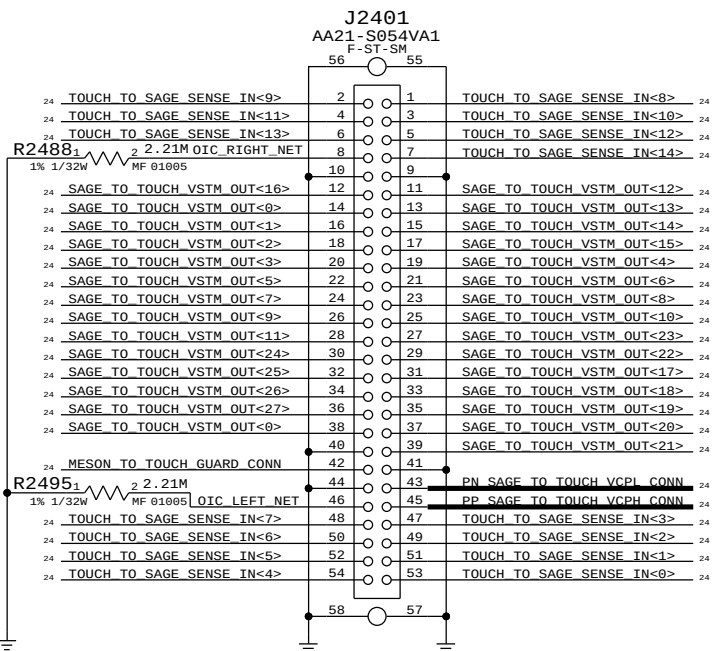
APN: 343S0638

Turn on is later than PP1V8_GRAPE
Turn off is same time as PP1V8_GRAPE

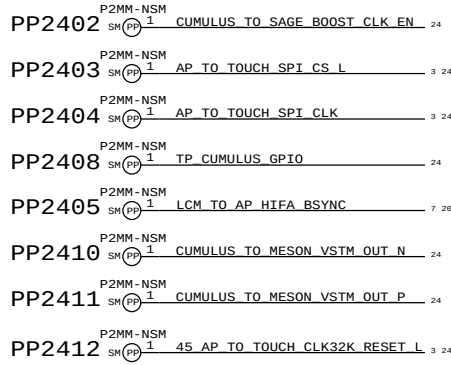


Touch B2B

MLB APN : 516S1183 (Receptacle)
Flex APN: 516S1182 (Plug)

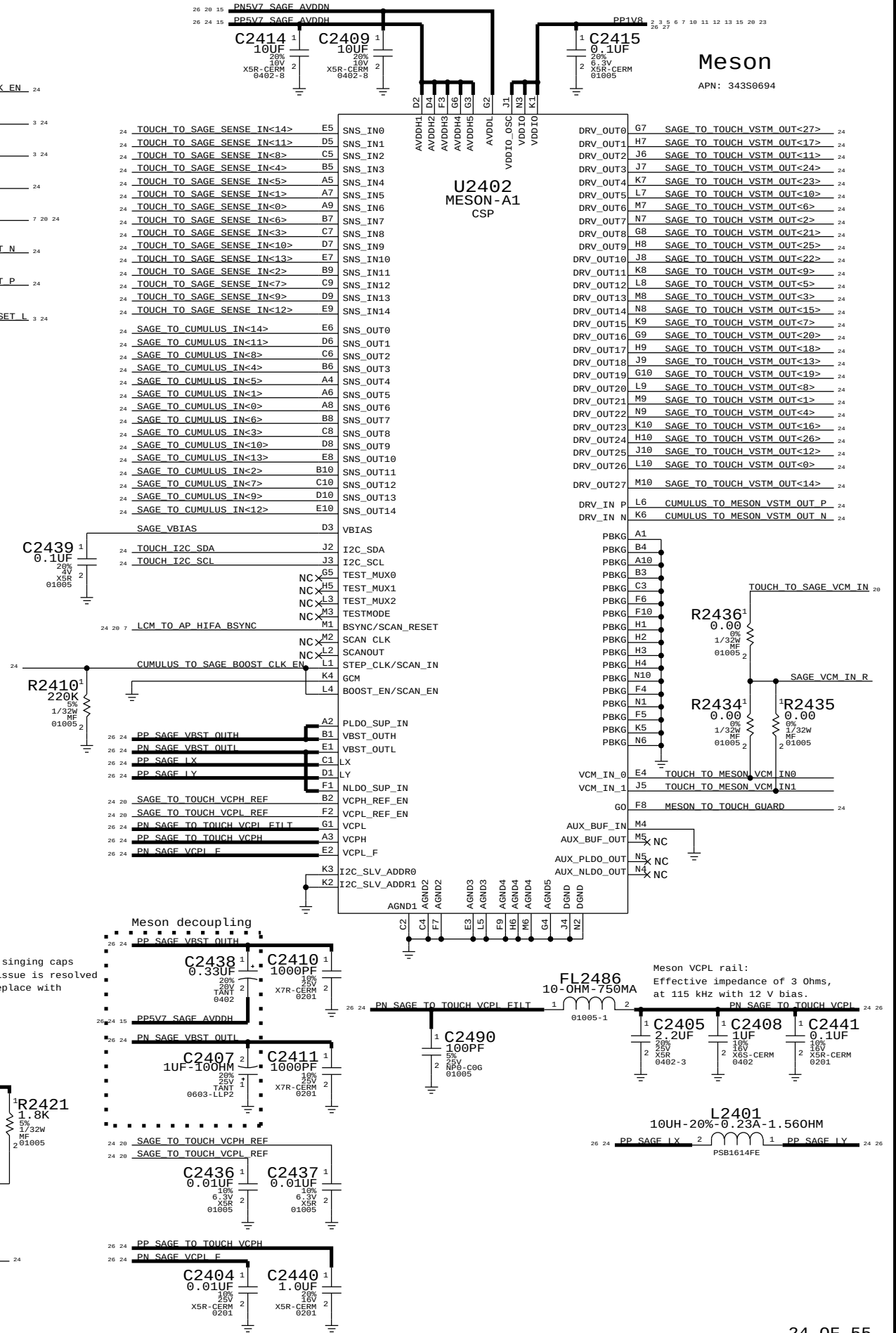


Touch probe points



Meson

APN: 343S0694

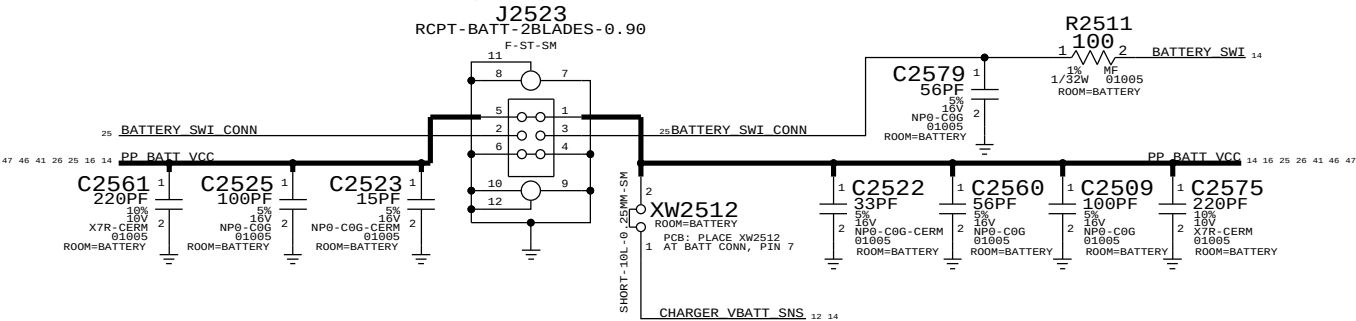


BATT CONN, TPS, STANDOFFS/SHIELDS/FIDUCIALS

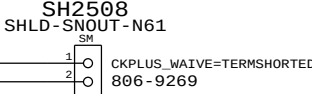
BATTERY CONN

THIS ONE ON MLB ----> 516S1080 RCPT

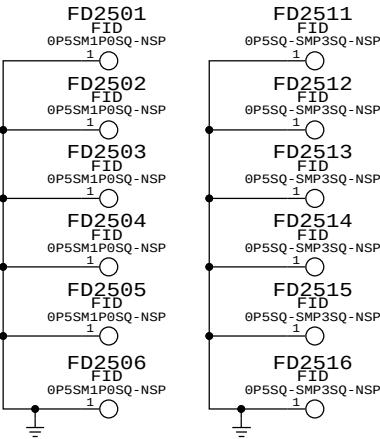
SCHEMATIC SYMBOL (PIN ORDER) IS WIERD !!



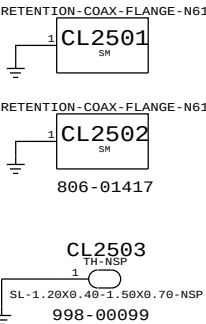
SHIELDS



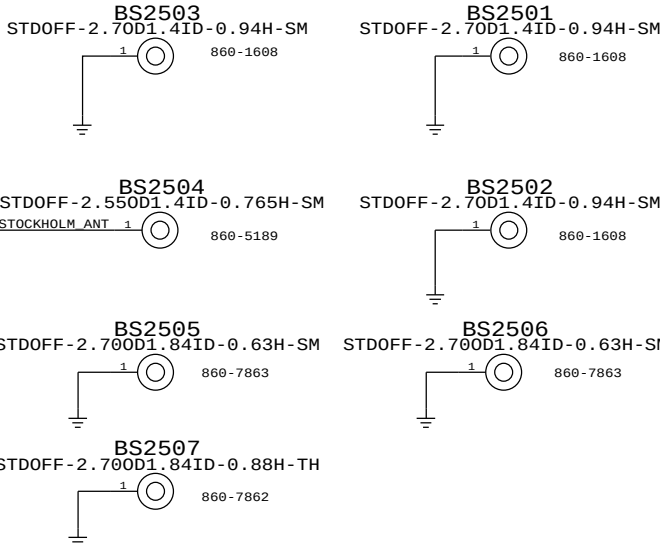
FIDUCIALS



RF CLIPS

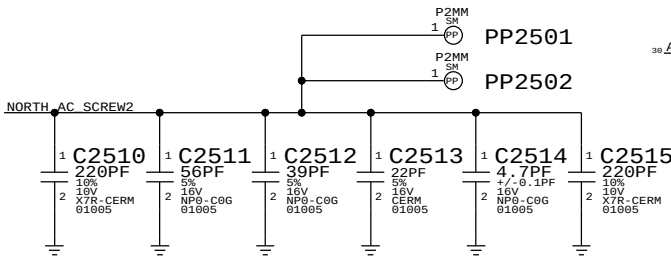


STANDOFFS



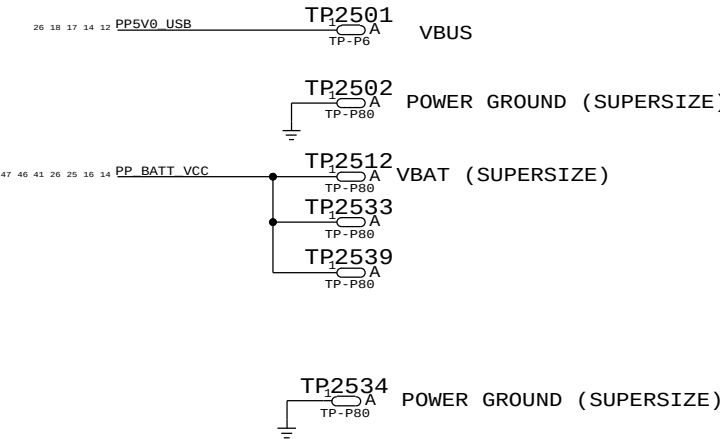
SCREW HOLES + STANDOFFS

PROBE POINTS TO BE PLACED NEAR THROUGH-HOLE IN UPPER LEFT OF BOARD. ONE FOR TOP, ONE FOR BOTTOM.

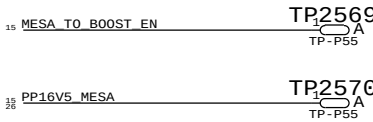


TESTPOINTS

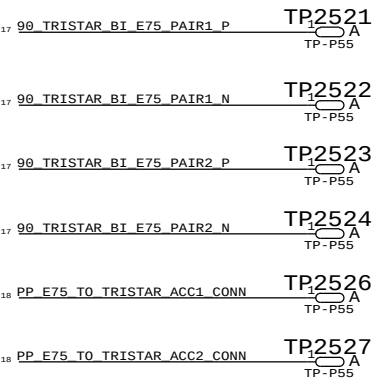
POWER TP



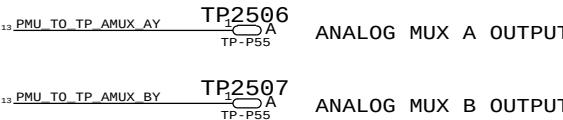
MOJAVE TP



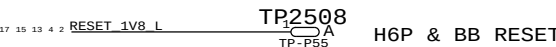
E75 - USB/UART/ID/POWER



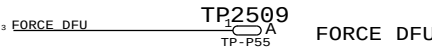
SUPER TP



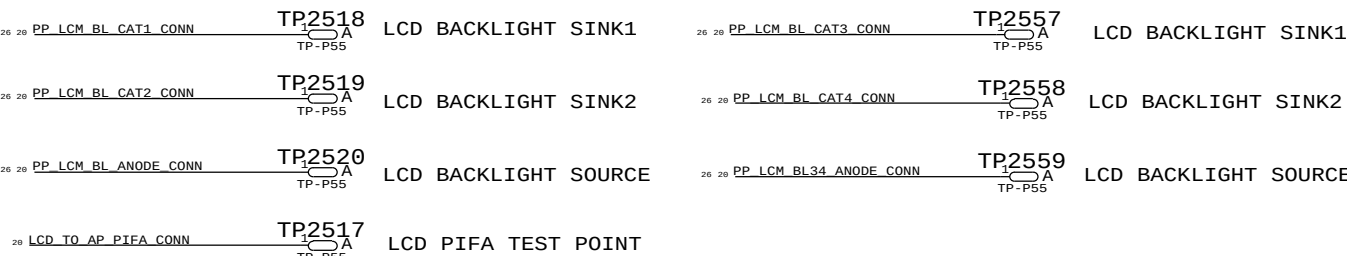
RESET



DFU



LCM BACKLIGHT



VOLTAGE PROPERTIES

1E55 VOLTAGE=3.3V PP3V3_USB 2 12
1E56 VOLTAGE=1.8V PP1V8_VA_I19_I67 10 12 16
1E57 VOLTAGE=3.0V PP3V0_TRISTAR 12 16 17 30
1E58 VOLTAGE=3.0V PP3V0_TMIJ 12 22
1E59 VOLTAGE=3.0V PP3V0_NAND 6 12
1E60 VOLTAGE=3.0V PP3V0_ACC 12 17
1E61 VOLTAGE=3.0V PP3V0_PROX_ALS 11 12
1E62 VOLTAGE=3.0V PP2V0_I1D09 12

1E64 VOLTAGE=4.6V PP_VCC_MAIN 18 12 14 15 16 17 23 30 32
1E65 VOLTAGE=1.0V PP1V0 7 12
1E66 VOLTAGE=3.0V PP3V0_PROX_TRIED 11 12
1E67 VOLTAGE=1.8V PP1V8_ALWAYS 1 5 12 14
1E68 VOLTAGE=3.0V PP3V0_MESA 12 21
1E69 VOLTAGE=1.1V PP_CPU 4 12
1E70 VOLTAGE=1.1V PP_GPU 4 12

1E71 VOLTAGE=1.2V PP1V2_SDRAM 2 4 12 23
1E72 VOLTAGE=1.8V PP1V8_SDRAM 3 4 10 12 13 14 15 17 30
1E73 VOLTAGE=1.8V PP1V8 2 4 5 11 12
1E74 VOLTAGE=1.8V PP1V8_GRAPE 12 24
1E75 VOLTAGE=1.8V PP1V8_OLSCAR 12 22
1E76 VOLTAGE=1.2V PP1V2_NAND_VDDT 6

1E74 VOLTAGE=1.8V PP_EXTMTC_BIAS_FILT_IN 10
1E75 VOLTAGE=1.8V BOARD_ID2 3 27
1E76 VOLTAGE=1.2V PP1V2 2 4 5 11 12
1E77 VOLTAGE=5.0V PP_E75_TO_TRISTAR_ACC1_CONN 18 25
1E78 VOLTAGE=5.0V PP_E75_TO_TRISTAR_ACC1_CONN 17 18
1E79 VOLTAGE=22.0V PP_LCM_BI_ANODE 15 20
1E80 VOLTAGE=0.2V PP_LCM_BI_CAT2 15 20
1E81 VOLTAGE=0.2V PP_LCM_BI_CAT1 15 20
1E82 VOLTAGE=0.2V PP_LCM_BI_CAT2_CONN 20 25
1E83 VOLTAGE=0.2V PP_LCM_BI_CAT1_CONN 20 25

1E85 VOLTAGE=-5.7V PN5V7_SAGE_AVDDN 15 20 24
1E86 VOLTAGE=1.2V PP1V2_OLSCAR 12 22
1E87 VOLTAGE=3.0V PP3V0_MESA_CONN 21
1E88 VOLTAGE=6V PP6V0_LCM_BOOST 15

1E92 VOLTAGE=5.0V PP_STRB_DRIVER_TO_LED_WARM 8 16
1E93 VOLTAGE=5.0V PP_STRB_DRIVER_TO_LED_COOL 8 16
1E94 VOLTAGE=1.8V PP_CODEC_TO_MIC1_BIAS 10 18
1E95 VOLTAGE=1.8V PP_EXTMTC_BIAS_FILT_IN 10
1E96 VOLTAGE=1.8V PP_EXTMTC_BIAS_FILT 10
1E97 VOLTAGE=1.8V PP_CODEC_TO_FRONTMIC3_BIAS 10 11
1E98 VOLTAGE=1.8V PP_CODEC_TO_REARMIC2_BIAS 8 10
1E99 VOLTAGE=1.8V PP_CODEC_FILT+ 10
1E100 VOLTAGE=2.2V PP_CODEC_SPKR_V0 10
1E101 VOLTAGE=2.5V PP_CODEC_VCPETILT- 10
1E102 VOLTAGE=2.5V PP_CODEC_VCPETILT+ 10
1E103 VOLTAGE=2.5V PP_CODEC_VHP_ELYN 10
1E104 VOLTAGE=0.2V PP_CODEC_VHP_ELYC 10
1E105 VOLTAGE=2.5V PP_CODEC_VHP_ELYP 10
1E106 VOLTAGE=1.8V PP1V8_ECAM_CONN 11
1E107 VOLTAGE=3.0V PP2V85_ECAM_AVDD_CONN 11
1E108 VOLTAGE=1.8V PP_CODEC_TO_FRONTMIC3_BIAS_CONN 11
1E109 VOLTAGE=3.0V PP3V0_ALS_CONN 11
1E110 VOLTAGE=1.2V PP1V2_ECAM_VDDIO_CONN 11
1E111 VOLTAGE=5.0V PP5V0_USB 12 14 17 18 25
1E112 VOLTAGE=5.0V PP5V0_USB_TO_PMI 12
1E113 VOLTAGE=4.6V PP_BUCK5_LX0 12
1E114 VOLTAGE=4.6V PP_BUCK3_LX 12
1E115 VOLTAGE=4.6V PP_BUCK4_LX 12
1E116 VOLTAGE=4.6V PP_BUCK2_LX 12
1E117 VOLTAGE=4.6V PP_BUCK1_LX1 12
1E118 VOLTAGE=4.6V PP_BUCK1_LX0 12
1E119 VOLTAGE=4.6V PP_BUCK0_LX3 12
1E120 VOLTAGE=4.6V PP_BUCK0_LX2 12
1E121 VOLTAGE=4.6V PP_BUCK0_LX1 12
1E122 VOLTAGE=4.6V PP_BUCK0_LX0 12
1E123 VOLTAGE=6.0V PP_CHESTNUT_LXP 15
1E124 VOLTAGE=6.0V PP_CHESTNUT_CP 15
1E125 VOLTAGE=6.0V PP_CHESTNUT_CN 15
1E126 VOLTAGE=5.7V PP5V7_SAGE_AVDDH 15 24
1E127 VOLTAGE=5.7V PP5V7_LCM_AVDDH 15 24
1E128 VOLTAGE=5.1V PP5V1_GRAPE_VDDH 15 24
1E129 VOLTAGE=22.0V PP_WLED_LX 15
1E130 VOLTAGE=18.0V PP18V0_MESA_SW 15
1E131 VOLTAGE=17.0V PP17V0_M0JAVE_I1D0IN 15
1E132 VOLTAGE=16.5V PP16V5_MESA 15 21 25
1E133 VOLTAGE=8.0V PP_SPKAMP_SW 16
1E134 VOLTAGE=8.0V PP_I19_VB00ST 16
1E135 VOLTAGE=1.8V PP_SPKAMP_FILT 16
1E136 VOLTAGE=1.8V PP_SPKAMP_I1D0_FILT 16

1E VOLTAGE=5.0V PP_LED_DRV_LX 16
1E VOLTAGE=5.0V PP_LED_BOOST_OUT 16

1E VOLTAGE=2.7V PP_BB_VDD_2V7_CONN 18

1E VOLTAGE=1.8V PP_CODEC_TO_MIC1_BIAS_CONN 18
1E VOLTAGE=4.6V PP_E75_TO_TRISTAR_ACC2 17 18
1E VOLTAGE=4.6V PP_E75_TO_TRISTAR_ACC2_CONN 18 25

1E10 VOLTAGE=1.8V PP1V8_LCM_CONN 20
1E11 VOLTAGE=22.0V PP_LCM_BI_ANODE_CONN 20 25
1E12 VOLTAGE=-5.7V PN5V7_LCM_AVDDN_CONN 20
1E13 VOLTAGE=5.7V PP5V7_LCM_AVDDH_CONN 20

1E16 VOLTAGE=1.8V PP1V8_MESA 21
1E17 VOLTAGE=16.5V PP16V5_MESA_CONN 21

1E135 VOLTAGE=5.0V PP_TRISTAR_PIN 17

1E20 VOLTAGE=1.2V PP1V2_RCAM_CONN 23
1E21 VOLTAGE=1.8V PP1V8_RCAM_CONN 23

1E23 VOLTAGE=3.0V PP2V85_CAM_VDD 11 23
1E24 VOLTAGE=1.8V PP2V85_RCAM_AVDD_CONN 23
1E25 VOLTAGE=1.8V PP_CUMULUS_VDDCORE 24
1E26 VOLTAGE=1.2V PP_CUMULUS_VDDANA 24
1E27 VOLTAGE=13.5V PP_SAGE_TO_TOUCH_VCPH_CONN 24
1E28 VOLTAGE=-12V PN_SAGE_TO_TOUCH_VCPH_CONN 24
1E29 VOLTAGE=13.5V PP_SAGE_TO_TOUCH_VCPH 24
1E30 VOLTAGE=-12V PN_SAGE_TO_TOUCH_VCPH 24

1E32 VOLTAGE=-12V PN_SAGE_VCPH_F 24
1E33 VOLTAGE=5.7V PP_SAGE_LX 24
1E34 VOLTAGE=17.0V PP_SAGE_LX 24

1E36 VOLTAGE=1.8V PP_PMI_VREF 13
1E37 VOLTAGE=14V PP_SAGE_VBST_OUTH 24

1E40 VOLTAGE=5.0V PP_TTGRTS_VBUS_DET 14

1E41 VOLTAGE=2.5V PP_PMI_VDD_REF 13
1E42 VOLTAGE=1.8V PP_EXTMTC_BIAS 10
1E43 VOLTAGE=1.8V PP1V8_XTAL 2
1E44 VOLTAGE=1.8V PP_PMI_VDD_RTC 13

1E45 VOLTAGE=3.80V PP_BATT_VCC 14 16 25 41 46 47
1E46 VOLTAGE=1.8V PP1V8_MESA_CONN 21
1E47 VOLTAGE=3.0V PP3V0_PROX_CONN 11

1E50 VOLTAGE=1.0V PP0V95_FIXED_SOC 4 7 12
1E51 VOLTAGE=1.0V PP0V95_FIXED_SOC_PCIE 7

1E52 VOLTAGE=1.2V PP1V2_PL1 2

1E53 VOLTAGE=1.0V PP_BUCK5_LX1 12

1E54 VOLTAGE=1.0V PP_VAR_SOC 5 12

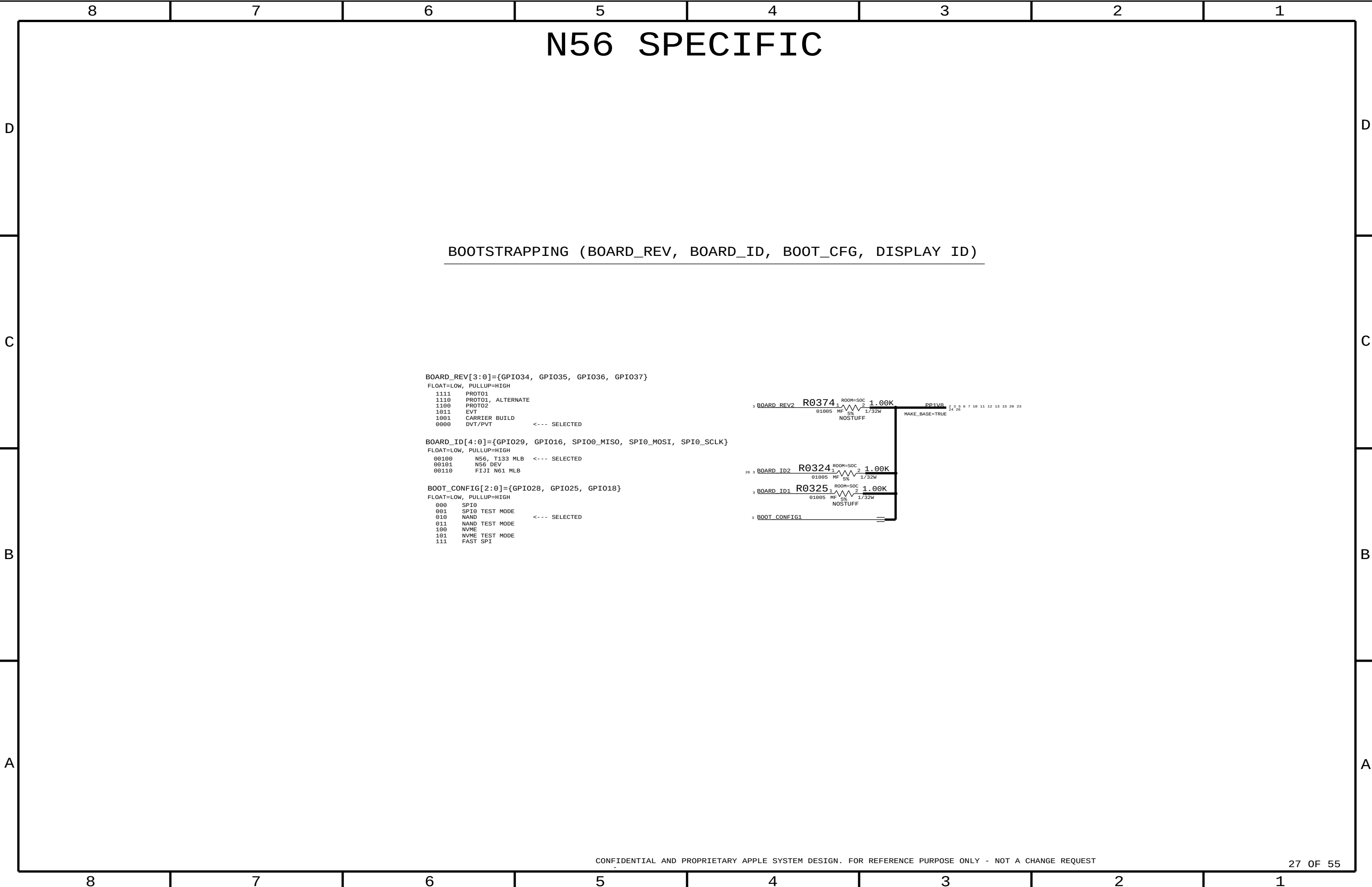
1E137 VOLTAGE=1.8V PP1V8_HALL_CONN 21
1E138 VOLTAGE=1.8V PP1V8_MESA_GND 21
1E139 VOLTAGE=5.0V CHARGER_I1D0 14
1E140 VOLTAGE=5.0V PMTD_CAP 14

N56 SPECIFIC VOLTAGE PROPERTIES

1E150 VOLTAGE=22.0V PP_WLED34_LX 15
1E151 VOLTAGE=22.0V PP_LCM_BI_34_ANODE 15 20
1E152 VOLTAGE=22.0V PP_LCM_BI_34_ANODE_CONN 20 25
1E153 VOLTAGE=0.2V PP_LCM_BI_CAT3 15 20
1E154 VOLTAGE=0.2V PP_LCM_BI_CAT4 15 20
1E155 VOLTAGE=0.2V PP_LCM_BI_CAT3_CONN 20 25
1E156 VOLTAGE=0.2V PP_LCM_BI_CAT4_CONN 20 25

1E158 VOLTAGE=-12V PN_SAGE_TO_TOUCH_VCPH_FILT 24
1E159 VOLTAGE=2.0V PP_RCAM_AF 12 23
1E160 VOLTAGE=2.0V PP_RCAM_AF_CONN 23

1E161 VOLTAGE=-14.0V PN_SAGE_VBST_OUTI 24
1E162 VOLTAGE=-5.7V PN5V7_SAGE_AVDDN_FILT 24
1E163 VOLTAGE=2.0V PP_BUCK6_LX 12



8

7

6

5

4

3

2

1

D

D

C

C

B

B

A

A

8

7

6

5

4

3

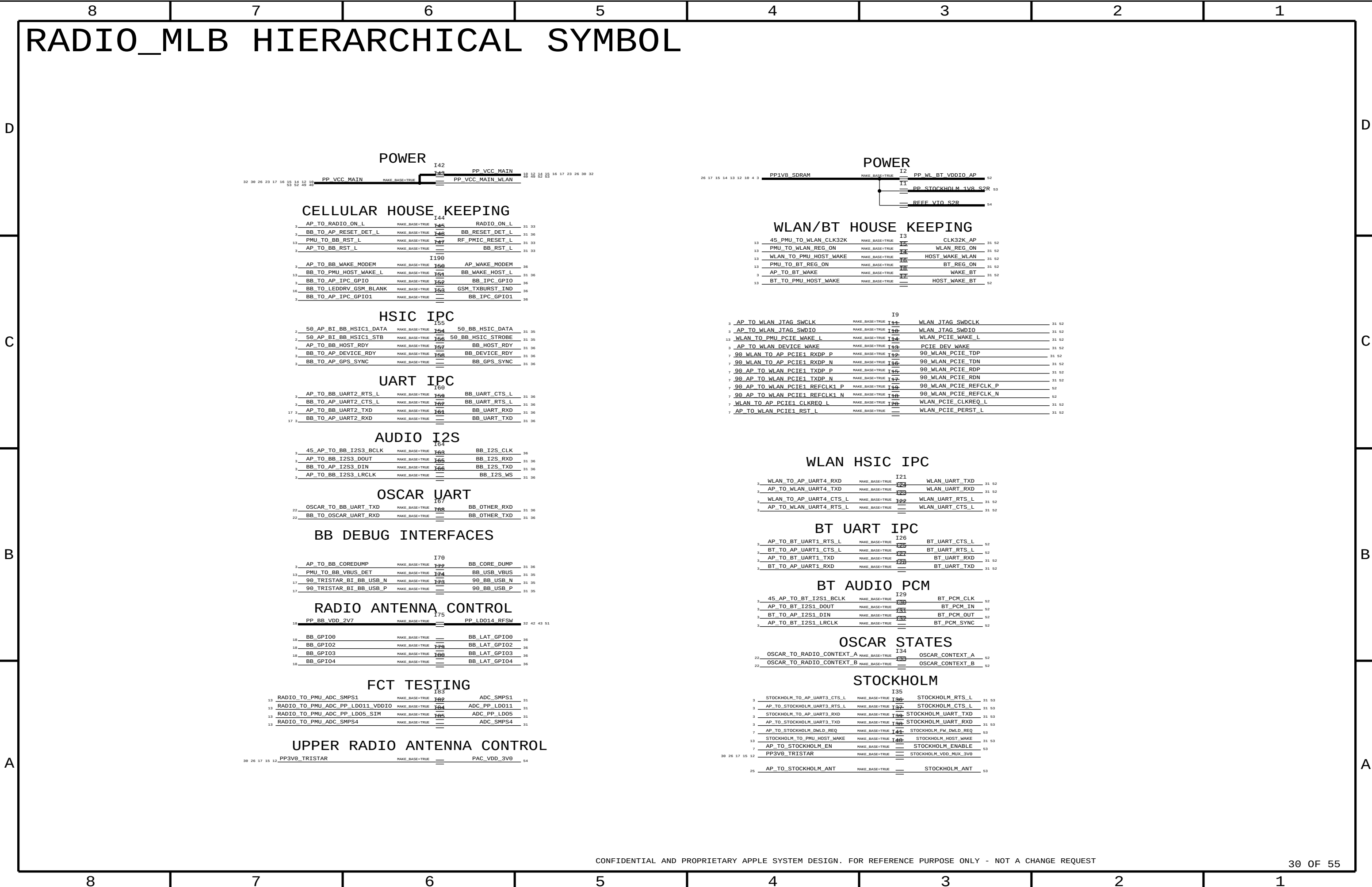
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1

8								7								6								5								4								3								2								1																							
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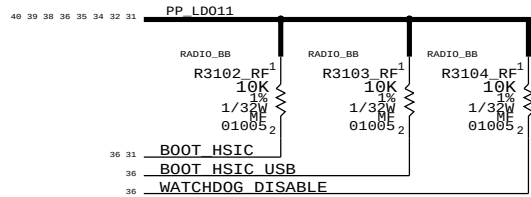
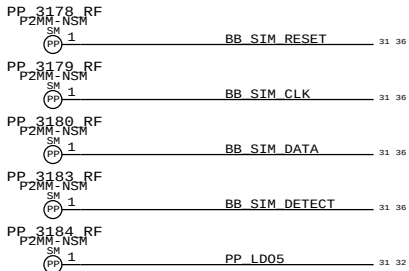
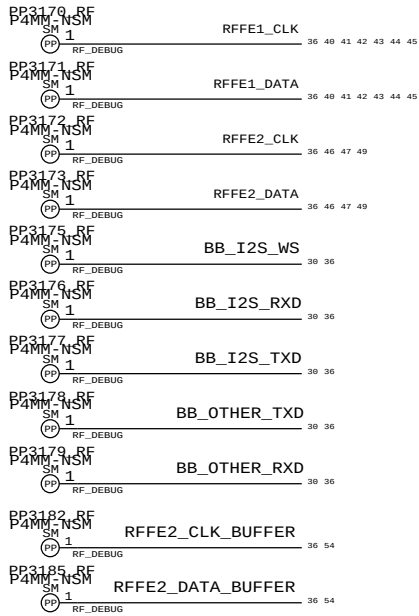
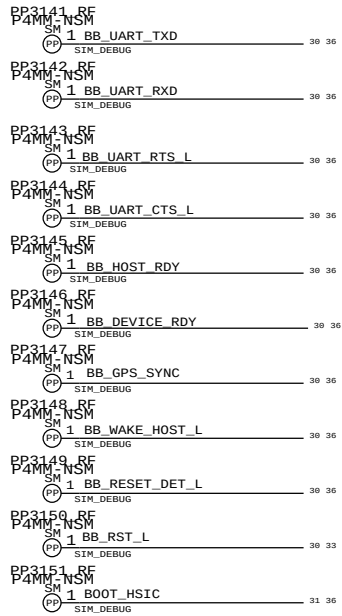
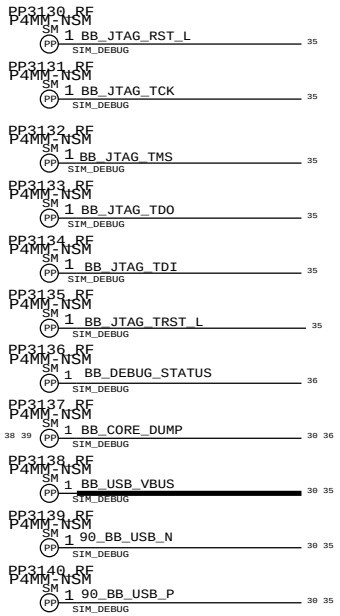
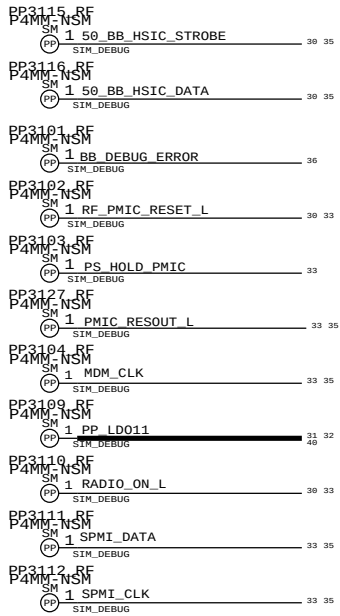
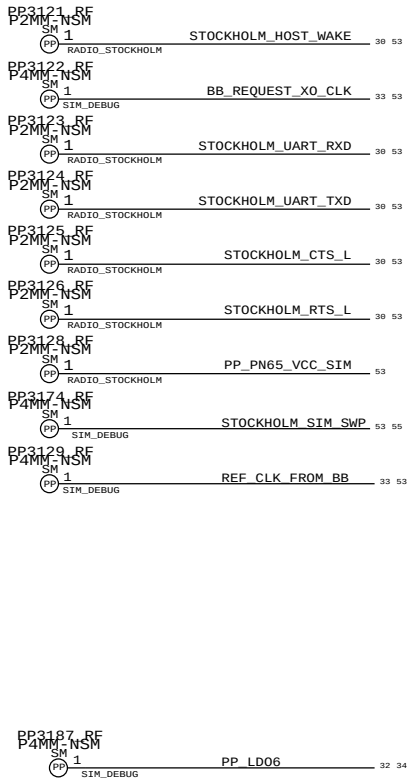
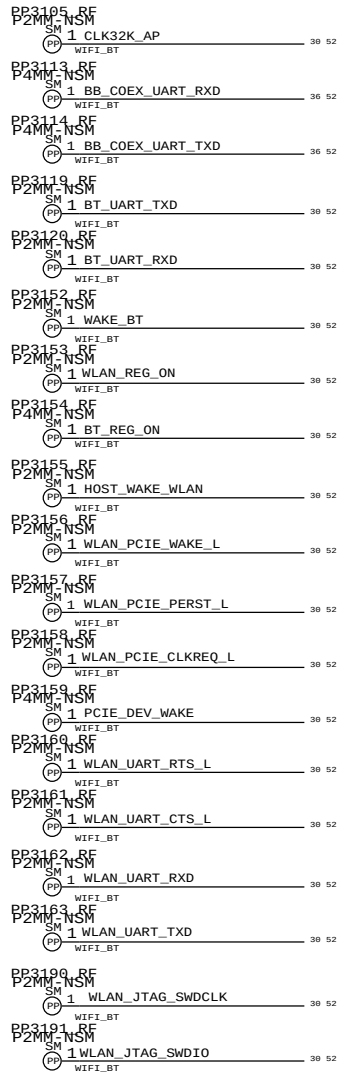
CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSE ONLY - NOT A CHANGE REQUEST

29 OF 55

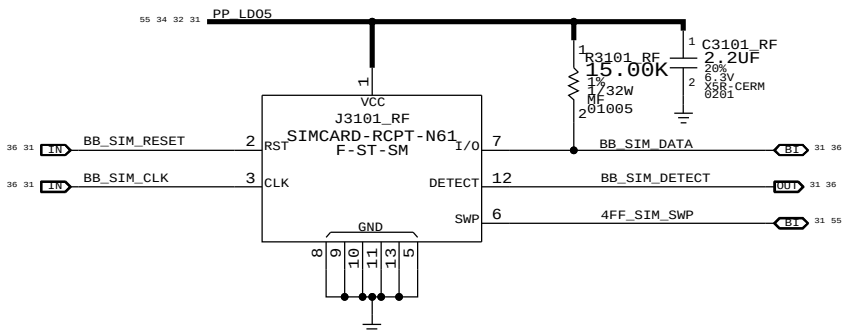


AP INTERFACE & DEBUG CONNECTORS

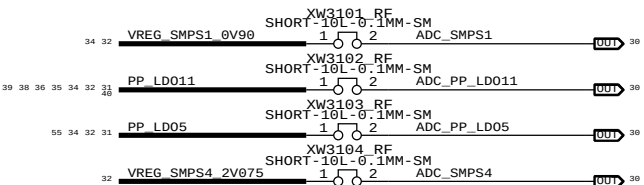
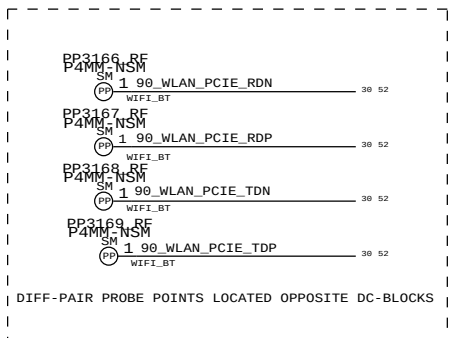
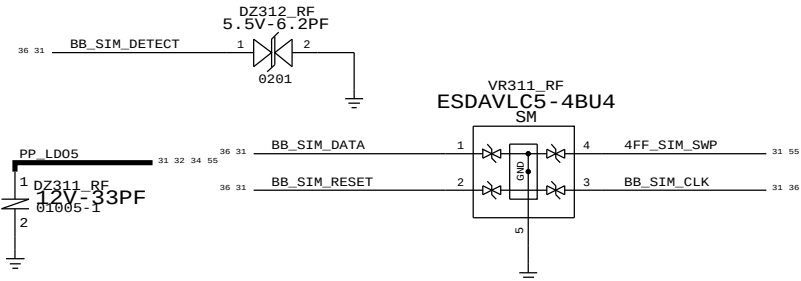
PROBE POINTS



SIM CARD CONNECTOR



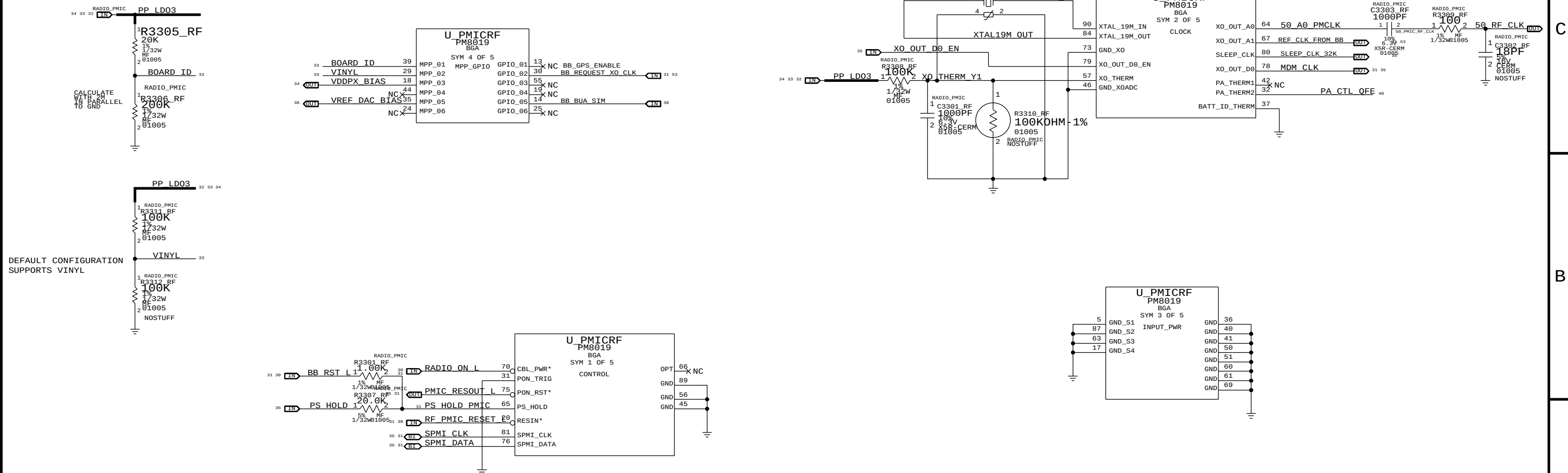
SIM CARD ESD PROTECTION



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C401
R411
L400
U404

BOARD_ID	REVISION
0.00V	N61 PROTO_MLB1
0.50V	N61 DEV3
0.70V	N61 DEV4
0.90V	N61 PROTO_MLB2
1.10V	N61/N56 PROTO1
1.30V	N61/N56 PROTO2
1.40V	N61/N56 EVT1
1.50V	N61/N56 EVT2 (CARRIER)
1.60V	N61/N56 DVT
1.70V	N61/N56 PVT



BASEBAND (1 OF 3)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

C538
R500
L500
U502

D

C

B

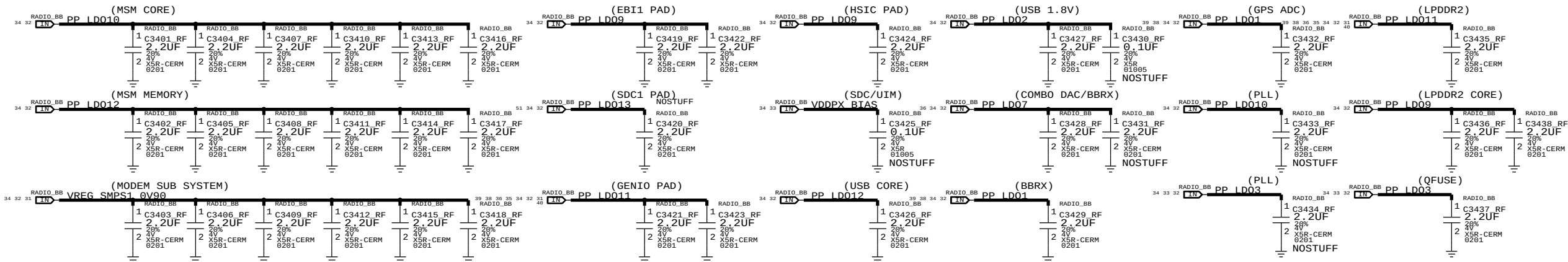
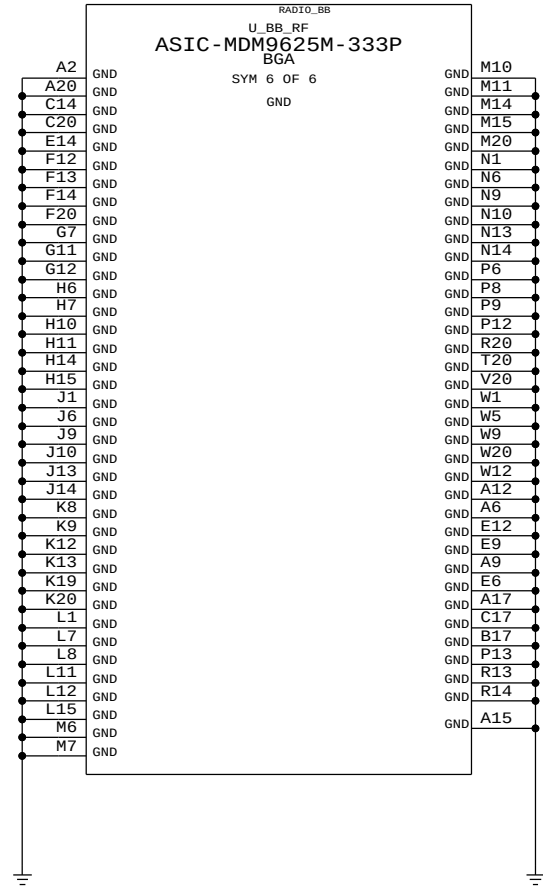
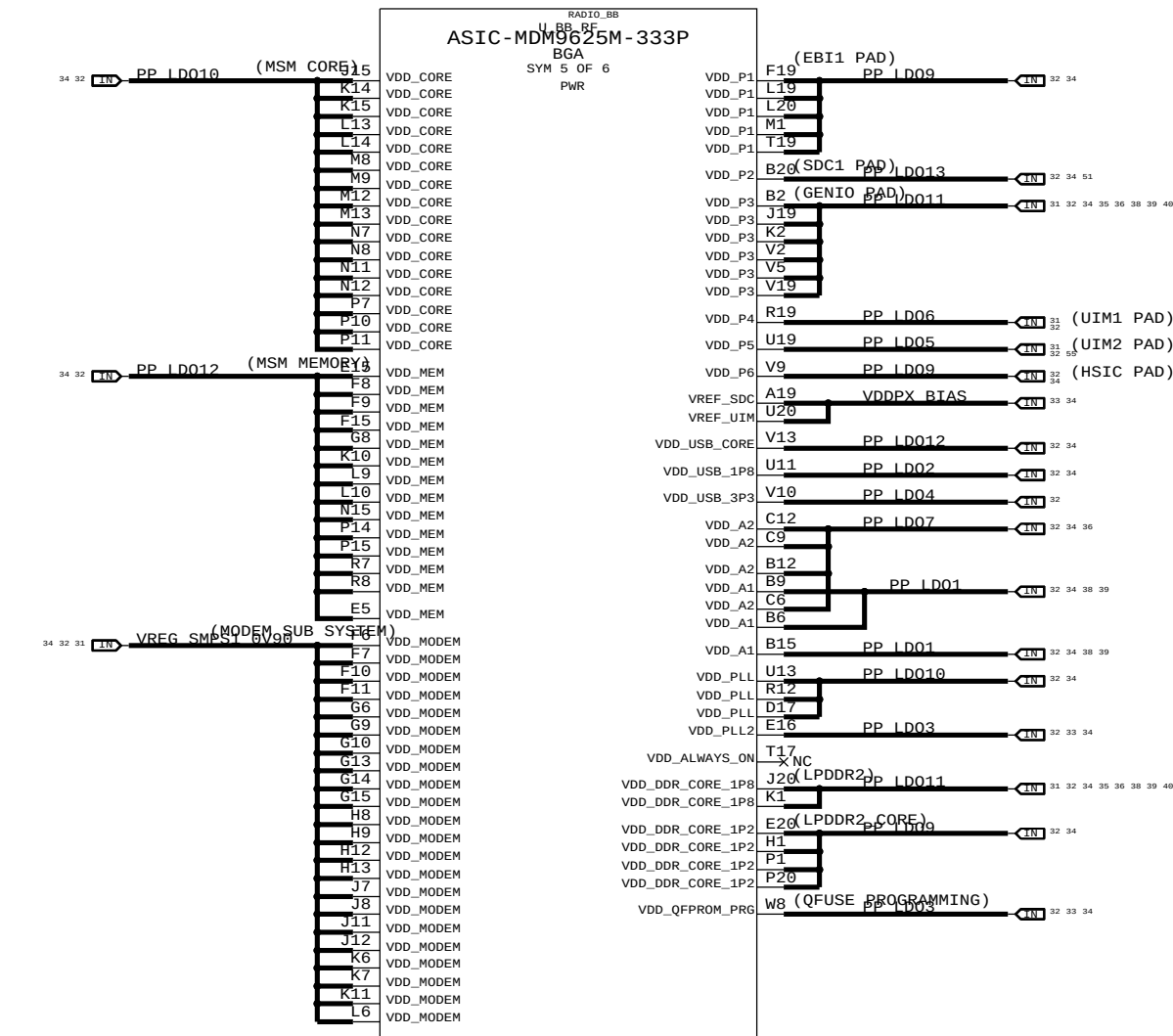
A

D

C

B

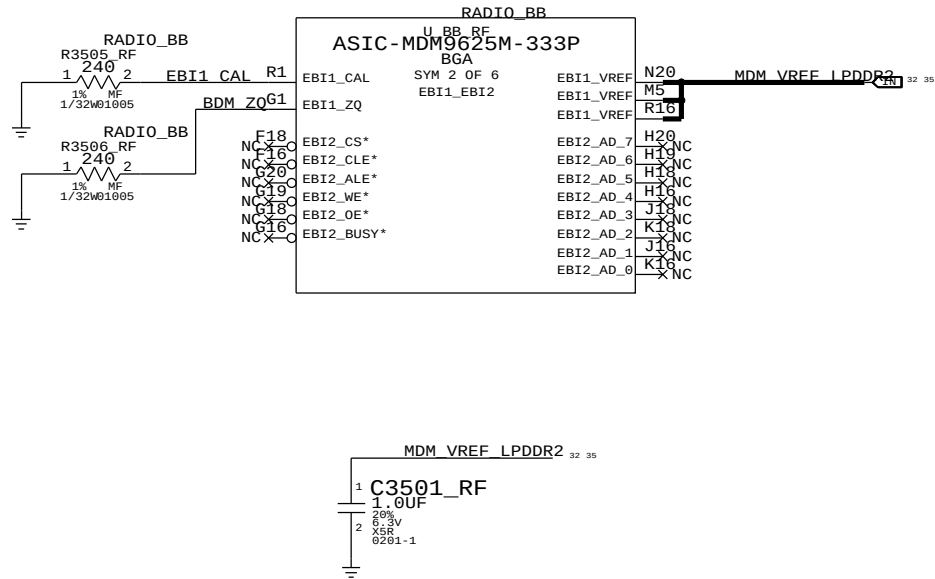
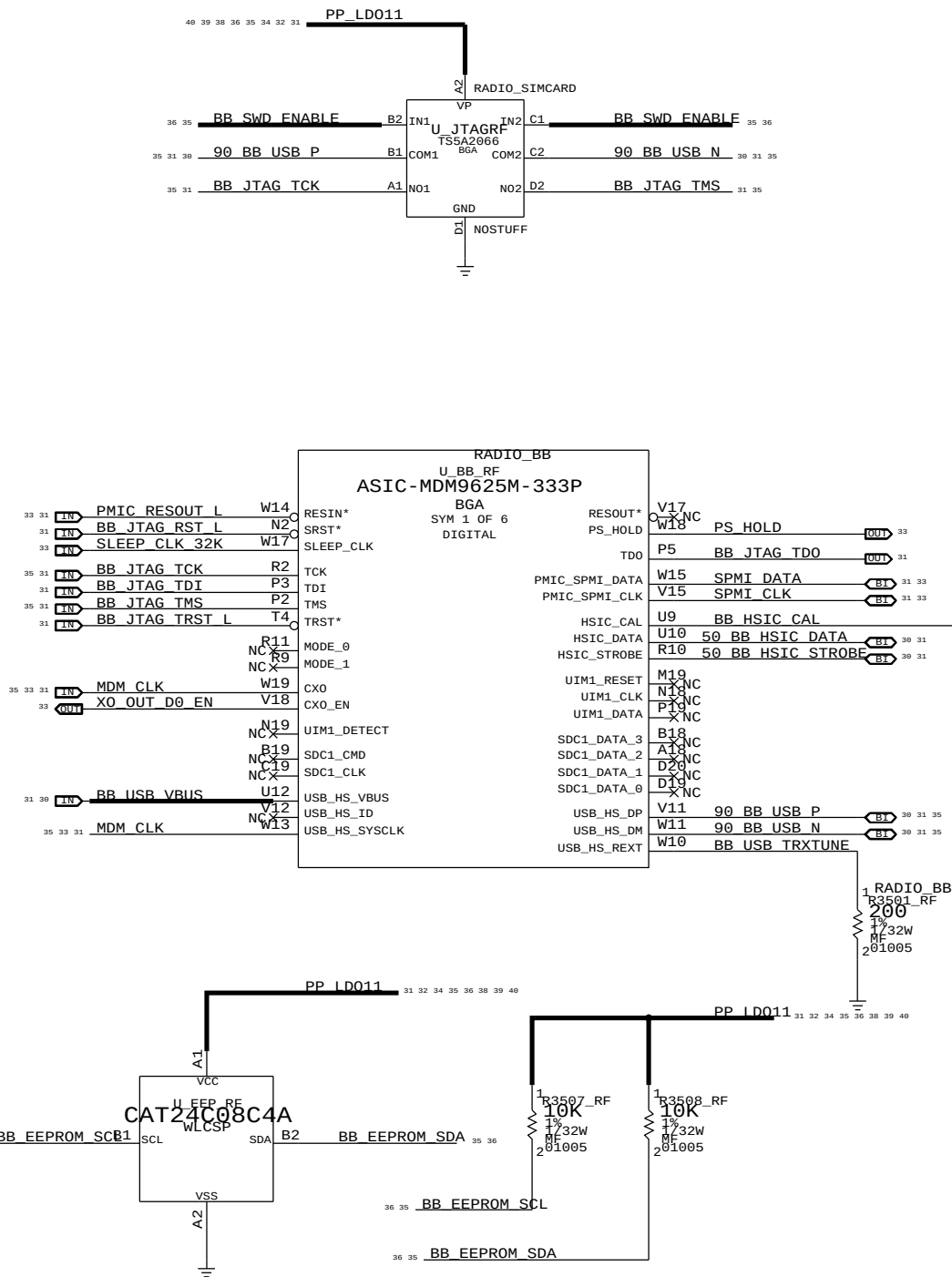
A



BASEBAND (2 OF 3)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

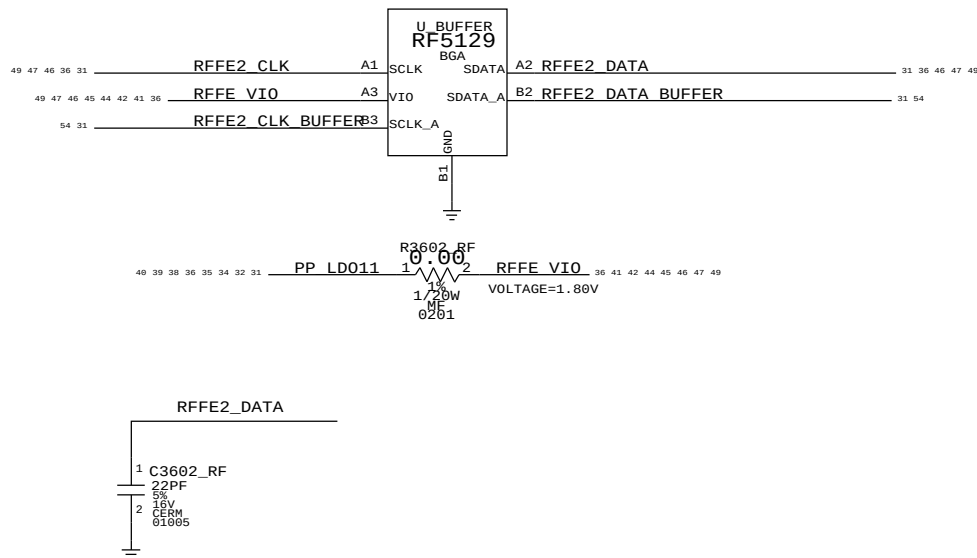
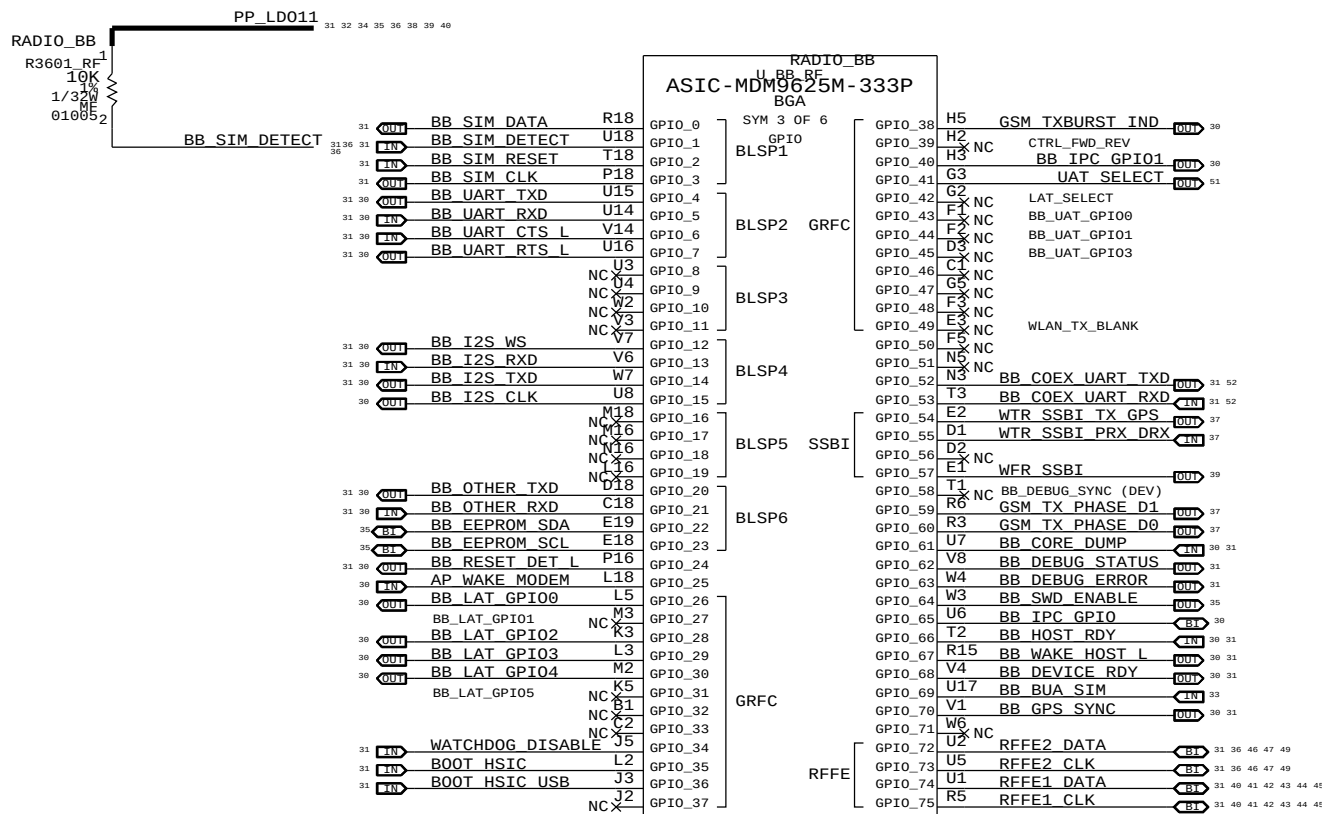
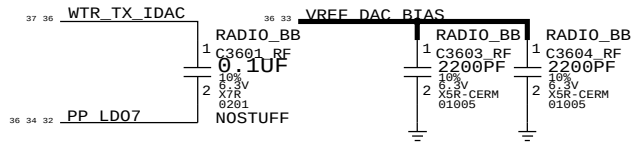
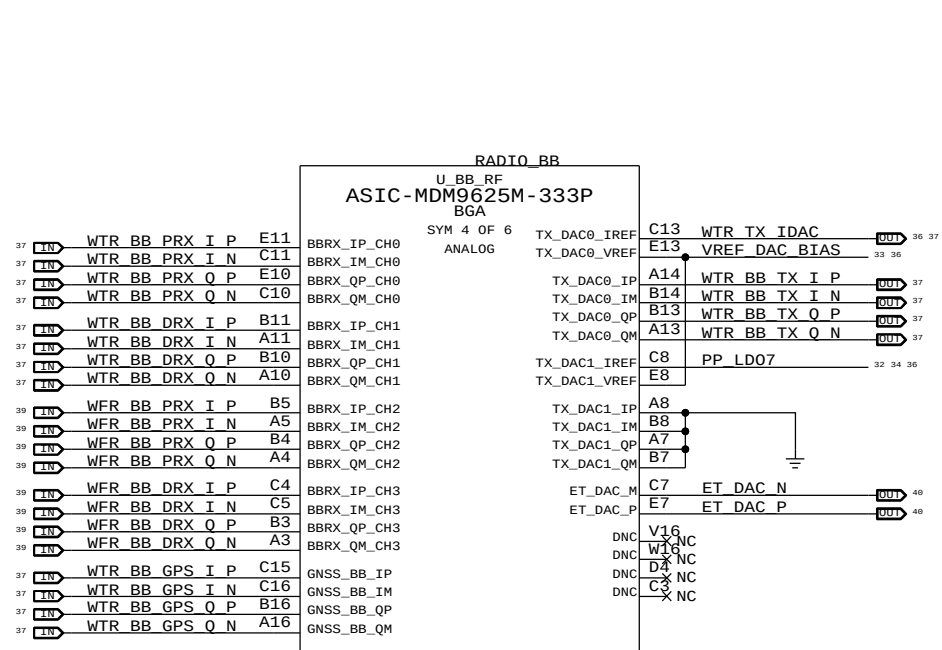
C600
R606
L600
U602



BASEBAND (3 OF 3)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

C704
R700
L700
U702



WTR TRANSCEIVER (1 OF 2)

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C802
R802
L800
U803

D

D

C

C

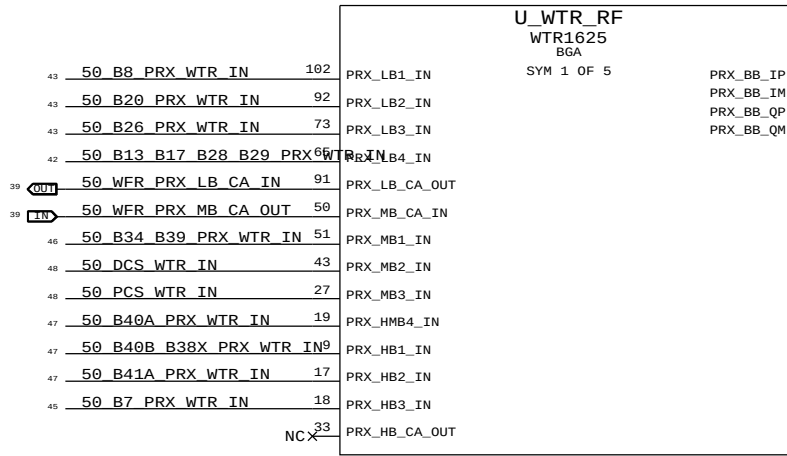
B

B

A

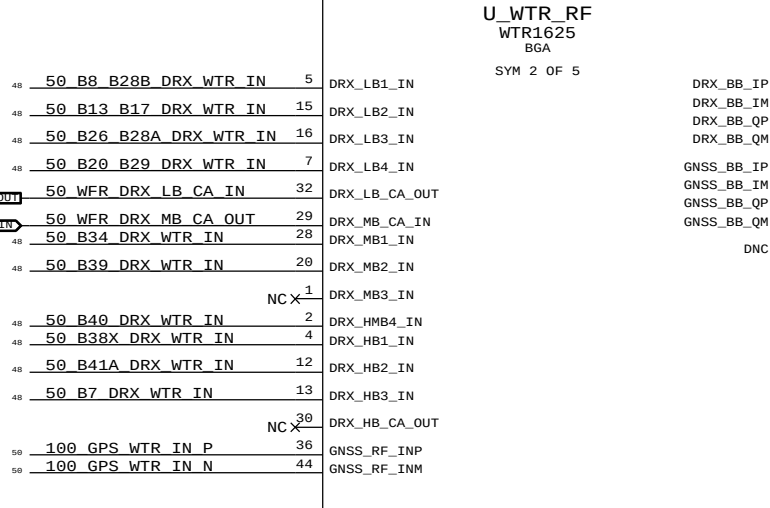
A

LB1	DC
LB2	DC
LB3	DC
LB4	DC
MB1	NO DC
MB2	DC
MB3	DC
HB1	NO DC
HB2	DC
HB3	DC
HBMB4	NO DC



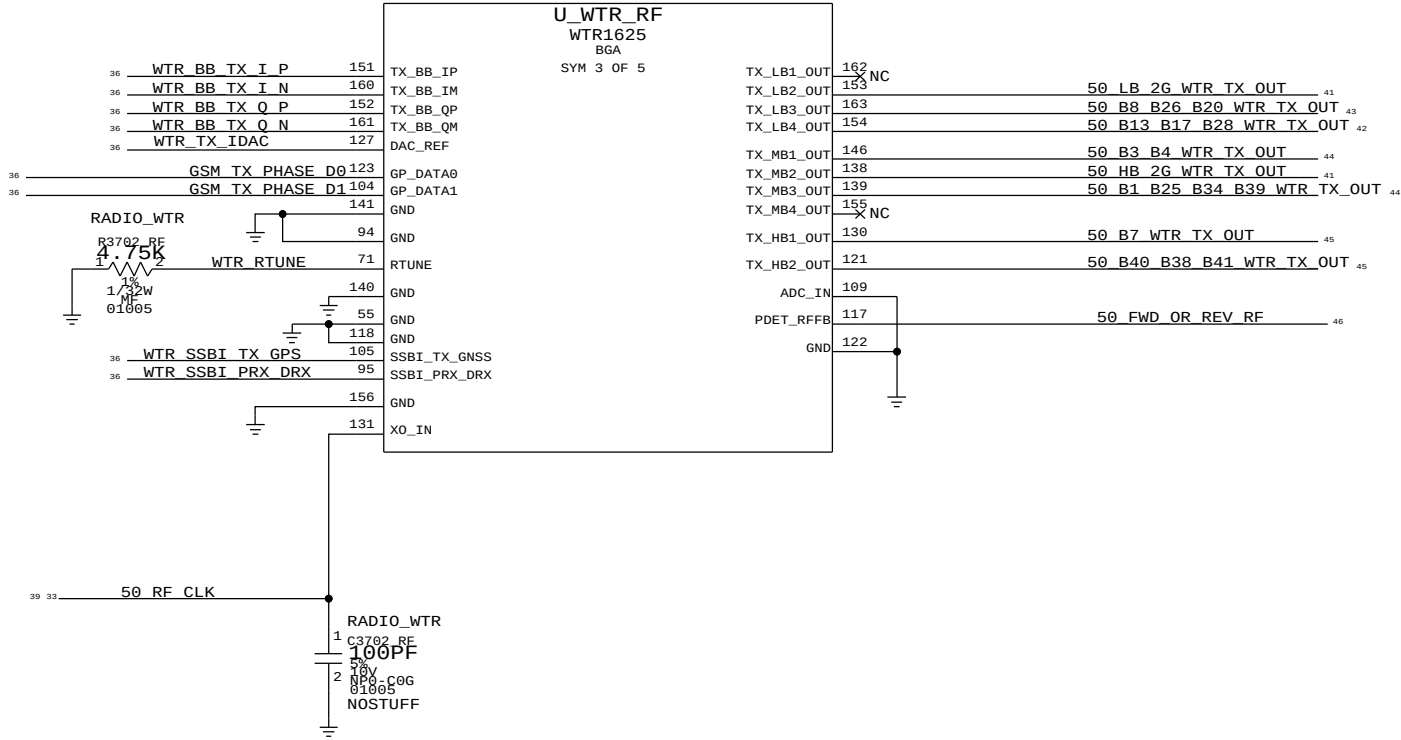
99	WTR_BB_PRX_I_P	36
108	WTR_BB_PRX_I_N	36
107	WTR_BB_PRX_Q_P	36
97	WTR_BB_PRX_Q_N	36

LB1	DC
LB2	DC
LB3	DC
LB4	DC
MB1	NO DC
MB2	DC
MB3	DC
HB1	NO DC
HB2	DC
HB3	DC
HBMB4	NO DC



76	WTR_BB_DRX_I_P	36	RADIO_WTR
86	WTR_BB_DRX_I_N	36	RADIO_WTR
61	WTR_BB_DRX_Q_P	36	RADIO_WTR
68	WTR_BB_DRX_Q_N	36	RADIO_WTR
60	WTR_BB_GPS_I_P	36	RADIO_WTR
53	WTR_BB_GPS_I_N	36	RADIO_WTR
67	WTR_BB_GPS_Q_P	36	RADIO_WTR
85	WTR_BB_GPS_Q_N	36	RADIO_WTR

DNC X NC



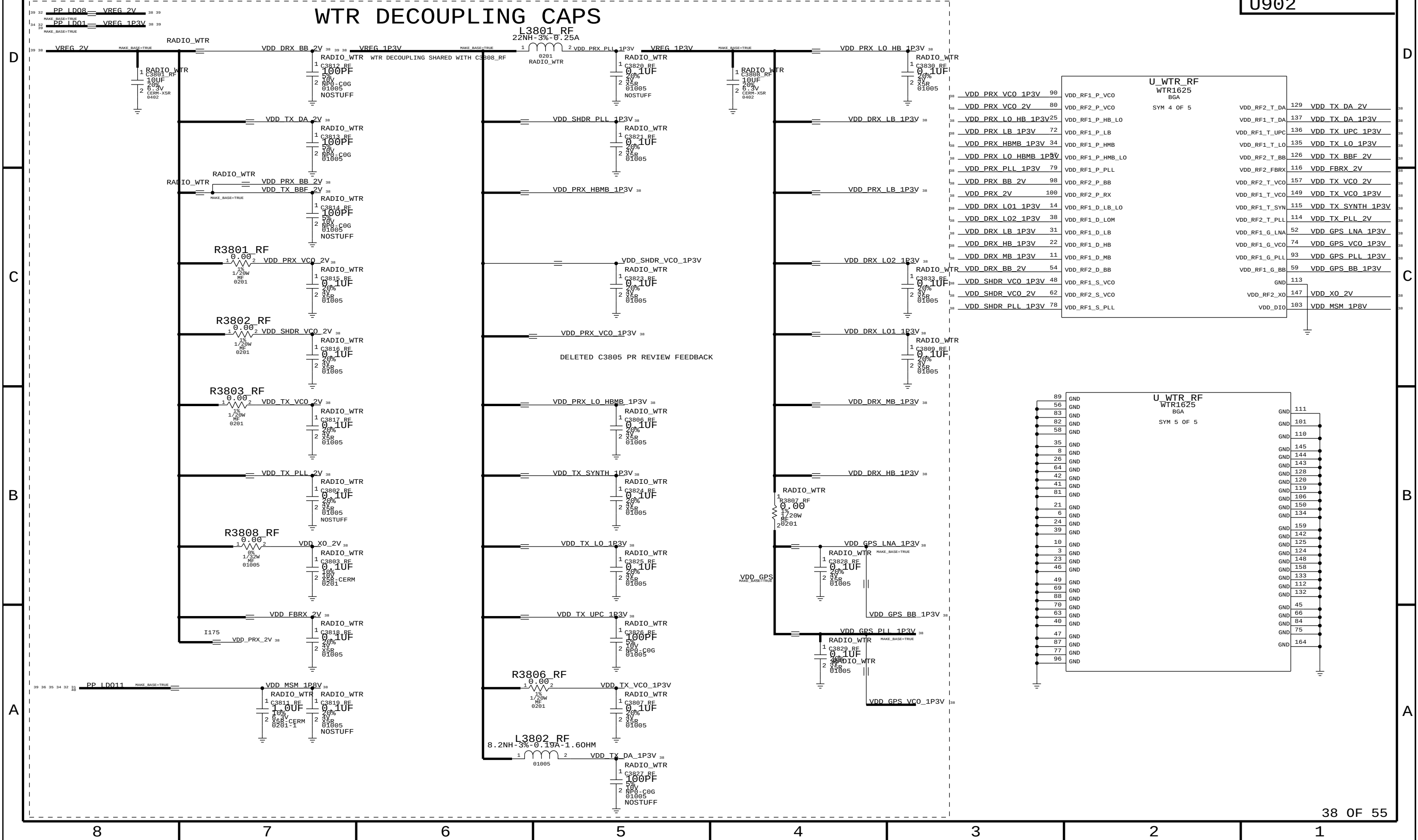
RF_CLK IS SHARED BETWEEN WTR AND WFR. LENGTH DIFFERENCE BETWEEN THE TWO SHOULD BE < 5MM.

WTR TRANSCEIVER (2 OF 2)

~~CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.~~

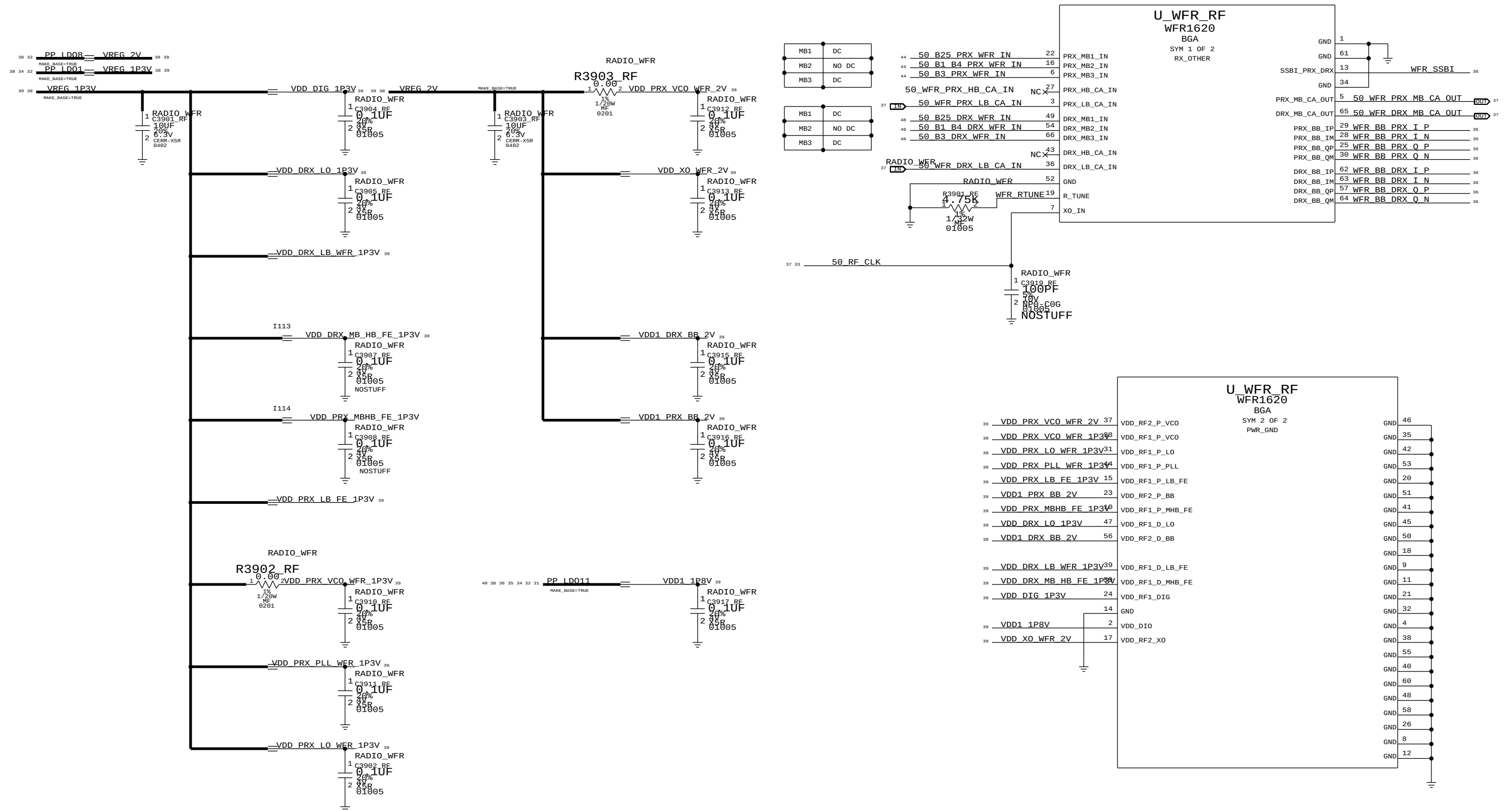
C934
R926
L3802_RF
U902

WTR DECOUPLING CAPS



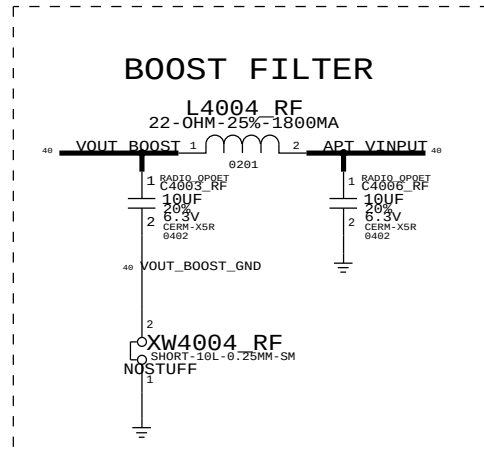
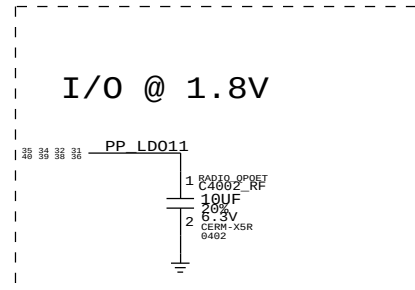
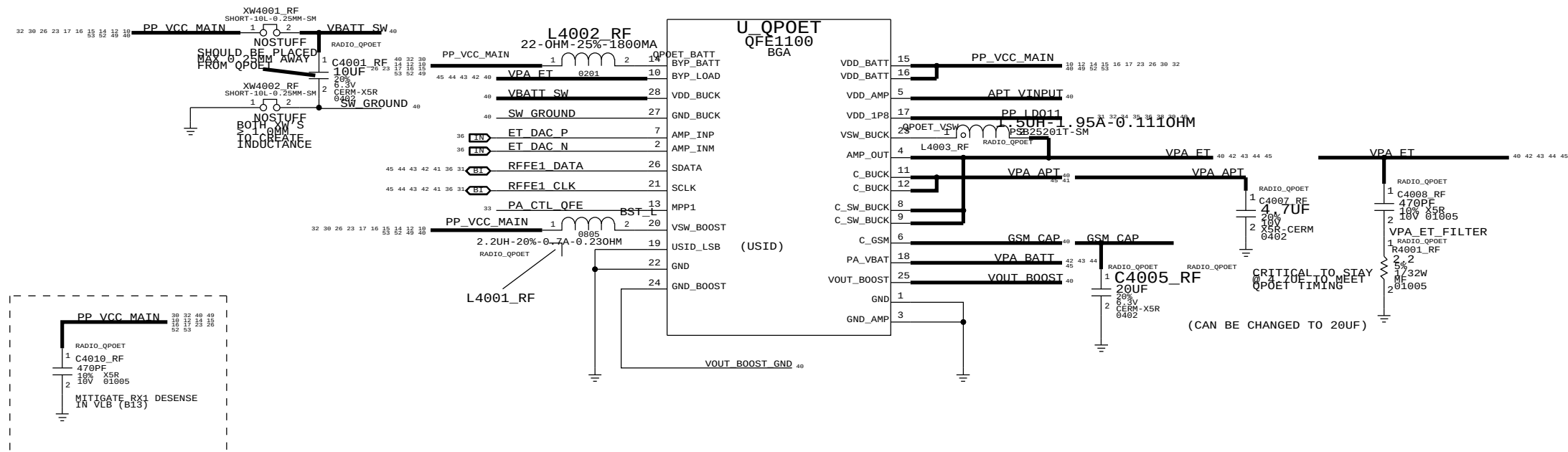
CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

C1019
R1016
L1000
U1002



~~CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.~~

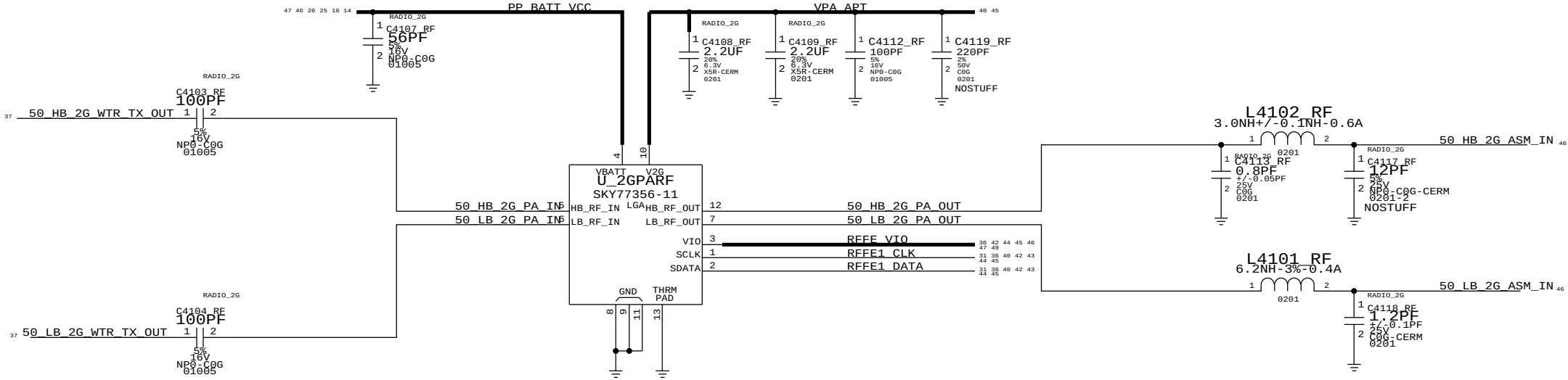
C11110
R11102
L11104
U11101



2G PA

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

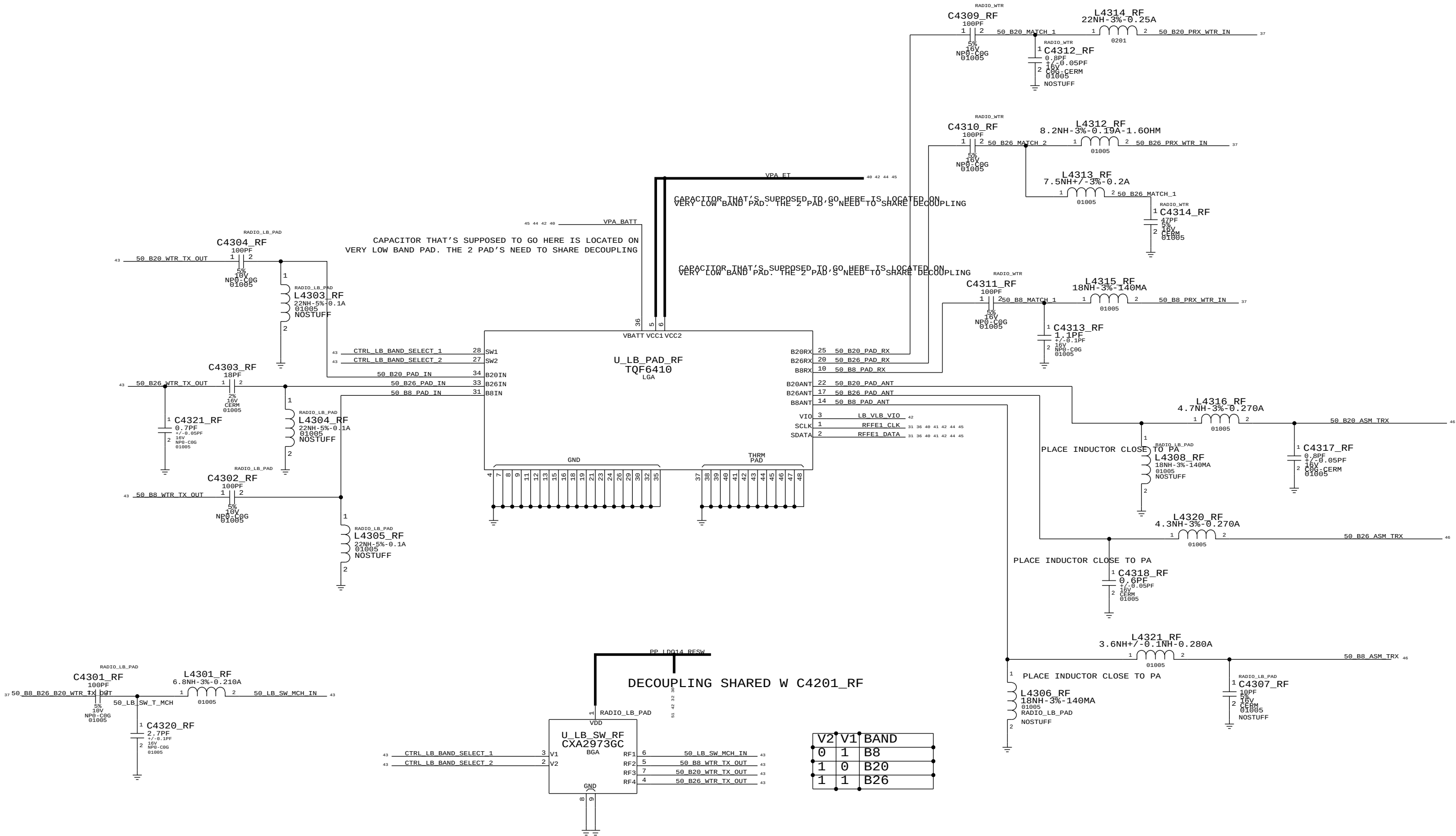
C1208
R1200
L1204
U1201



LOW BAND PAD (B8, B26, B20)

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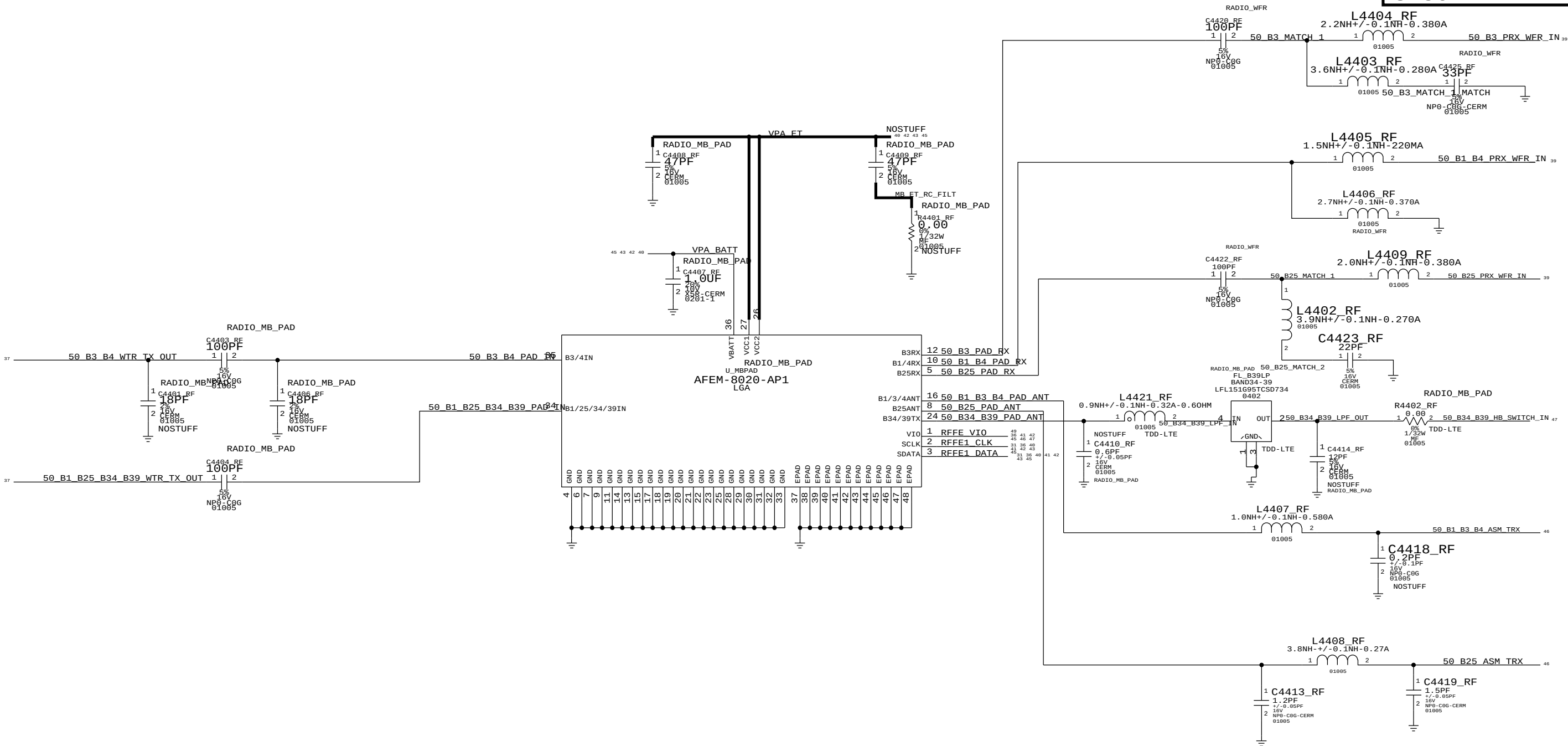
C4318_RF
R1400
L4322_RF
U1402



MID BAND PAD (B1, B25, B3, B4, B34, B39)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

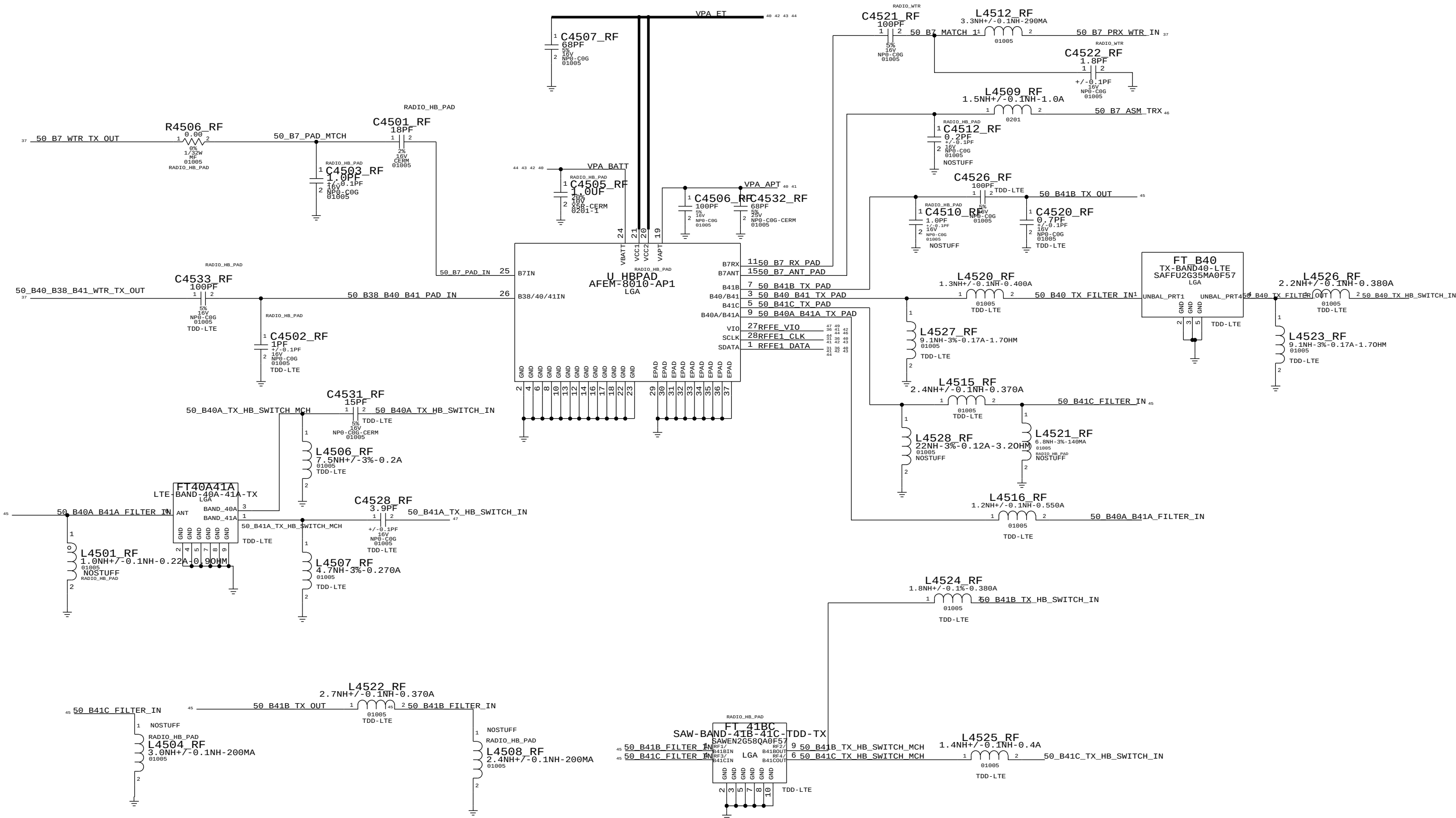
C4426_RF
R1500
L4409_RF
U1501



HIGH BAND PAD (B7, B38, B40, B41, XGP)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

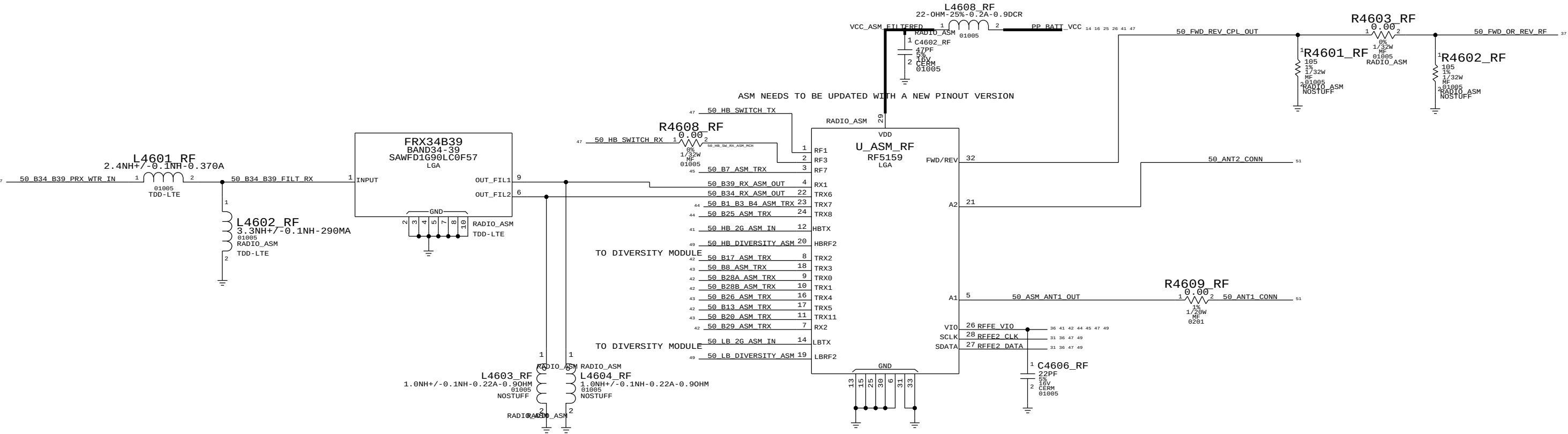
C4533_RF
R1600
L1616
U1601



ANTENNA SWITCH

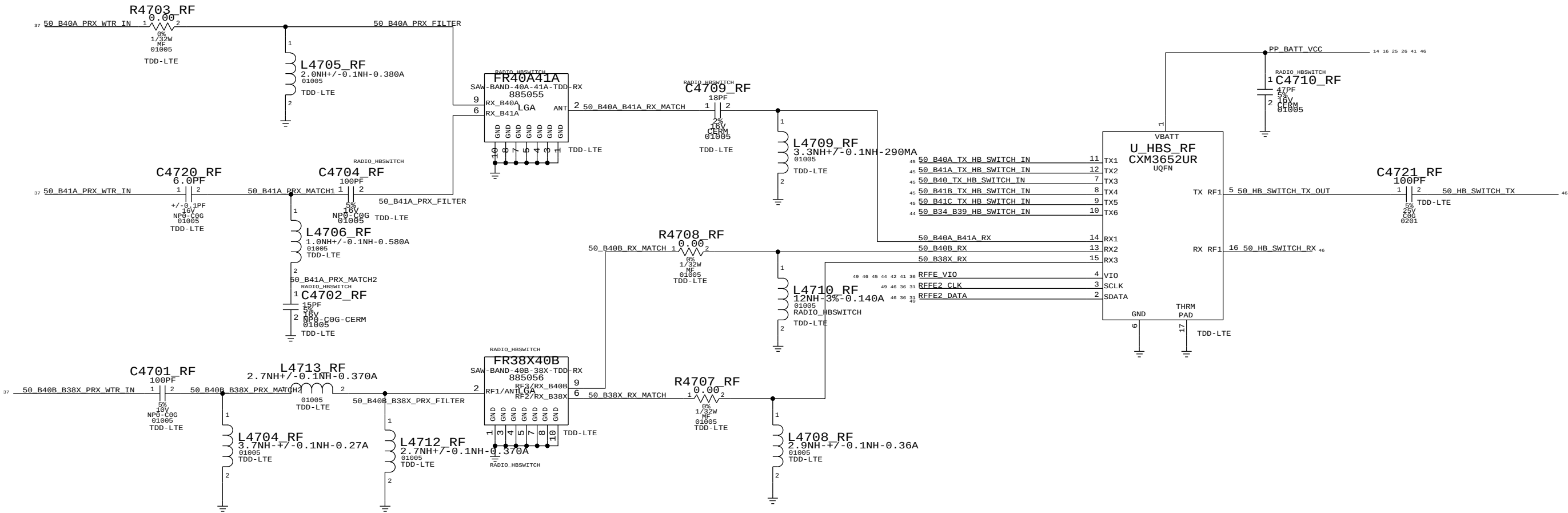
CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

C1702
R1700
L4608_RF
U1702



HIGH BAND SWITCH

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.



RX DIVERSITY (1)

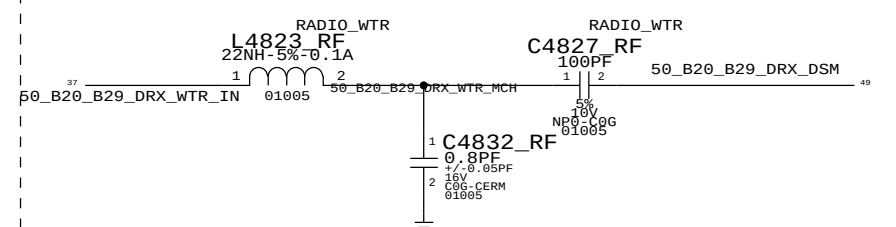
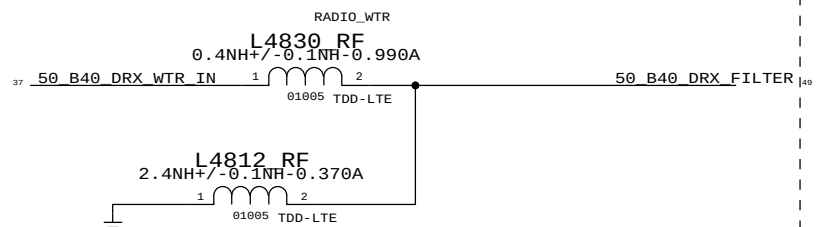
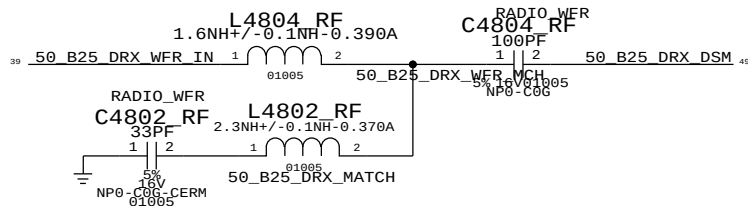
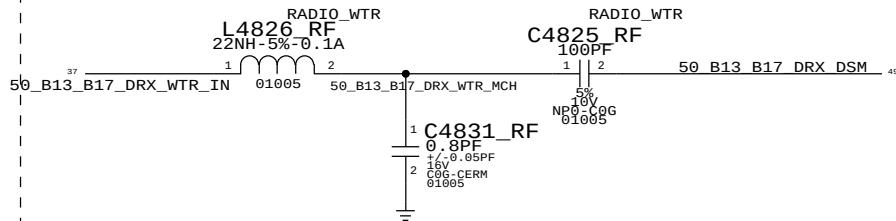
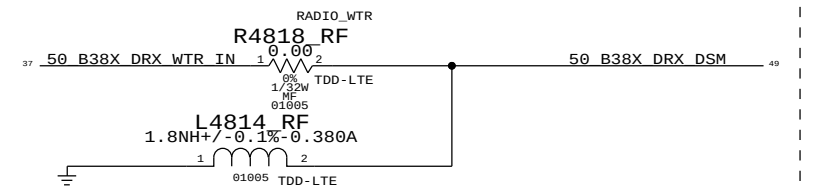
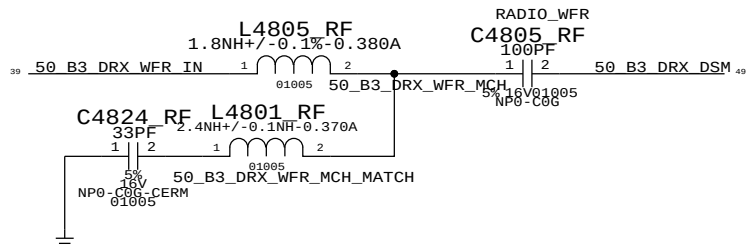
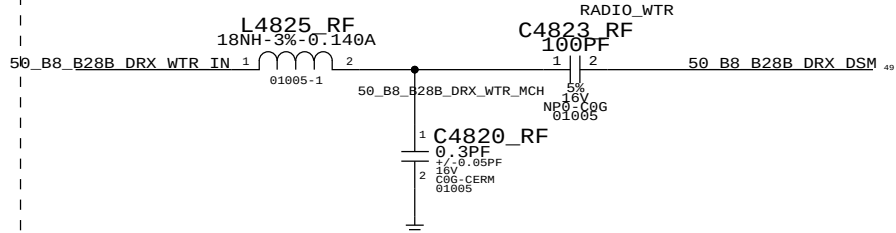
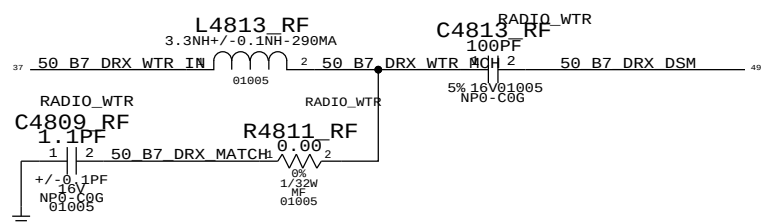
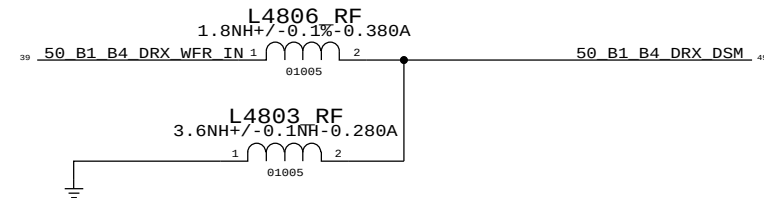
CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

C4826_RF
R1800
L1829
U1801

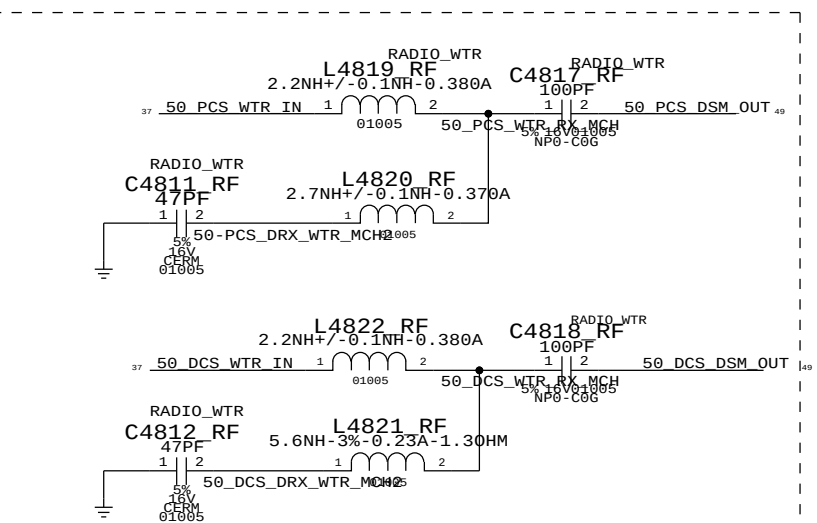
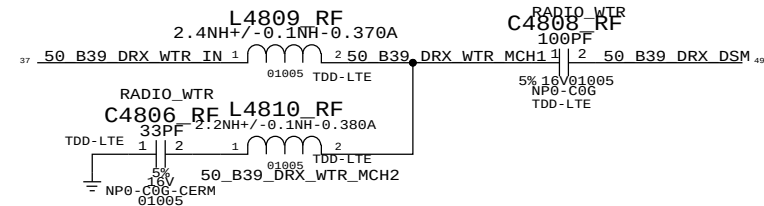
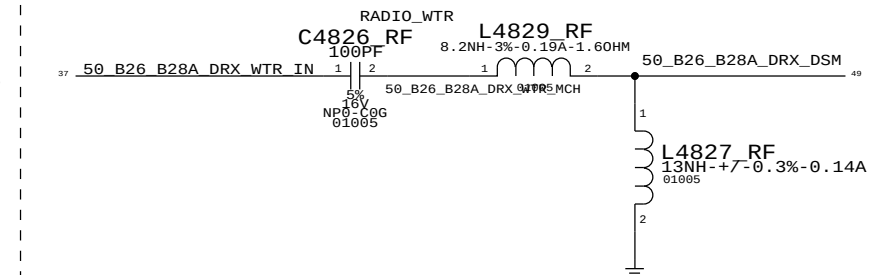
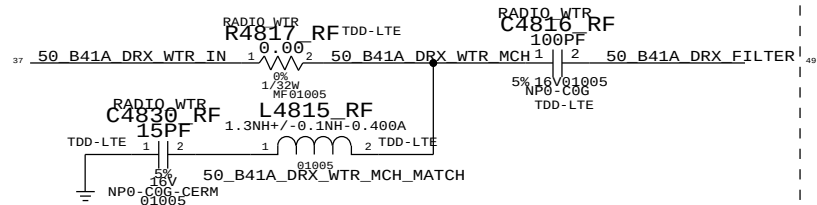
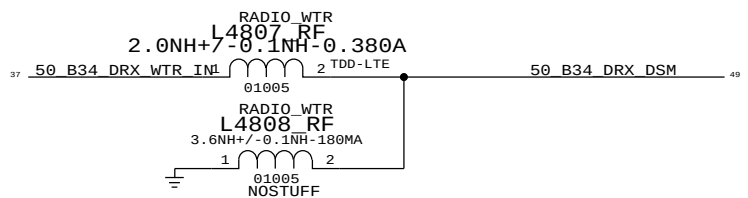
MIDBAND MIDBAND DIVERSITY - WFR

HIGHBAND DIVERSITY - WTR

LOWBAND DIVERSITY - WTR



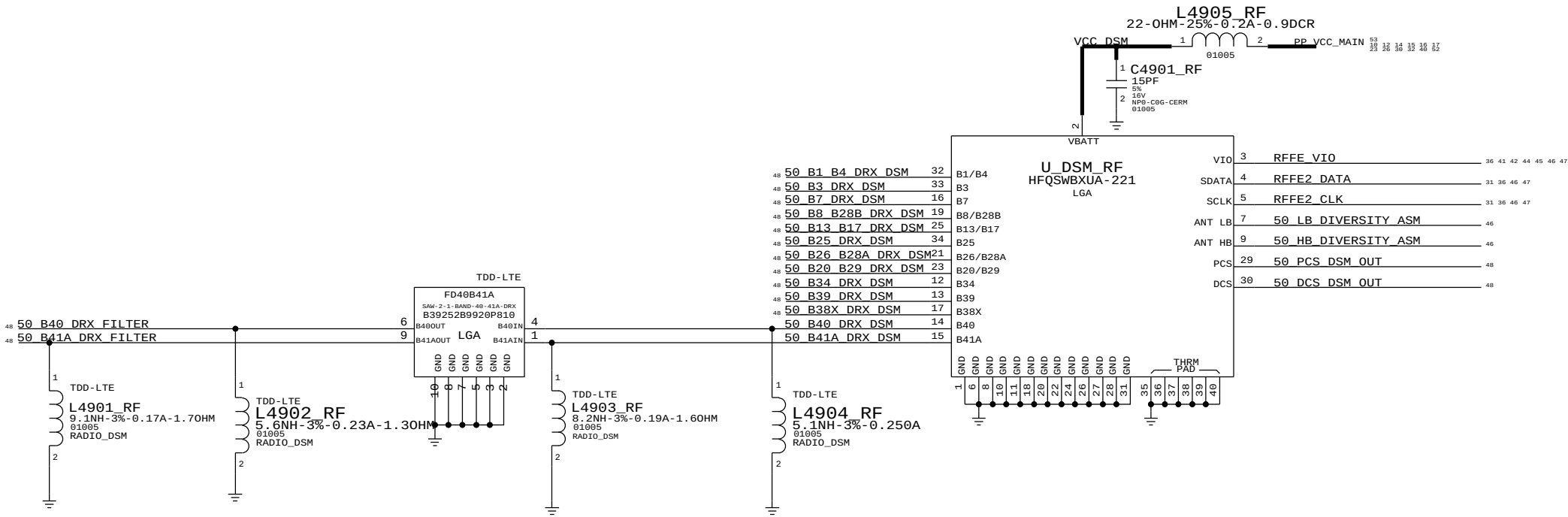
MIDBAND DIVERSITY - WTR



RX DIVERSITY (2)

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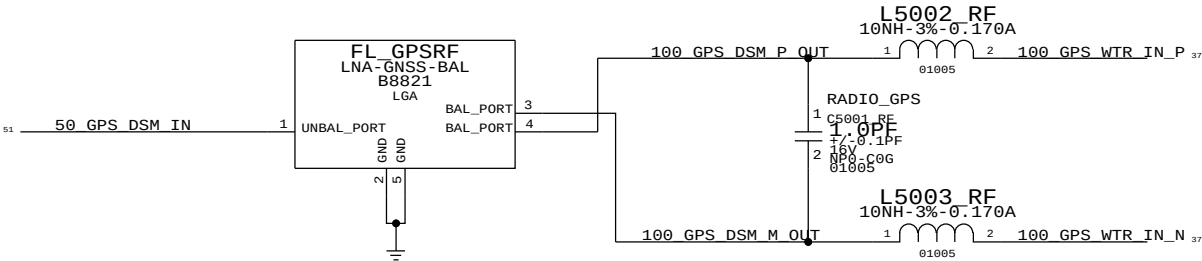
C1900
R1900
L1900
U1901



GPS

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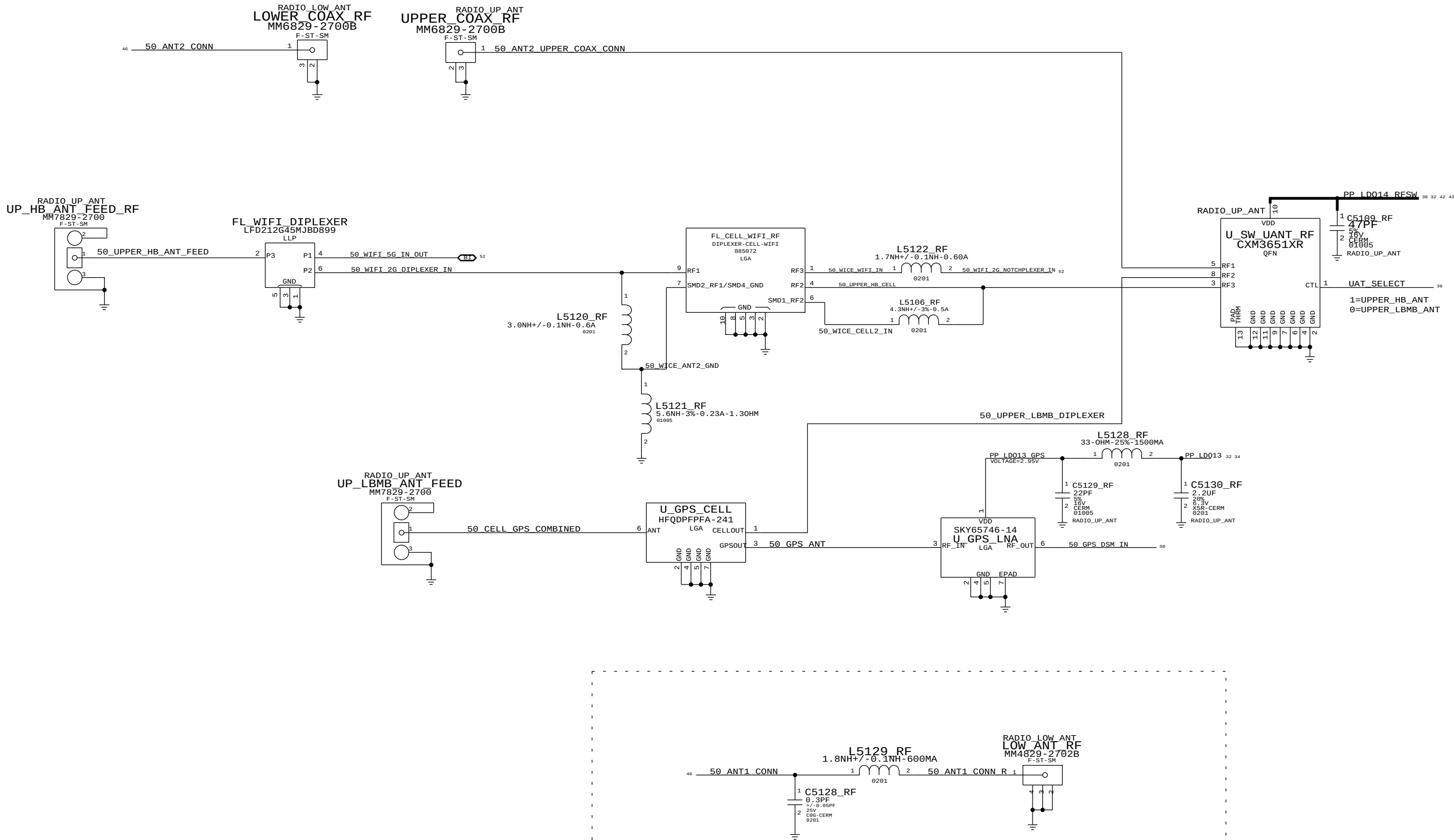
C1900
R1900
L1900
U1901



ANTENNA FEED'S

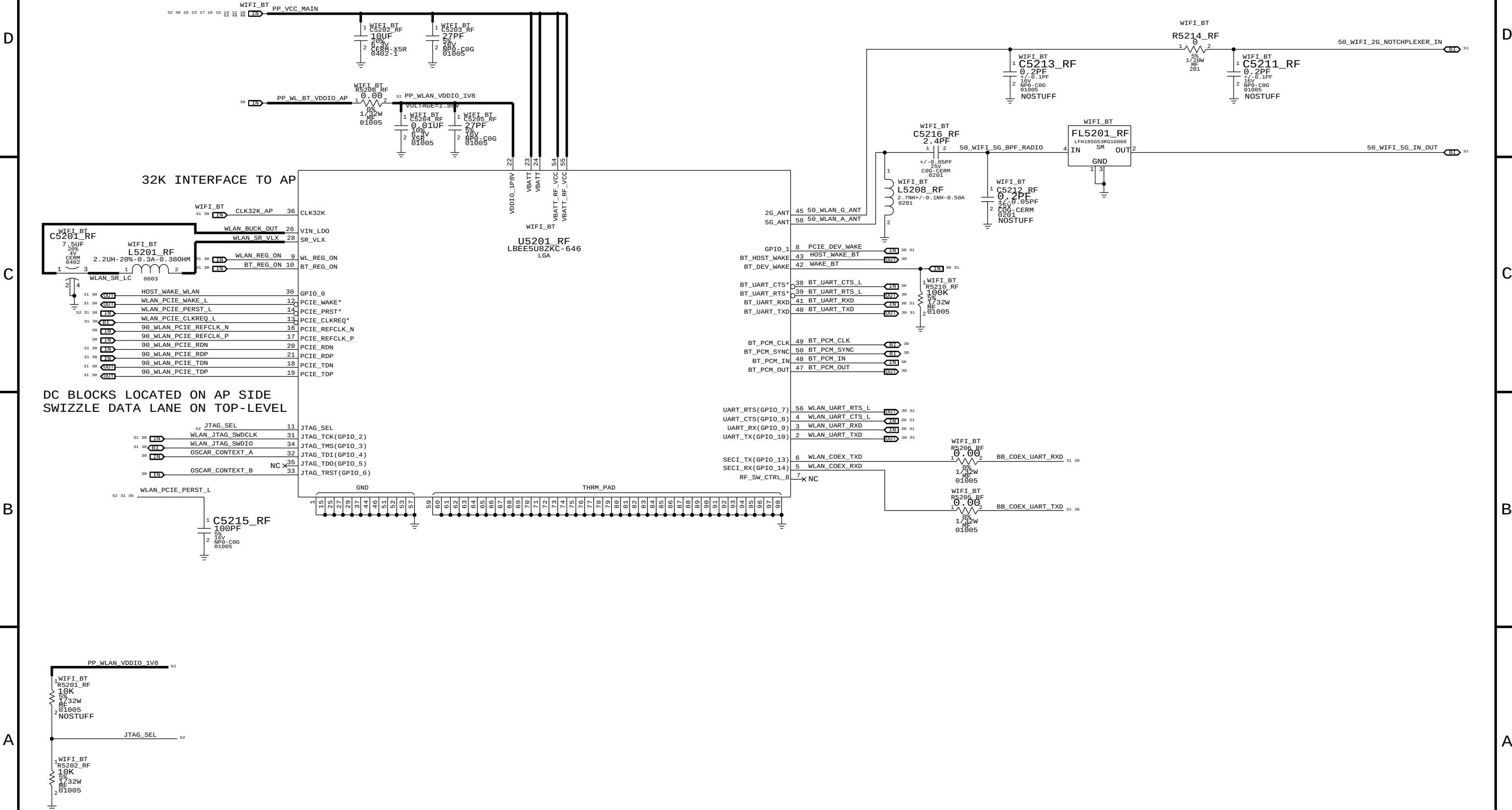
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TEST & COAX CONNECTOR FOR LOWER SECTION OF MLB



WLAN/BT

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MODULE BOOT-STRAPPED TO PCIe INTERNALLY

STOCKHOLM

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C2101
R2100
L2102
U2100

REMOVING BULK CAP 4.7UF 0402 -->
BECAUSE OF OTHER BULK CAPS IN LAYOUT

The diagram illustrates the electrical design for the STOCKHOLM system. It features several key components and connections:

- Power Rails:** PP_VCC_MAIN, PP_STOCKHOLM_VDD, PP_STOCKHOLM_ESE, PP_STOCKHOLM_1V8_S2R, RADIO_STOCKHOLM_VDD, RADIO_STOCKHOLM_ESE, RADIO_STOCKHOLM_RF, VDD, VDD_RF, VSS, VSS_RF, VSS_DMP.
- Capacitors:** C5302_RF (1.0UF), C5303_RF (0.1UF), C5304_RF (1.0UF), C5305_RF (0.1UF), C5306_RF (0.22UF), C5307_RF (0.22UF), C5308_RF (1.0UF), C5309_RF (0.022UF), C5310_RF (560PF), C5311_RF (560PF), C5312_RF (270PF), C5313_RF (33PF), C5314_RF (470PF), C5315_RF (330PF), C5316_RF (330PF).
- Resistors:** R5301_RF (100K), R5302_RF (100K), R5303 (10K), R5316_RF (0.00), R5317_RF (0.00), R5318_RF (0.00), R5319_RF (1.00K).
- Inductors:** L5301_RF (78NH-5%-0.97A-0.130HM), L5302_RF (78NH-5%-0.97A-0.130HM).
- Integrated Circuits:** PN65VEU3-B101D010, U5301_RF, U5302_RF (AS3923-B0-BWLT WLCSP), T5301_RF (ATB2012G-E-20011).
- Other Components:** TP5301_RF, TP5302_RF, TP5303_RF, TP5304_RF, NOSTUFF.
- Signal Lines:** STOCKHOLM_HOST_WAKE, STOCKHOLM_FW_DWLD_REQ, BB_REQUEST_XO_CLK, REF_CLK_FROM_BB, STOCKHOLM_UART_RXD, STOCKHOLM_UART_TXD, STOCKHOLM_CTS_L, STOCKHOLM_RTS_L, STOCKHOLM_ENABLE, STOCKHOLM_SIM_SWP, STOCKHOLM_SIM_VCC, STOCKHOLM_SIM_PMU_VCC, TX_PWR_REQ, ESE_DWPM_DBG, ESE_DWPS_DBG, ANT1, RXN/RX_CLK_RX, RXN/RX_CLK_TX, RF_DATA_IO, STOCKHOLM_VMID, STOCKHOLM_TIO, STOCKHOLM_CDMP2, STOCKHOLM_RF01, STOCKHOLM_RF02, STOCKHOLM_RFI1, STOCKHOLM_RFI2, STOCKHOLM_VSP_RF, STOCKHOLM_VSP.

The diagram is a detailed circuit board layout for a component labeled "STOCKHOLM". It features a coordinate grid with columns numbered 1 to 8 (from right to left) and rows labeled A to D (from bottom to top). The layout includes numerous electronic components and their interconnections:

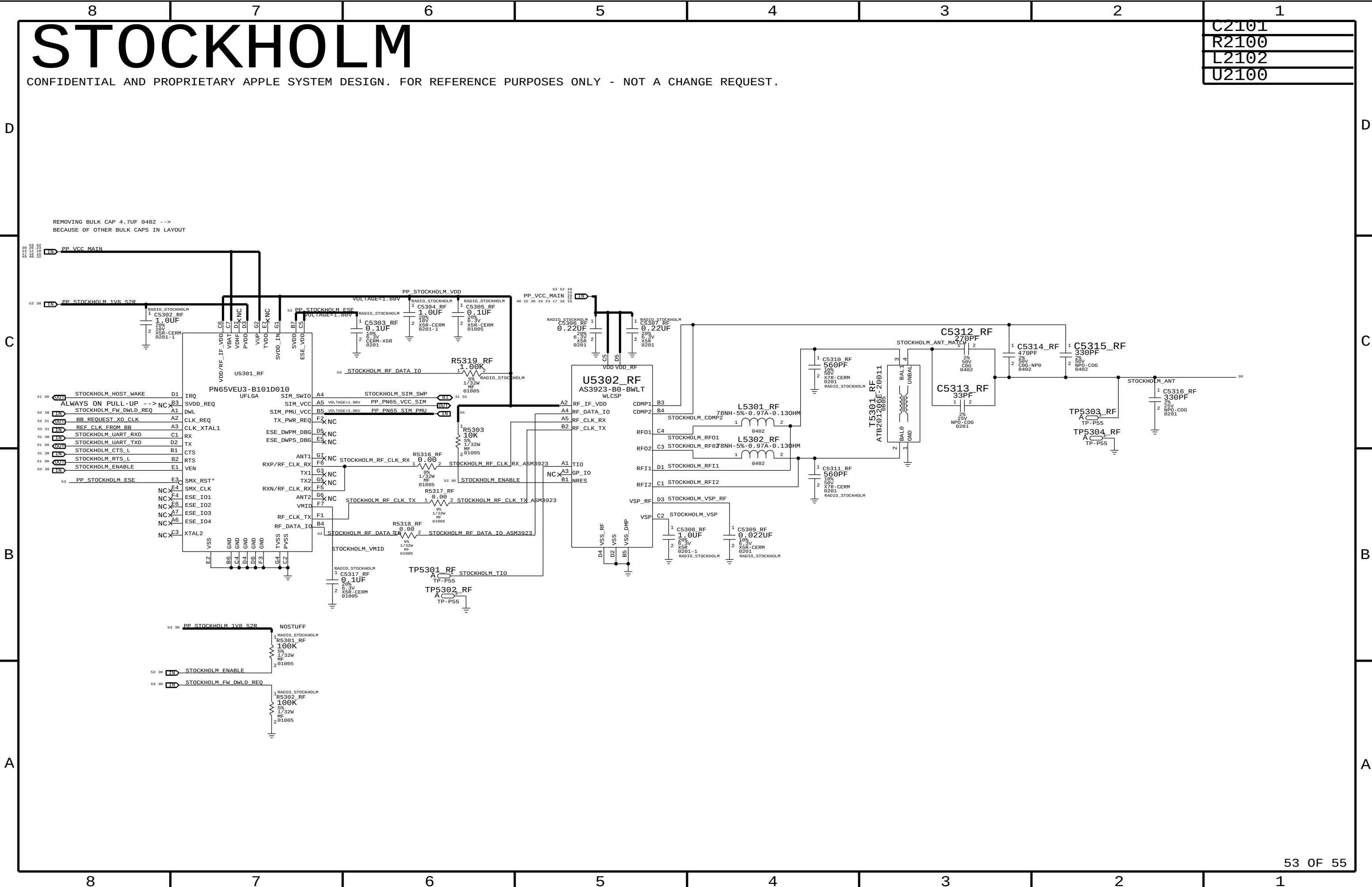
- Power and Grounding:** Includes connections for PP_VCC_MAIN, PP_STOCKHOLM_VDD, PP_STOCKHOLM_V1R_S2R, and various ground points (GND, TVSS, PVSS).
- Capacitors:** Various types are used, including electrolytic capacitors (e.g., C5302_RF, C5304_RF, C5305_RF, C5308_RF, C5309_RF, C5310_RF, C5311_RF, C5312_RF, C5313_RF, C5314_RF, C5315_RF, C5316_RF), ceramic capacitors (e.g., C5303_RF, C5306_RF, C5307_RF, C5317_RF), and NP0-COG capacitors (e.g., C5312_RF, C5313_RF, C5314_RF, C5315_RF, C5316_RF).
- Resistors:** Includes resistors like R5301_RF, R5302_RF, R5303, R5316_RF, R5317_RF, R5318_RF, R5319_RF, and R5301.
- Integrated Circuits (ICs):** Key components include the PN65VEU3-B101D010, U5301_RF, U5302_RF (AS3923-B0-BWLT WLCSP), and the T5301_RF (ATB20126E-20011).
- RF and Antenna:** Features RF components like L5301_RF, L5302_RF, and an antenna section with STOCKHOLM_ANT_MATCH, STOCKHOLM_ANT, and TP5303_RF, TP5304_RF.
- Signal and Control:** Includes various signal lines such as STOCKHOLM_HOST_WAKE, STOCKHOLM_FW_DWLD_REQ, BB_REQUEST_XO_CLK, REF_CLK_FROM_BB, STOCKHOLM_UART_RXD, STOCKHOLM_UART_TXD, STOCKHOLM_CTS_L, STOCKHOLM_RTS_L, STOCKHOLM_ENABLE, PP_STOCKHOLM_ESE, STOCKHOLM_SIM_SWP, STOCKHOLM_SIM_VCC, STOCKHOLM_SIM_PMU_VCC, TX_PWR_REQ, ESE_DWPM_DBG, ESE_DWPS_DBG, ANT1, TX1, TX2, RXN/RF_CLK_RX, RXM/RF_CLK_TX, RF_CLK_TX, RF_DATA_I0, STOCKHOLM_VMID, STOCKHOLM_RF_CLK_RX, STOCKHOLM_RF_CLK_TX, STOCKHOLM_RF_DATA_I0, STOCKHOLM_RF_DATA_I0, STOCKHOLM_TIO, and STOCKHOLM_VSP.

The layout is densely packed with components, particularly in the center and right-hand side, with a more open area on the left. The coordinate grid helps in locating specific components and traces across the board.

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The diagram is a detailed circuit board layout for a component labeled "STOCKHOLM". It features a coordinate grid with columns numbered 1 to 8 (from right to left) and rows labeled A to D (from bottom to top). The layout includes numerous electronic components and their interconnections:

- Capacitors:** Various types including electrolytic (e.g., C5302, C5304, C5305, C5308, C5309, C5310, C5311, C5312, C5313, C5314, C5315, C5316, C5317), ceramic (e.g., C5303, C5306, C5307, C5309, C5310, C5311, C5312, C5313, C5314, C5315, C5316, C5317), and NP0-COG (e.g., C5312, C5313, C5314, C5315, C5316, C5317).
- Resistors:** Various types including 100K (e.g., R5301, R5302), 10K (e.g., R5303), 1.00K (e.g., R5319), and 0.00 (e.g., R5316, R5317, R5318).
- Integrated Circuits (ICs):** U5301_RF, U5302_RF (AS3923-B0-BWLT WLCSP), PN65VEU3-B101D010, and various other components like C5302, C5304, C5305, C5308, C5309, C5310, C5311, C5312, C5313, C5314, C5315, C5316, C5317.
- Other Components:** T5301_RF (ATB201206E-20011), L5301_RF, L5302_RF, R5301, R5302, R5303, R5304, R5305, R5306, R5307, R5308, R5309, R5310, R5311, R5312, R5313, R5314, R5315, R5316, R5317, R5318, R5319, R5320, R5321, R5322, R5323, R5324, R5325, R5326, R5327, R5328, R5329, R5330, R5331, R5332, R5333, R5334, R5335, R5336, R5337, R5338, R5339, R5340, R5341, R5342, R5343, R5344, R5345, R5346, R5347, R5348, R5349, R5350, R5351, R5352, R5353, R5354, R5355, R5356, R5357, R5358, R5359, R5360, R5361, R5362, R5363, R5364, R5365, R5366, R5367, R5368, R5369, R5370, R5371, R5372, R5373, R5374, R5375, R5376, R5377, R5378, R5379, R5380, R5381, R5382, R5383, R5384, R5385, R5386, R5387, R5388, R5389, R5390, R5391, R5392, R5393, R5394, R5395, R5396, R5397, R5398, R5399, R5400, R5401, R5402, R5403, R5404, R5405, R5406, R5407, R5408, R5409, R5410, R5411, R5412, R5413, R5414, R5415, R5416, R5417, R5418, R5419, R5420, R5421, R5422, R5423, R5424, R5425, R5426, R5427, R5428, R5429, R5430, R5431, R5432, R5433, R5434, R5435, R5436, R5437, R5438, R5439, R5440, R5441, R5442, R5443, R5444, R5445, R5446, R5447, R5448, R5449, R5450, R5451, R5452, R5453, R5454, R5455, R5456, R5457, R5458, R5459, R5460, R5461, R5462, R5463, R5464, R5465, R5466, R5467, R5468, R5469, R5470, R5471, R5472, R5473, R5474, R5475, R5476, R5477, R5478, R5479, R5480, R5481, R5482, R5483, R5484, R5485, R5486, R5487, R5488, R5489, R5490, R5491, R5492, R5493, R5494, R5495, R5496, R5497, R5498, R5499, R5500, R5501, R5502, R5503, R5504, R5505, R5506, R5507, R5508, R5509, R5510, R5511, R5512, R5513, R5514, R5515, R5516, R5517, R5518, R5519, R5520, R5521, R5522, R5523, R5524, R5525, R5526, R5527, R5528, R5529, R5530, R5531, R5532, R5533, R5534, R5535, R5536, R5537, R5538, R5539, R5540, R5541, R5542, R5543, R5544, R5545, R5546, R5547, R5548, R5549, R5550, R5551, R5552, R5553, R5554, R5555, R5556, R5557, R5558, R5559, R5560, R5561, R5562, R5563, R5564, R5565, R5566, R5567, R5568, R5569, R5570, R5571, R5572, R5573, R5574, R5575, R5576, R5577, R5578, R5579, R5580, R5581, R5582, R5583, R5584, R5585, R5586, R5587, R5588, R5589, R5590, R5591, R5592, R5593, R5594, R5595, R5596, R5597, R5598, R5599, R5600, R5601, R5602, R5603, R5604, R5605, R5606, R5607, R5608, R5609, R5610, R5611, R5612, R5613, R5614, R5615, R5616, R5617, R5618, R5619, R5620, R5621, R5622, R5623, R5624, R5625, R5626, R5627, R5628, R5629, R5630, R5631, R5632, R5633, R5634, R5635, R5636, R5637, R5638, R5639, R5640, R5641, R5642, R5643, R5644, R5645, R5646, R5647, R5648, R5649, R5650, R5651, R5652, R5653, R5654, R5655, R5656, R5657, R5658, R5659, R5660, R5661, R5662, R5663, R5664, R5665, R5666, R5667, R5668, R5669, R5670, R5671, R5672, R5673, R5674, R5675, R5676, R5677, R5678, R5679, R5680, R5681, R5682, R5683, R5684, R5685, R5686, R5687, R5688, R5689, R5690, R5691, R5692, R5693, R5694, R5695, R5696, R5697, R5698, R5699, R5700, R5701, R5702, R5703, R5704, R5705, R5706, R5707, R5708, R5709, R5710, R5711, R5712, R5713, R5714, R5715, R5716, R5717, R5718, R5719, R5720, R5721, R5722, R5723, R5724, R5725, R5726, R5727, R5728, R5729, R5730, R5731, R5732, R5733, R5734, R5735, R5736, R5737, R5738, R5739, R5740, R5741, R5742, R5743, R5744, R5745, R5746, R5747, R5748, R5749, R5750, R5751, R5752, R5753, R5754, R5755, R5756, R5757, R5758, R5759, R5760, R5761, R5762, R5763, R5764, R5765, R5766, R5767, R5768, R5769, R5770, R5771, R5772, R5773, R5774, R5775, R5776, R5777, R5778, R5779, R5780, R5781, R5782, R5783, R5784, R5785, R5786, R5787, R5788, R5789, R5790, R5791, R5792, R5793, R5794, R5795, R5796, R5797, R5798, R5799, R5800, R5801, R5802, R5803, R5804, R5805, R5806, R5807, R5808, R5809, R5810, R5811, R5812, R5813, R5814, R5815, R5816, R5817, R5818, R5819, R5820, R5821, R5822, R5823, R5824, R5825, R5826, R5827, R5828, R5829, R5830, R5831, R5832, R5833, R5834, R5835, R5836, R5837, R5838, R5839, R5840, R5841, R5842, R5843, R5844, R5845, R5846, R5847, R5848, R5849, R5850, R5851, R5852, R5853, R5854, R5855, R5856, R5857, R5858, R5859, R5860, R5861, R5862, R5863, R5864, R5865, R5866, R5867, R5868, R5869, R5870, R5871, R5872, R5873, R5874, R5875, R5876, R5877, R58



ON-BOARD JUMPER FLEX

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UAT JUMPER

